

Project Data Page:





Table of Contents

Cover Page	1
Vessel Design Summary:	2
Nozzle Summary:	4
Nozzle Schedule:	5
Bill of Material:	6
MDMT Summary:	7
Warnings and Errors:	9
Input Echo:	10
XY Coordinate Calculations:	17
Internal Pressure Calculations:	18
Element: BOTTOM HEAD	19
Element: SHELL-1	22
Element: SHELL-2	24
Element: TOP HEAD	26
Shop Pressure Test Results:	29
Field Pressure Test Results:	31
External Pressure Calculations:	33
Element and Detail Weights:	34
Nozzle Flange MAWP:	37
Natural Frequency Calculation:	39
Wind Load Calculation:	40
Earthquake Load Calculation:	43
Wind/Earthquake Shear, Bending:	45
Wind/Earthquake Deflection:	46
Longitudinal Stress Constants:	47
Longitudinal Allowable Stresses:	48
Longitudinal Stresses due to:	49
Stress due to Combined Loads:	51
Center of Gravity Calculation:	55
Lifting Lug Calcs: Lifting Lug	57
Leg Check, (Operating Case):	65
Leg Check, (Filled w/Water):	74
Nozzle Calcs.: N3-4"	83
Nozzle Calcs.: N1-12"	93
Nozzle Calcs.: K1-2"	106
Nozzle Calcs.: K2-2"	111
Nozzle Calcs.: K4-2"	116
Nozzle Calcs.: K5-2"	121
Nozzle Calcs.: K3-8"	126
Nozzle Calcs.: N2-8"	140
Nozzle Calcs.: K6-2"	150
Nozzle Calcs.: K7-2"	157



DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\HP Steam Blowdown Drum (186-D)Rev0- 7%.pvdb

Date of Analysis : Sep 16,2026 2:19pm

PV Elite 28 U1, March 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 1075.000 mm. ID	
Vessel Design Length, Tangent to Tangent	3275.00 mm.
Distance of Bottom Tangent above Grade	300.00 mm.
Specified Datum Line Distance	0.00 mm.
Internal Design Temperature	200 °C
Internal Design Pressure	6.400 bars
External Design Temperature	200 °C
External Design Pressure	1.034 bars
Maximum Allowable Working Pressure	6.4 bars
External Max. Allowable Working Pressure	3.12 bars
Shop Test Pressure	22.555 bars
Field Test Pressure	8.320 bars
Required Minimum Design Metal Temperature	-12.0 °C
Warmest Computed Minimum Design Metal Temperature	-45.0 °C
Wind Design Code	ASCE/SEI 7-16
Earthquake Design Code	ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Nozzle	SA-106 B	...	...	K03006	
Nozzle	SA-105	...	...	K03504	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Leg Baseplate		...	...		
Leg Bolting	SA-307 A	...	6 ≤ t ≤ 100		

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (bars & mm.):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
BOTTOM HEAD	6.624	1.03	10.700	3.0000	No	No
SHELL-1	6.595	1.03	19.100	3.0000	N/A	No
SHELL-2	6.407	1.03	19.300	3.0000	N/A	No
TOP HEAD	6.407	1.03	10.900	3.0000	No	No

Liquid Level: 2418.75 mm. Dens.: 0.001 kg./cm³ Sp. Gr.: 0.917

Element Types and Properties:

Element Type	"To" Elev mm.	Element Length mm.	Nominal Thickness mm.	Finished Thickness mm.	Reqd Thk Internal mm.	Reqd Thk External mm.	Long Eff	Circ Eff
Ellipse	50.0	50.0	10.0	8.0	6.0	5.9	0.85	0.85
Cylinder	2550.0	2500.0	12.0	12.0	6.0	8.8	0.85	0.85
Cylinder	3225.0	675.0	12.0	12.0	6.0	8.8	0.85	0.85
Ellipse	3275.0	50.0	10.0	8.0	5.9	5.9	0.85	0.85



Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
Lift Lug	WRC-107/537	0.145	n/a	Passed
LEGS	WRC-107/537	0.281	n/a	Passed
LEGS	WRC-107/537	0.186	n/a	Passed
N3-4"	WRC-107/537	0.466	n/a	Passed
N1-12"	WRC-297	0.986	Shell	Passed
K3-8"	WRC-297	0.884	Shell	Passed
N2-8"	WRC-107/537	0.661	n/a	Passed

Loads for Foundation/Support Design:

Total Wind Shear on top of all Legs	244. Kgf
Total Earthquake Shear on top of all Legs	208. Kgf
Total Wind Moment at top of all Legs	307. Kg-m.
Total Earthquake Moment at top of all Legs	335. Kg-m.
Wind Load Case Test Moment on Support	153. Kg-m.
Test Case Wind Shear on Support	122. Kgf
Max. Wind Shear on one Leg (top & bottom)	97. Kgf
Max. Earthq. Shear on one Leg (top & bottom)	83. Kgf
Max. Wind Moment at base of one Leg	178. Kg-m.
Max. Earthquake Moment at base of one Leg	152. Kg-m.
Max. Vertical Load (Wt. + Wind) on one Leg	1347. Kgf
Max. Vertical Load (Wt. + Eq.) on one Leg	1370. Kgf

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	2408.4 kg.
Shop Test - Fabricated + water (full)	5704.7 kg.
Shipping - Fab. + rem. intls.+ shipping app.	2408.4 kg.
Erected - Fab. + rem. intls.+ insul. (etc)	2880.4 kg.
Empty - Fab. + intls. + details + wghts.	2880.4 kg.
Operating - Empty + operating liquid (no ca)	4818.0 kg.
Field Test - Empty weight + test medium (full)	6156.6 kg.

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Nozzle Calculation Summary (bars & mm.):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N3-4"	6.4	OK	...	OK	5.58	OK	Passed
N1-12"	6.4	OK	...	OK	5.60	OK	Passed
K1-2"	6.4	OK	...	OK	5.60	OK	Passed
K2-2"	6.4	OK	...	OK	5.60	OK	Passed
K4-2"	6.4	OK	...	OK	5.60	OK	Passed
K5-2"	6.4	OK	...	OK	5.60	OK	Passed
K3-8"	6.4	OK	...	OK	5.60	OK	Passed
N2-8"	6.4	OK	...	OK	5.58	OK	Passed
K6-2"	6.4	OK	...	OK	5.58	OK	Passed
K7-2"	6.4	OK	...	OK	5.58	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 6.4 Nozzle : K7-2"
 Minimum MAWP Shells/Flanges : 6.4 Element : BOTTOM HEAD
 Minimum MAPnc Shells/Flanges : 17.4 Element : TOP HEAD

Computed Vessel M.A.W.P. : 6.4 bars

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm.	Layout Angle deg	Dia. Limit mm.
10	N3-4"	0.000	0.000	196.099
20	N1-12"	2050.000	90.000	589.901
20	K1-2"	2300.000	270.000	113.600
20	K2-2"	175.000	270.000	113.600
20	K4-2"	1850.000	180.000	113.600
20	K5-2"	150.000	90.000	113.600
30	K3-8"	2850.000	270.000	399.350
40	N2-8"	0.000	0.000	399.350
40	K6-2"	0.000	270.000	113.600
40	K7-2"	0.000	90.000	113.600

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
K1-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	253.66	300
K2-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	253.66	300
K4-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	253.66	300
K5-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	253.66	300
K6-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	505.95	300
K7-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	505.95	300
N3-4"	4.000 in	120	WNF	4.500	11.125	220.00	8.00	169.68	300
K3-8"	8.000 in	XS	WNF	8.625	12.700	400.00	10.00	353.28	300
N2-8"	8.000 in	XS	WNF	8.625	12.700	350.00	8.00	344.18	300
N1-12"	12.000 in	80	WNF	12.750	17.450	750.00	10.00	366.97	600

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
K1-2"	SA-105	8.000	10.000	...	...	...
K2-2"	SA-105	8.000	10.000	...	...	...
K4-2"	SA-105	8.000	10.000	...	...	...
K5-2"	SA-105	8.000	10.000	...	...	...
K6-2"	SA-105	8.000	10.000	...	...	...
K7-2"	SA-105	8.000	10.000	...	...	...
N3-4"	SA-106 B	8.000	10.000	6.000	8.000	...
K3-8"	SA-106 B	8.000	10.000	8.000	10.000	...
N2-8"	SA-106 B	8.000	10.000	6.000	8.000	...
N1-12"	SA-106 B	8.000	10.000	8.000	10.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
K1-2"	2300.000	270.0	240.00	0.00	SHELL-1
K2-2"	175.000	270.0	240.00	0.00	SHELL-1
K4-2"	1850.000	180.0	240.00	0.00	SHELL-1
K5-2"	150.000	90.0	240.00	0.00	SHELL-1
K6-2"	...	270.0	480.00	0.00	TOP HEAD
K7-2"	...	90.0	480.00	0.00	TOP HEAD
N3-4"	...	0.0	160.00	0.00	BOTTOM HEAD
K3-8"	2850.000	270.0	330.00	0.00	SHELL-2
N2-8"	...	0.0	330.00	0.00	TOP HEAD
N1-12"	2050.000	90.0	330.00	0.00	SHELL-1



Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 10.0mm. THK X 1075.0mm. ID X 50.0mm.	SA-516 70
2	INSULATION: 318mm X 80mm THK	
1	INSULATION: 2500mm X 80mm THK	
4	LEGS, SHAPE: L200X200X16 1830.0mm	...
4	BASEPLATES: 350.0mm X 350.0mm X 20.0mm	SA-516 70
4	BASEPLATE BOLTS: 36.0mm	SA-307 A
4	LEG RE-PADS: 350.000mm X 350.000mm X 10.000mm	SA-516 70
1	INSULATION: 675mm X 80mm THK	
2	LIFTING LUGS:	SA-516 70
2	LIFTING LUG PADS: 455.0mm X 530.0mm X 12.0mm THK	
2	CYLINDRICAL SEGMENT 12.0mm. THK X 1343.7mm. X 2500.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 12.0mm. THK X 689.7mm. X 2500.0mm.	SA-516 70
4	CYLINDRICAL SEGMENT 12.0mm. THK X 727.2mm. X 675.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 12.0mm. THK X 468.5mm. X 675.0mm.	SA-516 70
1	NAMEPLATE	...

**Minimum Design Metal Temperature Results Summary (°C):**

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm.	E*	PWHT reqd
BOTTOM HEAD	[10]	B	-29	-48		0.498	8.000	0.85	No
BOTTOM HEAD	[7]	B	-29	-48		0.359	10.000	0.85	No
SHELL-1	[8]	B	-23	-104		0.279	12.000	0.85	No
SHELL-2	[8]	B	-23	-104		0.279	12.000	0.85	No
TOP HEAD	[10]	B	-29	-48		0.498	8.000	0.85	No
TOP HEAD	[7]	B	-29	-48		0.359	10.000	0.85	No
N3-4"	[1]	B	-29	-48		0.465	8.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N1-12"	[1]	B	4	-104		0.074	15.269	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K1-2"	[1]	A	-3	-104		0.289	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K2-2"	[1]	A	-3	-104		0.289	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K4-2"	[1]	A	-3	-104		0.289	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K5-2"	[1]	A	-3	-104		0.289	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K3-8"	[1]	B	-5	-104		0.076	11.113	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N2-8"	[1]	B	-29	-48		0.465	8.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K6-2"	[1]	A	-8	-45		0.465	8.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K7-2"	[1]	A	-8	-45		0.465	8.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
Nozl. Bolting[22]			-48			1.000	64.000	1.00	No

Warmest MDMT: 4 -45

Required Minimum Design Metal Temperature -12.0 °C

Warmest Computed Minimum Design Metal Temperature -45.0 °C

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.



UCS-66(i) applies to impact tested materials not by specification and
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (barsUniCode):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm2
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm2
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm3
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm3
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm3
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm2
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m2
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm3
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	6.3996	bars
Design Internal Temperature	200.0	°C
Projection of Nozzle from Vessel Top	0	mm.
Projection of Nozzle from Vessel Bottom	0	mm.
Minimum Design Metal Temperature	-12.0	°C
Type of Construction	Welded	
Special Service	None	
Degree of Radiography	RT-1	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW



Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW
Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm.
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.
Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.300
Table Value Fv	1.500
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	2.000
Site Class	C
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000
Overall M.A.W.P. + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N
Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/4
Element From Node	10



Element To Node	20
Element Type	Elliptical
Description	BOTTOM HEAD
Distance "FROM" to "TO"	50 mm.
Inside Diameter	1075 mm.
Element Thickness	8 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	10 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	6.3996 bars
Design Temperature Internal Pressure	200 °C
Design External Pressure	1.0342 bars
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.3
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm2
Allowable Stress, Operating	138 N./mm2
Allowable Stress, Hydrotest	234 N./mm2
Material Density	0.00775 kg./cm3
P Number Thickness	32.004 mm.
Yield Stress, Operating	225 N./mm2
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	10
Detail Type	Liquid
Detail ID	Liquid: 10
Dist. from "FROM" Node / Offset dist	-268.75 mm.
Height/Length of Liquid	318.75 mm.
Liquid Density	0.0009155 kg./cm3

Element From Node	10
Detail Type	Insulation
Detail ID	Insulation-2
Dist. from "FROM" Node / Offset dist	-268.75 mm.
Height/Length of Insulation	318.75 mm.
Thickness of Insulation	80 mm.
Density	0.000352 kg./cm3

Element From Node	10
Detail Type	Nozzle
Detail ID	N3-4"
Dist. from "FROM" Node / Offset dist	0 mm.
Nozzle Diameter	4 in.
Nozzle Schedule	120
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	48.218 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element From Node	10
Detail Type	Weight
Detail ID	Anti-Swirl Baff
Dist. from "FROM" Node / Offset dist	-268 mm.
Miscellaneous Weight	25 Kg
Offset from Element Centerline	0 mm.



Element Type	Cylinder
Description	SHELL-1
Distance "FROM" to "TO"	2500 mm.
Inside Diameter	1075 mm.
Element Thickness	12 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	12 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	6.3996 bars
Design Temperature Internal Pressure	200 °C
Design External Pressure	1.0342 bars
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.3
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No
Element From Node	20
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	2100 mm.
Liquid Density	0.0009155 kg./cm3
Element From Node	20
Detail Type	Insulation
Detail ID	Ins: 20
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	2500 mm.
Thickness of Insulation	80 mm.
Density	0.000352 kg./cm3
Element From Node	20
Detail Type	Nozzle
Detail ID	N1-12"
Dist. from "FROM" Node / Offset dist	2000 mm.
Nozzle Diameter	12 in.
Nozzle Schedule	80
Nozzle Class	600
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	184.69 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B
Element From Node	20
Detail Type	Nozzle
Detail ID	K1-2"
Dist. from "FROM" Node / Offset dist	2250 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	11.379 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105
Element From Node	20
Detail Type	Nozzle
Detail ID	K2-2"
Dist. from "FROM" Node / Offset dist	125 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	11.379 Kg
Grade of Attached Flange	GR 1.1



Nozzle Matl	SA-105	
Element From Node	20	
Detail Type	Nozzle	
Detail ID	K4-2"	
Dist. from "FROM" Node / Offset dist	1800	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	11.379	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	
Element From Node	20	
Detail Type	Nozzle	
Detail ID	K5-2"	
Dist. from "FROM" Node / Offset dist	100	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	11.379	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	
Element From Node	20	
Detail Type	Leg	
Detail ID	LEGS	
Dist. from "FROM" Node / Offset dist	330	mm.
Diameter at Leg Centerline	1228.8	mm.
Leg Orientation	3	
Number of Legs	4	
Section Identifier	L200X200X16	
Length of Legs	1830	mm.
Element From Node	20	
Detail Type	Weight	
Detail ID	Clip Weight	
Dist. from "FROM" Node / Offset dist	1250	mm.
Miscellaneous Weight	90	Kgf
Offset from Element Centerline	0	mm.

Element#	3/4	
Element From Node	30	
Element To Node	40	
Element Type	Cylinder	
Description	SHELL-2	
Distance "FROM" to "TO"	675	mm.
Inside Diameter	1075	mm.
Element Thickness	12	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	12	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	6.3996	bars
Design Temperature Internal Pressure	200	°C
Design External Pressure	1.0342	bars
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.3	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	No	

Element From Node	30
Detail Type	Insulation



Detail ID	Ins: 30	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Insulation	675	mm.
Thickness of Insulation	80	mm.
Density	0.000352	kg./cm ³

Element From Node	30	
Detail Type	Nozzle	
Detail ID	K3-8"	
Dist. from "FROM" Node / Offset dist	300	mm.
Nozzle Diameter	8	in.
Nozzle Schedule	XS	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	62.828	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	30	
Detail Type	Weight	
Detail ID	Demisting Pad	
Dist. from "FROM" Node / Offset dist	338	mm.
Miscellaneous Weight	20	Kgf
Offset from Element Centerline	0	mm.

Element From Node	30	
Detail Type	Weight	
Detail ID	Demistin Support	
Dist. from "FROM" Node / Offset dist	338	mm.
Miscellaneous Weight	28	Kgf
Offset from Element Centerline	0	mm.

Element#	4/4	
Element From Node	40	
Element To Node	50	
Element Type	Elliptical	
Description	TOP HEAD	
Distance "FROM" to "TO"	50	mm.
Inside Diameter	1075	mm.
Element Thickness	8	mm.
Internal Corrosion Allowance	3	mm.
Nominal Thickness	10	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	6.3996	bars
Design Temperature Internal Pressure	200	°C
Design External Pressure	1.0342	bars
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.3	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	

Element From Node	40	
Detail Type	Insulation	
Detail ID	Ins: 40	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Insulation	318.75	mm.
Thickness of Insulation	80	mm.
Density	0.000352	kg./cm ³

Element From Node	40	
Detail Type	Nozzle	
Detail ID	N2-8"	
Dist. from "FROM" Node / Offset dist	0	mm.
Nozzle Diameter	8	in.



Nozzle Schedule	XS	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	59.382	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	K6-2"	
Dist. from "FROM" Node / Offset dist	350	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	17.998	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	K7-2"	
Dist. from "FROM" Node / Offset dist	350	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	17.998	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

XY Coordinate Calculations:

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
BOTTOM HEAD		...	50	...	50
SHELL-1		...	2550	...	2500
SHELL-2		...	3225	...	675
TOP HEAD		...	3275	...	50

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd bars	Nominal Thickness mm.	Total Corr Allowance mm.	Element Diameter mm.	Allowable Stress(SE) N./mm ²	Element#
BOTTOM HEAD		6.624	10	3	1075	117.3	1
SHELL-1		6.5954	12	3	1075	117.3	2
SHELL-2		6.4066	12	3	1075	117.3	3
TOP HEAD		6.4066	10	3	1075	117.3	4

Element Required Thickness and MAWP :

From	To	Design Pressure bars	M.A.W.P. Corroded bars	M.A.P. New & Cold bars	Minimum Thickness mm.	Required Thickness mm.	Element#
BOTTOM HEAD		6.39963	10.7	17.4	8	6.03179	1
SHELL-1		6.39963	19.1	25.8	12	6.04957	2
SHELL-2		6.39963	19.3	25.8	12	5.96201	3
TOP HEAD		6.39963	10.9	17.4	8	5.93223	4

Minimum 6.4 17.4

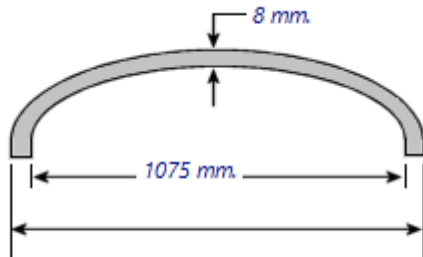
MAWP: 6.4 bars, limited by: DESIGN (user specified)

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: BOTTOM HEAD

Elliptical Head Design Information:							
Design Pressure	P	6.624	bars	Int. Design Temperature	200	°C	
Material		SA-516	70	Ext. Design Temperature	200	°C	
Length		50.000	mm.	External Pressure Chart	CS-2		
				UNS Number	K02700		
Int. Corr. All.	c	3.0000	mm.	Allowable Stress (ope)	S	138.0	N./mm^2
Ext. Corr. All.	ce	0.0000	mm.	Allowable Stress (amb)	Sa	138.0	N./mm^2
Inside Diameter	Di	1075.000	mm.	Actual Stress		83.7	N./mm^2
Spec. Min. Thk t,e, or sa		8.0000	mm.	Spec. Nominal Thk.		10.000	mm.
Surface Area		14717.3	cm^2	SG of Contents		0.917	
Element Volume		208034.8	cm3	Weight of Contents		190.7	Kgf
Empty Weight		122.4	Kgf	Operating Weight		313.1	Kgf
				Comp. All. Ope. Stress		97.0	N./mm^2
				Aspect Ratio		2.00	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 13.438 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 6.719 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (6.62 \cdot 1081 \cdot 0.99) / (2 \cdot 138 \cdot 0.85 - 0.2 \cdot 6.62)$$

$$= 3.0318 + 3.0000 = 6.0318 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.224 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 0.85 \cdot 5) / (0.99 \cdot 1081 + 0.2 \cdot 5)$$

$$= 10.9 - 0.22 = 10.7 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 0.85 \cdot 8) / (1 \cdot 1075 + 0.2 \cdot 8)$$

$$= 17.4 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (6.62 \cdot (0.99 \cdot 1081 + 0.2 \cdot 5)) / (2 \cdot 0.85 \cdot 5)$$

$$= 83.707 \text{ N./mm}^2$$

Straight Flange Required Thickness:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c$ per UG-27 (c)(1)
 $= (6.62 \cdot 540) / (138 \cdot 0.85 - 0.6 \cdot 6.62) + 3$
 $= 6.063 \text{ mm.}$

Straight Flange Maximum Allowable Working Pressure:
 Less Operating Hydrostatic Head Pressure of 0.200 bars
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (138 \cdot 0.85 \cdot 7) / (540 + 0.6 \cdot 7)$
 $= 15.1 - 0.2 = 14.9 \text{ bars}$

Factor K, corroded condition [Kcor]:
 $= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$
 $= (2 + (1081 / (2 \cdot 272))^2) / 6$
 $= 0.992661$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 3.995 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 8$, $t_r = 2.93$, $c = 3 \text{ mm.}$, $E^* = 0.85$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.5$, Temp. Reduction = 33 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 10$, $t_r = 2.96$, $c = 3 \text{ mm.}$, $E^* = 0.85$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.36$, Temp. Reduction = 71 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	6.400 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	10.000 mm.	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.359	Temperature Difference	19 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		Joint Efficiency E^*	85 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 13.438 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 6.719 mm.

EXTERNAL PRESSURE RESULTS for: BOTTOM HEAD

Elliptical Head From 10 to 20 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 193602528 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
5.000	1091.00	218.20	0.0006365	61.62

$MAEP = B / (K_0 \cdot Do / t) = 61.6 / (0.9 \cdot 218) = 3.14 \text{ bars}$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B

PV Elite 28 U1 Licensee: SL User
FileName : HP Steam Blowdown Drum (186-D)_Rev0- 7% Page 21 of 163
Element: BOTTOM HEAD Step: 4 2:19pm Sep 16,2026



HEXAGON

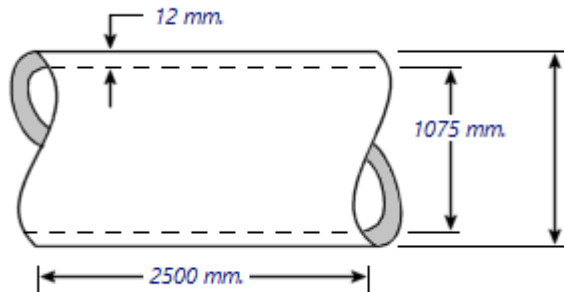
2.871	1091.00	380.06	0.0003654	35.38
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MAEP = $B/(K0*Do/t)$ = 35.4/(0.9 *380) = 1.03 bars

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: SHELL-1

Cylindrical Shell Design Information:					
Design Pressure	P	6.595 bars	Int. Design Temperature		200 °C
Material		SA-516 70	Ext. Design Temperature		200 °C
Length		2500.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	1075.000 mm.	Actual Stress		47.1 N./mm^2
Spec. Min. Thk t _e , or s _a		12.0000 mm.	Spec. Nominal Thk.		12.000 mm.
Surface Area		86315.3 cm^2	SG of Contents		0.917
Element Volume		2269471.0 cm3	Weight of Contents		1747.0 Kgf
Empty Weight		794.0 Kgf	Operating Weight		2541.0 Kgf
			Comp. All. Ope. Stress		111.4 N./mm^2

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (6.6 \cdot 540) / (138 \cdot 0.85 - 0.6 \cdot 6.6)$$

$$= 3.0496 + 3.0000 = 6.0496 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.196 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 9) / (540 + 0.6 \cdot 9)$$

$$= 19.3 - 0.2 = 19.1 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 12) / (538 + 0.6 \cdot 12)$$

$$= 25.8 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (6.6 \cdot (540 + 0.6 \cdot 9)) / (0.85 \cdot 9)$$

$$= 47.067 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.104 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 12, tr = 2.96, c = 3 mm., E* = 0.85



Thickness Ratio = $tr * E / (tg - c) = 0.28$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	6.400 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	12.000 mm.	Basic MDMT	-23 °C
Stress/Thk. Ratio	0.279	Temperature Difference	81 °C
UCS-66(c) red.	No	Computed Min. Temp.	-104 °C
		Joint Efficiency E*	85 %

EXTERNAL PRESSURE RESULTS for: SHELL-1

Cylindrical Shell From 20 to 30 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 193602528 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.000	1099.00	3454.17	122.11	3.1430	0.0002955	28.61

$$MAEP = (4*B)/(3*(Do/t)) = (4*28.6)/(3*122) = 3.12 \text{ bars}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
5.751	1099.00	3454.17	191.11	3.1430	0.0001531	14.82

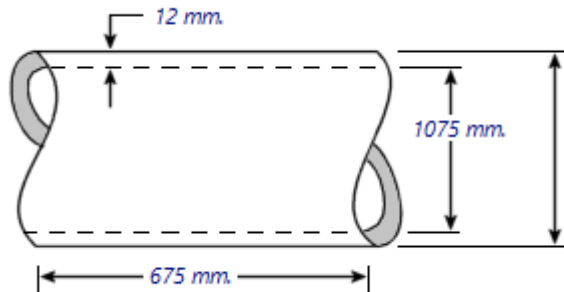
$$MAEP = (4*B)/(3*(Do/t)) = (4*14.8)/(3*191) = 1.03 \text{ bars}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.000	1099.00	9572.46	122.11	8.7102	0.0000979	9.48

$$MAEP = (4*B)/(3*(Do/t)) = (4*9.48)/(3*122) = 1.03 \text{ bars}$$

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: SHELL-2

Cylindrical Shell Design Information:					
Design Pressure	P	6.407 bars	Int. Design Temperature		200 °C
Material		SA-516 70	Ext. Design Temperature		200 °C
Length		675.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	1075.000 mm.	Actual Stress		45.7 N./mm^2
Spec. Min. Thk t _e , or s _a		12.0000 mm.	Spec. Nominal Thk.		12.000 mm.
Surface Area		23305.1 cm^2	SG of Contents		0.000
Element Volume		612757.1 cm3	Weight of Contents		0.0 Kgf
Empty Weight		214.4 Kgf	Operating Weight		214.4 Kgf
			Comp. All. Ope. Stress		111.4 N./mm^2

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (6.41 \cdot 540) / (138 \cdot 0.85 - 0.6 \cdot 6.41)$$

$$= 2.9620 + 3.0000 = 5.9620 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.007 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 9) / (540 + 0.6 \cdot 9)$$

$$= 19.3 - 0.007 = 19.3 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 12) / (538 + 0.6 \cdot 12)$$

$$= 25.8 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (6.41 \cdot (540 + 0.6 \cdot 9)) / (0.85 \cdot 9)$$

$$= 45.720 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.104 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 12, tr = 2.96, c = 3 mm., E* = 0.85



Thickness Ratio = $tr * E / (tg - c) = 0.28$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	6.400 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	12.000 mm.	Basic MDMT	-23 °C
Stress/Thk. Ratio	0.279	Temperature Difference	81 °C
UCS-66(c) red.	No	Computed Min. Temp.	-104 °C
		Joint Efficiency E*	85 %

EXTERNAL PRESSURE RESULTS for: SHELL-2

Cylindrical Shell From 30 to 40 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 193602528 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.000	1099.00	3454.17	122.11	3.1430	0.0002955	28.61

$$MAEP = (4*B)/(3*(Do/t)) = (4*28.6)/(3*122) = 3.12 \text{ bars}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
5.751	1099.00	3454.17	191.11	3.1430	0.0001531	14.82

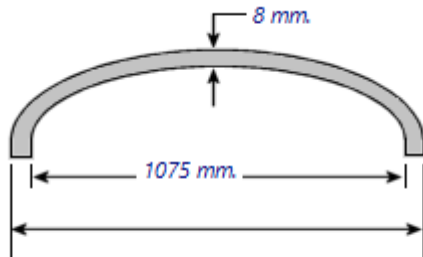
$$MAEP = (4*B)/(3*(Do/t)) = (4*14.8)/(3*191) = 1.03 \text{ bars}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.000	1099.00	9572.46	122.11	8.7102	0.0000979	9.48

$$MAEP = (4*B)/(3*(Do/t)) = (4*9.48)/(3*122) = 1.03 \text{ bars}$$

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: TOP HEAD

Elliptical Head Design Information:					
Design Pressure	P	6.407 bars	Int. Design Temperature		200 °C
Material		SA-516 70	Ext. Design Temperature		200 °C
Length		50.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	1075.000 mm.	Actual Stress		81.0 N./mm^2
Spec. Min. Thk t,e, or sa		8.0000 mm.	Spec. Nominal Thk.		10.000 mm.
Surface Area		14717.3 cm^2	SG of Contents		0.000
Element Volume		208034.8 cm3	Weight of Contents		0.0 Kgf
Empty Weight		122.4 Kgf	Operating Weight		122.4 Kgf
			Comp. All. Ope. Stress		97.0 N./mm^2
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 13.438 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 6.719 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (6.41 \cdot 1081 \cdot 0.99) / (2 \cdot 138 \cdot 0.85 - 0.2 \cdot 6.41)$$

$$= 2.9322 + 3.0000 = 5.9322 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.007 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 0.85 \cdot 5) / (0.99 \cdot 1081 + 0.2 \cdot 5)$$

$$= 10.9 - 0.007 = 10.9 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 0.85 \cdot 8) / (1 \cdot 1075 + 0.2 \cdot 8)$$

$$= 17.4 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (6.41 \cdot (0.99 \cdot 1081 + 0.2 \cdot 5)) / (2 \cdot 0.85 \cdot 5)$$

$$= 80.959 \text{ N./mm}^2$$

Straight Flange Required Thickness:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c$ per UG-27 (c)(1)
 $= (6.41 \cdot 540) / (138 \cdot 0.85 - 0.6 \cdot 6.41) + 3$
 $= 5.962 \text{ mm.}$

Straight Flange Maximum Allowable Working Pressure:
 Less Operating Hydrostatic Head Pressure of 0.007 bars
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (138 \cdot 0.85 \cdot 7) / (540 + 0.6 \cdot 7)$
 $= 15.1 - 0.007 = 15.1 \text{ bars}$

Factor K, corroded condition [Kcor]:
 $= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$
 $= (2 + (1081 / (2 \cdot 272))^2) / 6$
 $= 0.992661$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 3.995 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 8$, $t_r = 2.93$, $c = 3 \text{ mm.}$, $E^* = 0.85$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.5$, Temp. Reduction = 33 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 10$, $t_r = 2.96$, $c = 3 \text{ mm.}$, $E^* = 0.85$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.36$, Temp. Reduction = 71 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	6.400 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	10.000 mm.	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.359	Temperature Difference	19 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		Joint Efficiency E^*	85 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 13.438 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 6.719 mm.

EXTERNAL PRESSURE RESULTS for: TOP HEAD

Elliptical Head From 40 to 50 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 193602528 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
5.000	1091.00	218.20	0.0006365	61.62

$MAEP = B / (K_0 \cdot Do / t) = 61.6 / (0.9 \cdot 218) = 3.14 \text{ bars}$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B

PV Elite 28 U1 Licensee: SL User
FileName : HP Steam Blowdown Drum (186-D)_Rev0- 7% Page 28 of 163
Element: TOP HEAD Step: 7 2:19pm Sep 16,2026



2.871	1091.00	380.06	0.0003654	35.38
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MAEP = $B/(K0 \cdot Do/t) = 35.4/(0.9 \cdot 380) = 1.03$ bars

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	8.320 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	8.320 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	22.555 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	7.040 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	9.151 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	9.151 bars
App 27-4	= M.A.W.P.	6.400 bars

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
BOTTOM HEAD	179.4	234.0	0.767	22.66
SHELL-1	121.0	234.0	0.517	22.66
SHELL-2	121.0	234.0	0.517	22.66
TOP HEAD	179.4	234.0	0.767	22.66

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N3-4"	Nozzle	118.00	118.00	1.000
N3-4"	Pad	138.00	138.00	1.000
N1-12"	Nozzle	118.00	118.00	1.000
N1-12"	Pad	138.00	138.00	1.000
K1-2"	Nozzle	138.00	138.00	1.000
K2-2"	Nozzle	138.00	138.00	1.000
K4-2"	Nozzle	138.00	138.00	1.000
K5-2"	Nozzle	138.00	138.00	1.000
K3-8"	Nozzle	118.00	118.00	1.000
K3-8"	Pad	138.00	138.00	1.000
N2-8"	Nozzle	118.00	118.00	1.000
N2-8"	Pad	138.00	138.00	1.000
K6-2"	Nozzle	138.00	138.00	1.000
K7-2"	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
BOTTOM HEAD	138.00	138.00	1.000
SHELL-1	138.00	138.00	1.000
SHELL-2	138.00	138.00	1.000
TOP HEAD	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N3-4"	10.73	216.00	0.050
N1-12"	20.12	216.00	0.093
K1-2"	4.79	225.00	0.021
K2-2"	4.79	225.00	0.021
K4-2"	4.79	225.00	0.021
K5-2"	4.79	225.00	0.021
K3-8"	18.64	216.00	0.086



N2-8"	18.64	216.00	0.086
K6-2"	4.79	225.00	0.021
K7-2"	4.79	225.00	0.021

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	8.320 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	8.320 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	22.629 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	7.040 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	9.151 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	9.151 bars
App 27-4	= M.A.W.P.	6.400 bars

UG-99(b), Test Pressure Calculation:
 = Test Factor • MAWP • Stress Ratio
 = 1.3 • 6.4 • 1
 = 8.320 bars

Vertical Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
BOTTOM HEAD	109.9	234.0	0.469	8.69
SHELL-1	61.8	234.0	0.264	8.66
SHELL-2	60.1	234.0	0.257	8.42
TOP HEAD	105.5	234.0	0.451	8.35

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N3-4"	Nozzle	118.00	118.00	1.000
N3-4"	Pad	138.00	138.00	1.000
N1-12"	Nozzle	118.00	118.00	1.000
N1-12"	Pad	138.00	138.00	1.000
K1-2"	Nozzle	138.00	138.00	1.000
K2-2"	Nozzle	138.00	138.00	1.000
K4-2"	Nozzle	138.00	138.00	1.000
K5-2"	Nozzle	138.00	138.00	1.000
K3-8"	Nozzle	118.00	118.00	1.000
K3-8"	Pad	138.00	138.00	1.000
N2-8"	Nozzle	118.00	118.00	1.000
N2-8"	Pad	138.00	138.00	1.000
K6-2"	Nozzle	138.00	138.00	1.000
K7-2"	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
BOTTOM HEAD	138.00	138.00	1.000
SHELL-1	138.00	138.00	1.000
SHELL-2	138.00	138.00	1.000
TOP HEAD	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N3-4"	4.12	216.00	0.019
N1-12"	7.69	216.00	0.036



K1-2"	1.83	225.00	0.008
K2-2"	1.83	225.00	0.008
K4-2"	1.83	225.00	0.008
K5-2"	1.83	225.00	0.008
K3-8"	6.92	216.00	0.032
N2-8"	6.87	216.00	0.032
K6-2"	1.77	225.00	0.008
K7-2"	1.77	225.00	0.008

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm.	Outside Diameter mm.	Corroded Thickness mm.	Factor A	Factor B N./mm ²
10	20	No Calc	1091	5	0.00063652	61.6196
20	30	3454.17	1099	9	0.00029552	28.6085
30	40	3454.17	1099	9	0.00029552	28.6085
40	50	No Calc	1091	5	0.00063652	61.6196

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	8	5.87062	1.03415	3.14
20	30	12	8.75058	1.03415	3.12
30	40	12	8.75058	1.03415	3.12
40	50	8	5.87062	1.03415	3.14

Minimum

3.12

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	3454.17	9572.46	No Calc	No Calc
30	40	3454.17	9572.46	No Calc	No Calc
40	50	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm3	Corroded Metal Wgt. kg.	Corroded ID Volume cm3	Extra due Misc % kg.
10	20	122.411	208035	85.6878	211282	8.56878
20	30	794.018	2269471	597.156	2294875	55.5813
30	40	214.385	612757	161.232	619616	15.007
40	50	122.411	208035	85.6878	211282	8.56878
Total		1253	3298297.75	929	3337054.50	87

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset Dtl. Cent. mm.	Description
10	Liqd	190.652	...	-134.375	...	Liquid: 10
10	Insl	60.1825	...	-109.375	...	Insulation-2
10	Nozl	51.5937	...	-903.798	...	N3-4"
10	Wght	25	...	-268	...	Anti-Swirl Baff
20	Liqd	1747	...	1050	...	Liquid: 20
20	Insl	261.063	...	1250	...	Ins: 20
20	Nozl	197.619	702.498	2000	...	N1-12"
20	Nozl	12.1756	-657.502	2250	...	K1-2"
20	Nozl	12.1756	-657.502	125	...	K2-2"
20	Nozl	12.1756	...	1800	657.502	K4-2"
20	Nozl	12.1756	657.502	100	...	K5-2"
20	Legs	457.299	...	-585	...	LEGS
20	Wght	90	...	1250	...	Clip Weight
30	Insl	70.4869	...	337.5	...	Ins: 30
30	Nozl	67.2255	-702.498	300	...	K3-8"
30	Wght	20	...	338	...	Demisting Pad
30	Wght	28	...	338	...	Demistin Support
40	Insl	60.1825	...	159.375	...	Ins: 40
40	Nozl	63.5393	...	479.748	...	N2-8"
40	Nozl	19.2579	-350	489.969	...	K6-2"
40	Nozl	19.2579	350	489.969	...	K7-2"

Total Weight of Each Detail Type:

Liquid	1937.7
Insulation	451.9
Nozzles	467.2
Legs	457.3
Weights	163.0
Sum of the Detail Weights	
	3477.1 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	1341.0	1341.0	1341.0	1341.0	1341.0	1341.0
Nozzles	467.2	467.2	467.2	467.2	467.2	467.2
Legs	457.3	457.3	457.3	457.3	457.3	457.3
Wld Weights	143.0	143.0	143.0	143.0	143.0	143.0
Insulation	...	...	...	451.9	451.9	451.9
Empty Weights	...	...	...	20.0	20.0	...
Ope Weights	...	...	...	...	...	20.0
Ope. Liquid	...	...	...	...	...	1937.7
Test Liquid	...	3296.3	...	...	...	...
Totals	2408.4	5704.7	2408.4	2880.4	2880.4	4818.0

Field Installation Options:



Insulation installed before lifting.

Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	2408.4 kg.
Shop Test Wt.	- Fabricated weight + water (full)	5704.7 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	2408.4 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	2880.4 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	2880.4 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	4818.0 kg.
Field Test Wt.	- Empty weight + test medium (full)	6156.6 kg.

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	756.3 kg.
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	1895.4 kg.
Mass of the Upper 1/3 of the Vertical Vessel (empty)	883.4 kg.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	14717.3	14215.4
20	30	86315.3	84430.3
30	40	23305.1	22796.2
40	50	14717.3	14215.4
Total		139054.9	135657.3 cm ²
Total		149.7	146.0 ft ²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kg.	Total. Ele. Oper. Wgt. kg.	Total. Ele. Hydro. Wgt. kg.	Total Dtl. Offset Mom. Kg-m.	Oper. Wgt. No Liquid kg.
10	20	207.574	458.408	475.664	...	267.756
20	Legs	156.542	421.606	490.389	130.824	191.002
Legs	30	1029.38	2772.38	3224.68	...	1255.98
30	40	344.618	415.104	1007.49	47.2268	415.104
40	50	233.035	293.217	501.125	...	293.217

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kg.	Cumulative Oper. Wgt. kg.	Cumulative Hydro. Wgt. kg.
10	20	...	...	...
20	Legs	-267.756	-458.408	-475.664
Legs	30	1964.3	3480.7	4733.29
30	40	708.322	708.322	1508.61
40	50	293.217	293.217	501.125

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

		Cumulative	Cumulative	Cumulative
--	--	------------	------------	------------



From	To	Empty Mom. Kg-m.	Oper. Mom. Kg-m.	Hydro. Mom. Kg-m.
10	20	...	...	...
20	Legs	130.824	130.824	130.824
Legs	30	47.2268	47.2268	47.2268
30	40	47.2268	47.2268	47.2268
40	50	...	...	...

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Nozzle Flange MAWP Results: (bars & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
N3-4"	43.80	51.00	200	300	GR 1.1	38.210	27.491	43.800	27.491
N1-12"	87.60	102.00	200	600	GR 1.1	19.996	87.601	87.601	87.601
K1-2"	43.80	51.00	200	300	GR 1.1	...	...	...	...
K2-2"	43.80	51.00	200	300	GR 1.1	...	...	...	...
K4-2"	43.80	51.00	200	300	GR 1.1	...	...	...	...
K5-2"	43.80	51.00	200	300	GR 1.1	...	...	...	...
K3-8"	43.80	51.00	200	300	GR 1.1	26.262	39.439	43.800	39.439
N2-8"	43.80	51.00	200	300	GR 1.1	26.262	39.439	43.800	39.439
K6-2"	43.80	51.00	200	300	GR 1.1	...	...	...	...
K7-2"	43.80	51.00	200	300	GR 1.1	...	...	...	...
Min Rating	27.491	51.000, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm.	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kg-f	MAWP PD bars	Maximum Pressure Pmax bars	Pass Fail
N3-4"	138.176	0.50	202	0	6.400	27.491	Pass
N1-12"	350.901	0.50	1730	0	6.400	87.601	Pass
K3-8"	248.539	0.50	807	0	6.400	39.439	Pass
N2-8"	248.539	0.50	807	0	6.400	39.439	Pass

Evaluating Flange on Nozzle [N3-4"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(202) + 4 \cdot \text{abs}(0) 138.176 \leq \pi \cdot 138.176^3 [(43.800 - 6.400) + .500 \cdot 43.800]$$

$$= 0.32294E+04 \leq 0.50120E+04 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N3-4" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(43.800, 0.500 \cdot 43.800 + 43.800 - 3229.4 / (\pi \cdot 138.176^3))$$

$$= \min(43.800, 27.491)$$

$$= 27.491 \text{ bars}$$

Evaluating Flange on Nozzle [N1-12"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(1730) + 4 \cdot \text{abs}(0) 350.901 \leq \pi \cdot 350.901^3 [(87.601 - 6.400) + .500 \cdot 87.601]$$

$$= 0.27679E+05 \leq 0.17303E+06 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N1-12" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(87.601, 0.500 \cdot 87.601 + 87.601 - 27679.4 / (\pi \cdot 350.901^3))$$

$$= \min(87.601, 111.405)$$

$$= 87.601 \text{ bars}$$

Evaluating Flange on Nozzle [K3-8"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(807) + 4 \cdot \text{abs}(0) 248.539 \leq \pi \cdot 248.539^3 [(43.800 - 6.400) + .500 \cdot 43.800]$$

$$= 0.12917E+05 \leq 0.29168E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle K3-8" [Pmax]:



$$\begin{aligned}
 &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\
 &= \min(43.800, 0.500 \cdot 43.800 + 43.800 - 12917.1 / (\pi \cdot 248.539^3)) \\
 &= \min(43.800, 39.439) \\
 &= 39.439 \text{ bars}
 \end{aligned}$$

Evaluating Flange on Nozzle [N2-8"]:

$$\begin{aligned}
 &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\
 &= 16(807) + 4 \cdot \text{abs}(0) 248.539 \leq \pi \cdot 248.539^3 [(43.800 - 6.400) + .500 \cdot 43.800] \\
 &= 0.12917E+05 \leq 0.29168E+05 \text{ Check Passes}
 \end{aligned}$$

Maximum Pressure for Flange on Nozzle N2-8" [Pmax]:

$$\begin{aligned}
 &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\
 &= \min(43.800, 0.500 \cdot 43.800 + 43.800 - 12917.1 / (\pi \cdot 248.539^3)) \\
 &= \min(43.800, 39.439) \\
 &= 39.439 \text{ bars}
 \end{aligned}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 23.9764 Hz.

The Natural Frequency for the Vessel (Ope...) is 19.5123 Hz.

The Natural Frequency for the Vessel (Filled) is 15.8968 Hz.

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm.
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot g_Q \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot g_V \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.23 \cdot 0.96) / (1 + 1.7 \cdot 3.4 \cdot 0.23)))$
 $= \min(0.85, 0.91)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	19.512 Hz
Natural Frequency of Vessel (Empty)	23.976 Hz
Natural Frequency of Vessel (Test)	15.897 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.532
Structure Height to Diameter ratio	2.949
Height to top of Structure	3631.750 mm.

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
 Exposure Category: C from Table 26.11-1
 $\alpha = 9.5 : Zg = 274320 \text{ mm.}$

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 232 + 300 = 532 \text{ mm.}$
 $= 1.75 \text{ ft. Imperial Units}$

Velocity Pressure coefficient evaluated at height z $[Kz]$:
 Because $z (1.75 \text{ ft.}) < 15 \text{ ft.}$
 $= 2.01(15 / Zg)^{2/\alpha}$
 $= 2.01(15/900)^{2/9.5}$
 $= 0.85$

Type of Hill: No Hill

Wind Directionality Factor $[Kd]$:
 $= 1, \text{ per Table 26.6-1}$

As there is No Hill Present: $[Kzt]$:

K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0 \cdot 0 \cdot 0)^2$$

$$= 1$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 0.85 \cdot 1 \cdot 1 \cdot 1 \cdot 60.4^2)$$

$$= 16 \text{ psf } [78.1] \text{ Kgs/m}^2$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 16 \cdot 0.85 \cdot 0.53 \cdot 5.78$$

$$= 41.9 \text{ lbs. } [19] \text{ Kg}$$

Element	Hgt (z) mm.	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
BOTTOM HEAD	532.1	0.000	0.000	0.000	0.849	1.000	78.120
SHELL-1	1868.8	0.000	0.000	0.000	0.849	1.000	78.120
SHELL-2	3456.2	0.000	0.000	0.000	0.849	1.000	78.120
TOP HEAD	3968.4	0.000	0.000	0.000	0.849	1.000	78.120

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

Cf - Correction factor for natural frequency
 D - Average outer insulated diameter of vessel, mm.
 Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
 Dr - Average outer insulated of top half of vessel, mm.
 f - Natural frequency of vibration (Hertz)
 f1 - Natural frequency of bare vessel based on a unit value of (D/L²)(10⁴)
 L - Total height of structure mm.
 Lc - Total length of conical section(s) of vessel, mm.
 tb - Uncorroded plate thickness at bottom of vessel, mm.
 V30 - Design Wind Speed provided by user, Km/hr
 Vc - Critical wind velocity, Km/hr
 Vw - Maximum wind speed at top of structure, Km/hr
 W - Total corroded weight of structure, Kg
 Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. Kg
 Z - Maximum amplitude of vibration at top of vessel, mm.
 DL - Logarithmic decrement (taken as 0.03 for Welded Structures)
 Vp - Vib. Chance, <= 0.32037E-06 (High); 0.32037E-06 < 0.40047E-06 (Probable)
 P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5
 :: 0/3275 = 0

:: [D / L²] • 10⁴ < 8 (English Units, ft.)
 :: [4.13/10.7²] • 10⁴ = 358 [Geometry Violation]

Compute the vibration possibility. If Vp > 0.40047E-06 no chance. [Vp]:

$$= W / (L \cdot Dr^2)$$

$$= 4472 / (3275 \cdot 1205^2)$$

$$= 0.93973E-06$$

Since Vp is > 0.40047E-06 no further vibration analysis is required !

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm.	Pressure Kgs/m ²	Force Kg
----	------------------------------	------------------	--------------------------------	-------------



Anti-Swirl Ba	0.00	32.00	78.12	0.00
Clip Weight	0.00	1600.00	78.12	0.00
Demisting Pad	0.00	3188.00	78.12	0.00
Demistin Supp	0.00	3188.00	78.12	0.00

The Natural Frequency for the Vessel (Ope...) is 19.5123 Hz.

Wind Load Calculation:

From	To	Wind Height mm.	Wind Diameter mm.	Wind Area cm^2	Wind Pressure Kgs/m^2	Element Wind Load Kgf
10	20	532.054	1626.3	5369.89	78.12	20.8854
20	30	1868.75	1636.7	40917.5	78.12	159.143
30	40	3456.25	1636.7	11047.7	78.12	42.9685
40	50	3968.45	1626.3	5369.89	78.12	20.8854

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.300
Table Value Fv	1.500
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	2.000
Site Class	C
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned}
 SMS &= Fa * Ss = 1.3 * 0.15 = 0.19 \\
 SM1 &= Fv * S1 = 1.5 * 0.065 = 0.097 \\
 SDS &= 2/3 * Sms = 2/3 * 0.19 = 0.13 \\
 SD1 &= 2/3 * Sm1 = 2/3 * 0.097 = 0.065
 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(16.40.75)$
 = 0.163 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/19.5)
 = 0.051

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of ($1.7 \cdot 0.16$, 0.051) per 12.8.2
 = 0.051

As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 = $0.3 \cdot Sds \cdot W \cdot I$
 = $0.3 \cdot 0.13 \cdot 4361 \cdot 1.25$
 = 208.333 Kgf

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot Sds \cdot W$
 = $0.2 \cdot 0.13 \cdot 4361$
 = 111.1 Kgf

Distribute the Base shear force to each element according to the equations
 $F_x = C_{vx} \cdot V$ (eqn. 12.8-11) and the vertical distribution factor
 $C_{vx} = W_x \cdot h_x^k / (\text{Sum of } W_i \cdot h_i^k)$ and k is an exponent which is related
 to the period of Vibration.

In this case, the value of k was 1

The Natural Frequency for the Vessel (Ope...) is 19.5123 Hz.

Earthquake Load Calculation:

	Earthquake	Earthquake	Element	Element	Element
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From	To	Height mm.	Weight Kgf	Ope Load Kgf	Emp Load Kgf	Vertical Load Kgf
10	20	25	458.408	0.37394	0.14697	-11.6802
20	Legs	380	421.606	5.22761	1.68469	-22.4227
Legs	30	1465	2772.38	132.526	42.7091	88.6882
30	40	2887.5	415.104	39.1104	28.1816	18.048
40	50	3250	293.217	31.0947	21.4492	7.47118

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The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm.	Cumulative Wind Shear Kgf	Earthquake Shear Kgf	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	504.696	...	...	...	...
20	Legs	165	20.8854	0.37394	3.64867	0.065328
Legs	30	1085	243.882	208.333	306.688	334.712
30	40	2507.5	63.8539	70.205	32.2488	39.6215
40	50	3019.7	20.8854	31.0947	3.64867	5.43222

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

Note:

Review the Vessel Design Summary for the cumulative shear force and bending moment on the support.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear Kgf	Centroid Deflection mm.	Elem. End Deflection mm.	Elem. Ang. Rotation
10	20	...	0.25642	0.25642	0.00020913
20	Legs	20.8854	0.25648	0.25663	0.00021035
Legs	30	243.882	0.25984	0.26512	0.00021438
30	40	63.8539	0.26691	0.26872	0.00021449
40	50	20.8854	0.26885	0.26899	0.00021449

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed Km/hr	2nd Crit. Wind Speed Km/hr
10	20	569.654	3560.34
20	30	573.297	3583.1
30	40	573.297	3583.1
40	50	569.654	3560.34

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
 Allowable deflection : 16.375 Actual deflection : 0.269 mm.

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Longitudinal Stress Constants:

From	To	Metal Area New cm ²	Metal Area Corroded cm ²	Section Modulus New mm. ³	Section Modulus Corroded mm. ³
10	20	272.188	170.588	7315835	4610338
20	30	409.789	308.19	11015770	8329965
30	40	409.789	308.19	11015770	8329965
40	50	272.188	170.588	7315835	4610338

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Longitudinal Allowable Stresses:

From	To	Tensile N./mm ²	Hydrotest Tensile N./mm ²	Compressive N./mm ²	Hydrotest Compressive N./mm ²
10	20	140.757	238.676	-96.9635	-107.247
20	Legs	140.757	238.676	-111.395	-125.135
Legs	30	140.757	238.676	-111.395	-125.135
30	40	140.757	238.676	-111.395	-125.135
40	50	140.757	238.676	-96.9635	-107.247

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Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure N./mm ²	Longitudinal Stress External Pressure N./mm ²	Longitudinal Stress Hydrotest Pressure N./mm ²
10	20	34.5017	-5.6676	44.8032
20	30	19.1107	-3.18329	24.8167
30	40	19.1107	-3.18329	24.8167
40	50	34.5017	-5.6676	44.8032

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty N./mm ²	Wght. Str. Operating N./mm ²	Wght. Str. Hydrotest N./mm ²	Wght. Str. Emp. Mom. N./mm ²	Wght. Str. Opr. Mom. N./mm ²
10	20	...	...	...	...	...
20	Legs	0.085202	0.14587	0.15136	0.15402	0.15402
Legs	30	-0.62506	-0.55168	-0.62506	0.055599	0.055599
30	40	-0.22539	-0.22539	-0.22539	0.055599	0.055599
40	50	-0.16857	-0.16857	-0.16857	...	...

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. N./mm ²	Bend. Str. Oper. Wind N./mm ²	Bend. Str. Oper. Equ. N./mm ²	Bend. Str. Hyd. Wind N./mm ²	Bend. Str. Hyd. Equ. N./mm ²
10	20	...	...	...	...	...
20	Legs	0.15402	0.0042955	0.76908E-04	0.0021477	...
Legs	30	0.055599	0.36105	0.39405	0.18053	...
30	40	0.055599	0.037966	0.046645	0.018983	...
40	50	...	0.0077611	0.011555	0.0038805	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case N./mm ²	Vortex Shedding Empty Case N./mm ²	Vortex Shedding Test Case N./mm ²	Earthquake Empty Case N./mm ²
10	20	...	...	...	...
20	Legs	...	...	...	0.30226E-04
Legs	30	...	...	...	0.21359
30	40	...	...	...	0.032654
40	50	...	...	...	0.0079706

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind N./mm ²	Longitudinal Stress Y Forces Seismic N./mm ²
10	20	...	-0.0067148
20	Legs	...	-0.0071351
Legs	30	...	0.028221
30	40	...	0.005743

40	50	...	0.0042951
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Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded N./mm ²	Earthquake For/Mom Corroded N./mm ²	Wind For/Mom No Corrosion N./mm ²	Earthquake For/Mom No Corrosion N./mm ²
10	20	...	...	...	...
20	Legs	...	...	...	...
Legs	30	...	...	...	...
30	40	...	...	...	...
40	50	...	...	...	...

Bending Stresses due to Blast Load:

From	To	Bending Stress N./mm ²
10	20	...
20	Legs	...
Legs	30	...
30	40	...
40	50	...

Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	140.76	0.00	96.96	0.0000	-0.0000
20	0.24	140.76	-0.07	111.40	0.0017	0.0007
20		140.76	-1.04	111.40		0.0094
30		140.76	-0.32	111.40		0.0029
40		140.76	-0.18	96.96		0.0018

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		140.76	-0.01	96.96		0.0001
20	0.23	140.76	-0.08	111.40	0.0016	0.0007
20		140.76	-0.87	111.40		0.0078
30		140.76	-0.31	111.40		0.0028
40		140.76	-0.17	96.96		0.0018

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	140.76	0.00	96.96	0.0000	-0.0000
20	0.30	140.76	-0.01	111.40	0.0022	0.0001
20		140.76	-0.97	111.40		0.0087
30		140.76	-0.32	111.40		0.0029



40 140.76 -0.18 96.96 0.0018

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		140.76	-0.01	96.96		0.0001
20	0.29	140.76	-0.02	111.40	0.0021	0.0001
20		140.76	-0.97	111.40		0.0087
30		140.76	-0.32	111.40		0.0029
40		140.76	-0.18	96.96		0.0018

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	238.68	0.00	107.25	0.0000	-0.0000
20	0.31	238.68	-0.00	125.14	0.0013	0.0000
20		238.68	-0.86	125.14		0.0069
30		238.68	-0.30	125.14		0.0024
40		238.68	-0.17	107.25		0.0016

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	238.68	0.00	107.25	0.0000	-0.0000
20	0.31	238.68	-0.00	125.14	0.0013	0.0000
20		238.68	-0.68	125.14		0.0054
30		238.68	-0.28	125.14		0.0022
40		238.68	-0.17	107.25		0.0016

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	34.50	140.76		96.96	0.2451	
20	19.41	140.76		111.40	0.1379	
20	18.98	140.76		111.40	0.1348	
30	18.98	140.76		111.40	0.1348	
40	34.34	140.76		96.96	0.2440	

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	34.49	140.76		96.96	0.2451	
20	19.40	140.76		111.40	0.1379	
20	19.04	140.76		111.40	0.1352	
30	18.99	140.76		111.40	0.1349	
40	34.35	140.76		96.96	0.2440	

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		140.76	-5.67	96.96		0.0585
20		140.76	-3.20	111.40		0.0287
20		140.76	-4.15	111.40		0.0373
30		140.76	-3.50	111.40		0.0314
40		140.76	-5.84	96.96		0.0603

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		140.76	-5.67	96.96		0.0585
20		140.76	-3.20	111.40		0.0287
20		140.76	-4.16	111.40		0.0373
30		140.76	-3.51	111.40		0.0315



40	140.76	-5.84	96.96	0.0603
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Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	44.80	238.68		107.25	0.1877	
20	25.12	238.68		125.14	0.1053	
20	24.43	238.68		125.14	0.1023	
30	24.67	238.68		125.14	0.1033	
40	44.64	238.68		107.25	0.1870	

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	44.80	238.68		107.25	0.1877	
20	25.12	238.68		125.14	0.1053	
20	24.25	238.68		125.14	0.1016	
30	24.65	238.68		125.14	0.1033	
40	44.63	238.68		107.25	0.1870	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	34.50	140.76		96.96	0.2451	
20	19.35	140.76		111.40	0.1375	
20	18.54	140.76		111.40	0.1317	
30	18.94	140.76		111.40	0.1346	
40	34.33	140.76		96.96	0.2439	

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	34.50	140.76		96.96	0.2451	
20	19.22	140.76		111.40	0.1365	
20	18.70	140.76		111.40	0.1328	
30	18.94	140.76		111.40	0.1346	
40	34.40	140.76		96.96	0.2444	

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	34.50	140.76		96.96	0.2451	
20	19.41	140.76		111.40	0.1379	
20	18.61	140.76		111.40	0.1322	
30	18.94	140.76		111.40	0.1346	
40	34.33	140.76		96.96	0.2439	

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	34.50	140.76		96.96	0.2451	
20	19.35	140.76		111.40	0.1375	
20	18.54	140.76		111.40	0.1317	
30	18.94	140.76		111.40	0.1346	
40	34.33	140.76		96.96	0.2439	

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	140.76	0.00	96.96	0.0000	-0.0000
20	0.30	140.76	-0.01	111.40	0.0021	0.0001
20		140.76	-0.61	111.40		0.0055
30		140.76	-0.28	111.40		0.0025



40 140.76 -0.17 96.96 0.0017

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	34.49	117.30		80.80	0.2941	
20	19.40	117.30		92.83	0.1654	
20	18.64	117.30		92.83	0.1589	
30	18.95	117.30		92.83	0.1615	
40	34.34	117.30		80.80	0.2927	

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		117.30	-5.67	80.80		0.0702
20		117.30	-3.20	92.83		0.0345
20		117.30	-3.76	92.83		0.0405
30		117.30	-3.46	92.83		0.0373
40		117.30	-5.83	80.80		0.0722

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	34.50	140.76		96.96	0.2451	
20	19.41	140.76		111.40	0.1379	
20	18.61	140.76		111.40	0.1322	
30	18.94	140.76		111.40	0.1346	
40	34.33	140.76		96.96	0.2439	

Absolute Maximum of the all of the Stress Ratio's 0.2941

Governing Element: BOTTOM HEAD
 Governing Load Case 18 : FS+BS+IP+OW

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
BOTTOM HEAD	34.5	...	...	...	34.5	+ Tens
BOTTOM HEAD	34.5	...	...	...	...	- Comp
SHELL-1	19.1	0.1	0.2	...	19.4	+ Tens
SHELL-1	19.1	0.1	-0.2	...	...	- Comp
SHELL-1	19.1	-0.5	0.1	...	18.6	+ Tens
SHELL-1	19.1	-0.5	-0.1	...	...	- Comp
SHELL-2	19.1	-0.2	0.1	...	18.9	+ Tens
SHELL-2	19.1	-0.2	-0.1	...	...	- Comp
TOP HEAD	34.5	-0.2	...	...	34.3	+ Tens
TOP HEAD	34.5	-0.2	...	...	...	- Comp

Shop/Field Installation Options :

Insulation is installed in the Field before being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Liquid	978.546 mm.
Center of Gravity of the Insulation	1637.500 mm.
Center of Gravity of the Nozzles	2104.148 mm.
Center of Gravity of the Legs	-535.000 mm.
Center of Gravity of the Added Weights (Operating)	1527.141 mm.
Center of Gravity of the Added Weights (Empty)	1527.141 mm.
Center of Gravity of Bare Shell New and Cold	1637.500 mm.
Center of Gravity of Bare Shell Corroded	1637.500 mm.
Vessel CG in the Operating Condition	1174.548 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	1310.767 mm.
Vessel CG in the Test Condition	1498.899 mm.

Rigging Analysis Results:

- Lifting weight based on the user selected calculated empty weight

Total Effective Length of Vessel for this analysis	3275.00 mm.
Total vessel weight (No Liquid) Twt	2880.36 Kg
Impact weight multiplication factor Imp	2.00
Design lifting weight, DWT = Imp * Twt	5760.72 Kg
Elevation of the Tailing Lug (bottom)	0.00 mm.
Elevation of the Lifting Lug (top)	2887.50 mm.
Design Reaction force at the tailing lug	3145.67 Kg
Design Reaction force at the lifting lug	2615.05 Kg
CG distance during field lifting operation	1310.77 mm.
CG distance from tailing lug	1310.77 mm.
CG distance from the nearer lifting lug/trunnion	1310.77 mm.
Compressive Allowable Increase Factor	1.00

Critical Values: (N./mm², mm.)

	Max Stress	Elevation	Allowables
Bending	0.41	1550.00	110.18 (UG-23)
Shear	0.53	2820.00	96.60 (0.7*S)

The compressive bending allowable stress is the minimum of the tensile allowable and the compressive allowable times the specified compressive allowable increase factor.

Forces and Moments at selected elevations (not all analysis points shown):

Distance mm.	Bending Moment Kg-m.	Bending Stress N./mm ²	Shear Force Kg	Shear Stress N./mm ²
0.00	0.0	0.0	473.6	0.2
550.00	300.6	0.3	107.7	0.0
2050.00	313.1	0.3	824.4	0.2
2820.00	348.7	0.3	2230.4	0.5
3225.00	98.0	0.1	91.7	0.0

Unity Check (Actual Stress / Allowable Stress):

Maximum Unity Check is 0.0038 at elevation 1550 mm. - Must be <=1

Note: The rigging analysis is performed using a uniformly distributed load.

--- Plot data successfully generated ---



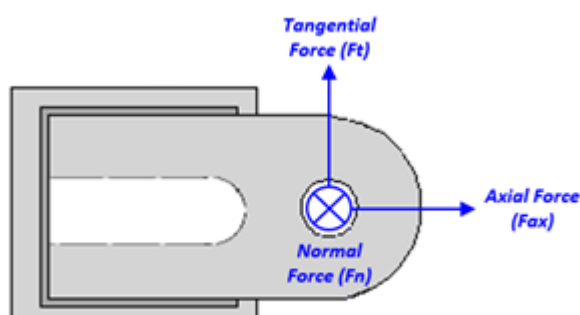
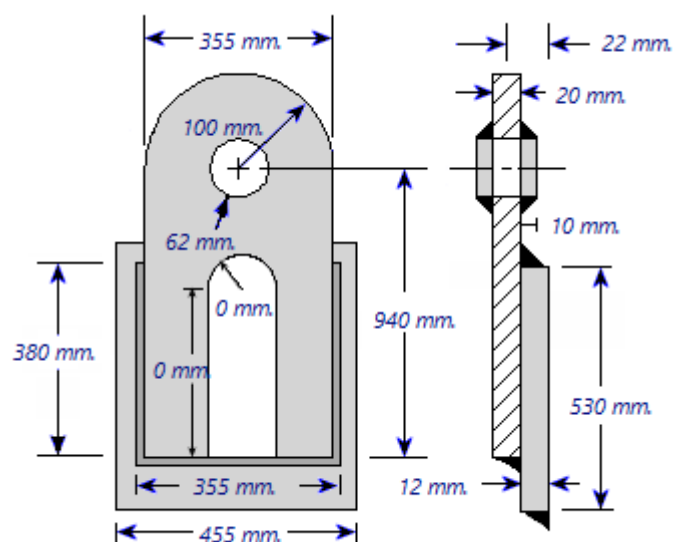


Lifting Lug Calculations:

Input Values:

Lifting Lug Material		SA-516 70
Lifting Lug Yield Stress	Yield	262.00 N./mm2
Width of Lifting Lug	w	355.0000 mm.
Thickness of Lifting Lug	t	20.0000 mm.
Diameter of Hole in Lifting Lug	dh	62.0000 mm.
Radius of Semi-Circular Arc of Lifting Lug	r	100.0000 mm.
Height of Lug from Bottom to Center of Hole	h	940.0000 mm.
Offset from Vessel OD to Center of Hole	off	22.0000 mm.
Lug Fillet Weld Size	tw	10.0000 mm.
Length of weld along side of Lifting Lug	wl	380.0000 mm.
Length of Weld along Bottom of Lifting Lug	wb	355.0000 mm.
Thickness of Collar (if any)	tc	0.0000 mm.
Diameter of Collar (if any)	dc	0.0000 mm.
Impact Factor	Impfac	2.00
Number of Lugs in Group		2
Reinforcing Pad Circumferential Width	wpb	455.00 mm.
Reinforcing Pad Longitudinal Length	wpl	530.00 mm.
Reinforcing Pad Thickness	thkp	12.00 mm.
Reinforcing Pad Fillet Weld Size	tfp	10.00 mm.

Lifting Lug Orientation to Vessel: Flat
 Lift Orientation : From Horizontal to Vertical





increments. The table below provides the results in 5 degree increments.

The 0 degree angle position is in the horizontal direction and the 90 degree angle position is in the vertical direction.

PV Elite does not compute weak axis bending forces on the lugs. It is assumed that a spreader bar is used.

Lifting Lug and Weld Stresses at various Angles: N./mm²

Angle	----- Weld Stresses			----- Lug Stresses		Maximum
	Primary	Secondary	Allowable	Combined	Allowable	Ratio
0	1.9	11.3	104.8	17.1	172.9	0.108
5	2.0	11.5	104.8	17.6	172.9	0.110
10	2.1	11.6	104.8	17.9	172.9	0.111
15	2.2	11.6	104.8	18.1	172.9	0.111
20	2.2	11.5	104.8	18.2	172.9	0.110
25	2.3	11.4	104.8	18.1	172.9	0.108
30	2.4	11.1	104.8	17.8	172.9	0.106
35	2.5	10.7	104.8	17.5	172.9	0.102
40	2.6	10.3	104.8	16.9	172.9	0.098
45	2.7	9.7	104.8	16.3	172.9	0.094
50	2.8	9.1	104.8	15.5	172.9	0.090
55	2.9	8.3	104.8	14.5	172.9	0.084
60	3.0	7.5	104.8	13.5	172.9	0.078
65	3.1	6.6	104.8	12.3	172.9	0.071
70	3.3	5.6	104.8	11.0	172.9	0.063
75	3.4	4.5	104.8	9.5	172.9	0.055
80	3.7	3.3	104.8	7.9	172.9	0.046
85	4.1	1.8	104.8	6.1	172.9	0.039
90	4.8	0.4	104.8	4.0	172.9	0.046

Computed Results:

Total vessel weight (no liquid) 2880.36 Kg
 Design reaction force at the tailing lug 3145.67 Kg
 Design reaction force at the lifting lug 2615.05 Kg

Worst case lift angle for primary weld stresses 90 deg
 Tangential force at this angle 0.0 Kg
 Axial force at this angle 2880.4 Kg

Worst case lift angle for secondary weld stresses 13 deg
 Tangential force at this angle 1348.5 Kg
 Axial force at this angle 311.3 Kg

Worst case lift angle for lug stresses 19 deg
 Tangential force at this angle 1341.6 Kg
 Axial force at this angle 462.0 Kg

Converting the weld leg dimension (tw) to the weld throat dimension.

Weld Group Inertia Calculations:

Weld group inertia about the longitudinal axis I_{ll} 12874.534 cm⁴
 Weld group centroid distance in the long. direction Y_{ll} 251.619 mm.
 Dist. of weld group centroid from lug bottom Y_{llb} 128.381 mm.
 Weld group inertia about the circumferential axis I_{lc} 18899.443 cm⁴
 Weld group centroid distance in the circ. direction Y_{lc} 177.500 mm.

Note: The Impact Factor is applied to the Forces acting on the Lug.

Primary Shear Stress in the Welds due to Shear Loads [S_{sl}]:

$$= \sqrt{(F_{ax}^2 + F_t^2 + F_n^2)} / ((2 \cdot w_l + w_b) \cdot t_w)$$

$$= \sqrt{(2880^2 + 0^2 + 0^2)} / ((2 \cdot 380 + 355) \cdot 7.07)$$

$$= 3.58 \text{ N./mm}^2$$

Shear Stress in the Welds due to Bending Loads [S_{blf}]:



$$\begin{aligned}
 &= (F_n(h-Y_{IIb}))Y_{III/III} + (F_{ax} \cdot \text{off} \cdot Y_{II/III}) + (F_t \cdot \text{off} \cdot Y_{Ic/IIc}) \\
 &= (0(940-128))252/12875 + \\
 &\quad (2880 \cdot 22 \cdot 252/12875) + \\
 &\quad (0 \cdot 22 \cdot 178/18899) \\
 &= 1.21 \text{ N./mm}^2
 \end{aligned}$$

Total Shear Stress for Combined Loads [St]:

$$\begin{aligned}
 &= S_{sII} + S_{bIf} \\
 &= 3.58 + 1.21 \\
 &= 4.80 \text{ N./mm}^2
 \end{aligned}$$

Allowable Shear Stress for Combined Loads [Sta]:

$$\begin{aligned}
 &= 0.4 \cdot \text{Yield} \cdot \text{Occfac} \text{ (AISC Shear Allowable)} \\
 &= 0.4 \cdot 262 \cdot 1 \\
 &= 104.80 \text{ N./mm}^2
 \end{aligned}$$

Secondary Shear Stress in the Welds due to Shear Loads:

Unit Weld Polar Section Modulus [Uwsm]:

$$\begin{aligned}
 &= (2 \cdot w_l + w)^3 / 12 - w_l^2(w_l + w)^2 / (2 \cdot w_l + w) \\
 &= 45553568.00 \text{ mm}^3
 \end{aligned}$$

Loads on Welds due to Torsional Moment:

Shear Stress in Normal Direction [Fth]:

$$\begin{aligned}
 &= F_t \cdot (h - (w_l - \text{Cent})) \cdot (w_b/2) / U_{wsm} \\
 &= 4.26 \text{ Kg/mm.}
 \end{aligned}$$

Shear Stress in Circumferential Direction [Ftv]:

$$\begin{aligned}
 &= F_t \cdot (h - (w_l - \text{Cent})) \cdot \text{Cent} / U_{wsm} \\
 &= 6.01 \text{ Kg/mm.}
 \end{aligned}$$

Primary Shear Stress [Fsv]:

$$\begin{aligned}
 &= F_t / (2 \cdot w_l + w) \\
 &= 1.21 \text{ Kg/mm.}
 \end{aligned}$$

Resultant Load on Weld Group [Fr]:

$$\begin{aligned}
 &= \sqrt{(F_{th}^2 + (F_{tv} + F_{sv})^2)} \\
 &= \sqrt{(4.26^2 + (6.01 + 1.21)^2)} \\
 &= 8.38 \text{ Kg/mm.}
 \end{aligned}$$

Resultant Secondary Weld Stress [Fws]:

$$\begin{aligned}
 &= F_r / T_{weld} \\
 &= 8.38/7.07 \\
 &= 11.63 \text{ N./mm}^2
 \end{aligned}$$

Allowable Resultant Secondary Weld Stress [Psa]:

$$\begin{aligned}
 &= 0.4 \cdot \text{Yield} \cdot \text{Occfac} \\
 &= 0.4 \cdot 262 \cdot 1 \\
 &= 104.80 \text{ N./mm}^2
 \end{aligned}$$

Shear Stress in Lug above Hole [Shs]:

$$\begin{aligned}
 &= \sqrt{(P_l^2 + F_n^2)} / S_{ha} \\
 &= \sqrt{(2880^2 + 0^2)} / 27.6 \\
 &= 10.23 \text{ N./mm}^2
 \end{aligned}$$

Allowable Shear Stress in Lug above Hole [Sas]:

$$\begin{aligned}
 &= 0.4 \cdot \text{Yield} \cdot \text{Occfac} \\
 &= 0.4 \cdot 262 \cdot 1 \\
 &= 104.80 \text{ N./mm}^2
 \end{aligned}$$

Pin Hole Bearing Width [Pbw]:

$$\begin{aligned}
 &= \text{Lug Thickness} + 2 \cdot \text{Collar Thickness} \\
 &= 20 + 2 \cdot 0 \\
 &= 20.000 \text{ mm.}
 \end{aligned}$$

Pin Hole Bearing Stress [Pbs]:

$$\begin{aligned}
 &= \sqrt{(P_l^2 + F_n^2)} / (P_{bw} \cdot d_h) \\
 &= \sqrt{(2880^2 + 0^2)} / (20 \cdot 62) \\
 &= 22.78 \text{ N./mm}^2
 \end{aligned}$$

Allowable Bearing Stress [Pba]:
 = min(0.75 • Yield • Occfac, 0.9 • Yield) AISC Bearing All.
 = min(0.75 • 262 • 1, 236)
 = 196.50 N./mm²

Bending Stress at the Base of the Lug [Fbs]:
 = $F_n \cdot (h-w) / (w \cdot t^2 / 6) + F_t \cdot (h-w) / (w^2 \cdot t / 6)$
 = 0 • (940 - 380) / (355 • 20 (²) / 6) +
 1342 • (940 - 380) / (355 (²) • 20 / 6)
 = 17.54 N./mm²

Tensile Stress at the Base of the Lug [Fa]:
 = Fax / (w • t)
 = 462 / (355 • 20)
 = 0.64 N./mm²

Total Combined Stress at the Base of the Lug:
 = Fbs + Fa
 = 17.5 + 0.64
 = 18.18 N./mm²

Lug Allowable Stress for Bending and Tension:
 = min(0.66 • Yield • Occfac, 0.75 • Yield)
 = min(0.66 • 262 • 1, 196)
 = 172.92 N./mm²

Required Shackle Pin Diameter [Spd]:
 = $\sqrt{((2(P_l^2 + F_n^2))^{1/2} / (\pi \cdot Sta))}$
 = $\sqrt{(2(2880^2 + 0^2))^{1/2} / (\pi \cdot 105)}$
 = 13.0994 mm.

WRC 107/537 Stress Analysis for the Lifting Lug to Shell Junction in the new and Cold Condition (no corrosion applied).

The full lug load is applied in both the circ. and long. directions simultaneously.

Attachment dimensions (C11P, C22P, etc.) may be modified accounting for attachment welds where applicable.

Input Echo, WRC107/537 Item 1, Description: Lift Lug

Diameter Basis for Vessel	Vbasis	ID
Cylindrical or Spherical Vessel	Cylsph	Cylindrical
Internal Corrosion Allowance	Cas	0.0000 mm.
Vessel Diameter	Dv	1075.000 mm.
Vessel Thickness	Tv	12.000 mm.

Note: Lifting lug calculations are performed at ambient temperature conditions.

Attachment Type	Type	Rectangular
Parameter C11	C11	375.00 mm.
Parameter C22	C22	390.00 mm.
Thickness of Reinforcing Pad	Tpad	12.000 mm.
Pad Parameter C11P	C11p	475.000 mm.
Pad Parameter C22P	C22p	540.000 mm.
Design Internal Pressure	Dp	0.000 bars
Include Pressure Thrust		No

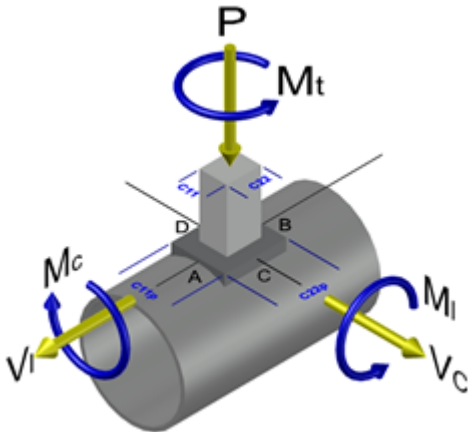
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	0.0 Kg-f
Longitudinal Shear (SUS)	Vl	2880.4 Kg-f
Circumferential Shear (SUS)	Vc	2880.4 Kg-f
Circumferential Moment (SUS)	Mc	0.0 Kg-m.
Longitudinal Moment (SUS)	Ml	63.4 Kg-m.
Torsional Moment (SUS)	Mt	2160.3 Kg-m.

Use Interactive Control No
 WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No
 Compute Pressure Stress per WRC-368 No
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

WRC 107 Stress Calculation for SUSTained loads:

Radial Load	P	0.0	Kgf
Circumferential Shear	VC	2880.4	Kgf
Longitudinal Shear	VL	2880.4	Kgf
Circumferential Moment	MC	0.0	Kg-m.
Longitudinal Moment	ML	63.4	Kg-m.
Torsional Moment	MT	2160.3	Kg-m.

Dimensionless Parameters used : Gamma = 22.90

Dimensionless Loads for Cylindrical Shells at Attachment Junction:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.357	4C	2.657	(A,B)
N(PHI) / (P/Rm)	0.357	3C	1.449	(C,D)
M(PHI) / (P)	0.346	2C1	0.021	(A,B)
M(PHI) / (P)	0.346	1C !	0.059	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.346	3A	0.941	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.387	1A	0.071	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.350	3B	1.880	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.360	1B	0.016	(A,B,C,D)
N(x) / (P/Rm)	0.352	3C	1.473	(A,B)
N(x) / (P/Rm)	0.352	4C	2.683	(C,D)
M(x) / (P)	0.352	1C1	0.042	(A,B)
M(x) / (P)	0.352	2C !	0.030	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.346	4A	2.098	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.426	2A	0.030	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.350	4B	0.885	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	0.373	2B	0.026	(A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Circ. Bend. MC		0.0	0.0	0.0	0.0	-0.0	0.0	0.0	-0.0
Circ. Memb. ML		-0.4	-0.4	0.4	0.4	0.0	0.0	0.0	0.0
Circ. Bend. ML		-0.5	0.5	0.5	-0.5	0.0	0.0	0.0	0.0
Tot. Circ. Str.		-0.9	0.1	0.9	-0.1	0.0	0.0	0.0	0.0
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Long. Bend. MC		0.0	0.0	0.0	0.0	-0.0	0.0	0.0	-0.0
Long. Memb. ML		-0.2	-0.2	0.2	0.2	0.0	0.0	0.0	0.0
Long. Bend. ML		-0.8	0.8	0.8	-0.8	0.0	0.0	0.0	0.0
Tot. Long. Str.		-1.0	0.6	1.0	-0.6	0.0	0.0	0.0	0.0
Shear VC		1.6	1.6	-1.6	-1.6	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-1.5	-1.5	1.5	1.5
Shear MT		24.1	24.1	24.1	24.1	24.1	24.1	24.1	24.1
Tot. Shear		25.7	25.7	22.6	22.6	22.6	22.6	25.7	25.7
Str. Int.		51.4	51.4	45.1	45.1	45.3	45.3	51.3	51.3

Dimensionless Parameters used : Gamma = 45.29

Dimensionless Loads for Cylindrical Shells at Pad Edge:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.502	4C !	2.834	(A,B)
N(PHI) / (P/Rm)	0.502	3C !	1.020	(C,D)
M(PHI) / (P)	0.457	2C1	0.006	(A,B)
M(PHI) / (P)	0.457	1C !	0.064	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.456	3A	1.023	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.503	1A !	0.058	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.476	3B	1.622	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.482	1B	0.004	(A,B,C,D)
N(x) / (P/Rm)	0.481	3C	1.088	(A,B)
N(x) / (P/Rm)	0.481	4C	2.983	(C,D)
M(x) / (P)	0.485	1C1	0.013	(A,B)
M(x) / (P)	0.485	2C !	0.033	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.456	4A	4.212	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.559	2A !	0.024	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.476	4B	1.003	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	0.505	2B !	0.007	(A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0

Circ. Memb. MC	0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Circ. Bend. MC	0.0	0.0	0.0	0.0	-0.0	0.0	0.0	-0.0
Circ. Memb. ML	-0.5	-0.5	0.5	0.5	0.0	0.0	0.0	0.0
Circ. Bend. ML	-0.4	0.4	0.4	-0.4	0.0	0.0	0.0	0.0
Tot. Circ. Str.	-1.0	-0.1	1.0	0.1	0.0	0.0	0.0	0.0
Long. Memb. P	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P	-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC	0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Long. Bend. MC	0.0	0.0	0.0	0.0	-0.0	0.0	0.0	-0.0
Long. Memb. ML	-0.4	-0.4	0.4	0.4	0.0	0.0	0.0	0.0
Long. Bend. ML	-0.7	0.7	0.7	-0.7	0.0	0.0	0.0	0.0
Tot. Long. Str.	-1.0	0.3	1.0	-0.3	0.0	0.0	0.0	0.0
Shear VC	2.5	2.5	-2.5	-2.5	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-2.2	-2.2	2.2	2.2
Shear MT	27.5	27.5	27.5	27.5	27.5	27.5	27.5	27.5
Tot. Shear	30.0	30.0	25.1	25.1	25.4	25.4	29.7	29.7
Str. Int.	60.0	60.0	50.1	50.1	50.7	50.7	59.4	59.4

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pl (SUS)		-0.4	-0.4	0.4	0.4	0.0	0.0	0.0	0.0
Circ. Q (SUS)		-0.5	0.5	0.5	-0.5	0.0	0.0	0.0	0.0
Long. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pl (SUS)		-0.2	-0.2	0.2	0.2	0.0	0.0	0.0	0.0
Long. Q (SUS)		-0.8	0.8	0.8	-0.8	0.0	0.0	0.0	0.0
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		1.6	1.6	-1.6	-1.6	-1.5	-1.5	1.5	1.5
Shear Q (SUS)		24.1	24.1	24.1	24.1	24.1	24.1	24.1	24.1
Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+Pl (SUS)		3.1	3.1	3.1	3.1	3.0	3.0	3.0	3.0
Pm+Pl+Q (Total)		51.4	51.4	45.1	45.1	45.3	45.3	51.3	51.3

Vessel Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	138.00	Passed
Pm+Pl (SUS)	3.15	207.00	Passed
Pm+Pl+Q (TOTAL)	51.43	413.99	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

WRC 107/537 Stress Summations:



Vessel Stress Summation at Reinforcing Pad Edge (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pl (SUS)		-0.5	-0.5	0.5	0.5	0.0	0.0	0.0	0.0
Circ. Q (SUS)		-0.4	0.4	0.4	-0.4	0.0	0.0	0.0	0.0
Long. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pl (SUS)		-0.4	-0.4	0.4	0.4	0.0	0.0	0.0	0.0
Long. Q (SUS)		-0.7	0.7	0.7	-0.7	0.0	0.0	0.0	0.0
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		2.5	2.5	-2.5	-2.5	-2.2	-2.2	2.2	2.2
Shear Q (SUS)		27.5	27.5	27.5	27.5	27.5	27.5	27.5	27.5
Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+Pl (SUS)		5.0	5.0	5.0	5.0	4.4	4.4	4.4	4.4
Pm+Pl+Q (Total)		60.0	60.0	50.1	50.1	50.7	50.7	59.4	59.4

Vessel Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	138.00	Passed
Pm+Pl (SUS)	4.96	207.00	Passed
Pm+Pl+Q (TOTAL)	60.02	413.99	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

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RESULTS FOR LEGS : Operating Case Description: LEGS

Legs attached to: SHELL-1

Section Properties : Single Angle L200X200X16

European Structural Steel Data

Overall Leg Length		1830.000	mm.
Effective Leg Length	Leglen	1600.000	mm.
Distance Leg Up Side of Vessel		330.000	mm.
Number of Legs	Nleg	4	
Cross Sectional Area for L200X200X16	Aleg	61.800	cm2
Section Inertia (strong axis)		2339.988	cm4
Section Inertia (weak axis)		2339.988	cm4
Section Modulus (strong axis)		162000.672	mm. ³
Section Modulus (weak axis)		162000.672	mm. ³
Radius of Gyration (strong axis)		61.600	mm.
Radius of Gyration (weak axis)		61.600	mm.

Leg Orientation - Diagonal

Overturning Moment at top of Legs		334.7	Kg-m.
Total Weight Load at top of Legs	W	4390.6	Kgf
Total Shear force at top of Legs		243.9	Kgf
Additional force in Leg due to Bracing	Fadd	0.0	Kgf
Occasional Load Factor	Occfac	1.333	
Effective Leg End Condition Factor	k	1.000	

Note: The Legs are Not Cross Braced

The Leg Shear Force includes Wind, Seismic, and Applied Load Effects.

Pad Width along Circumference	C11P	350.000	mm.
Pad Length along Vessel Axis	C22P	350.000	mm.
Pad Thickness	Tpad	10.000	mm.

Maximum Shear at top of one Leg [Vleg]:
 $= (\max(\text{Wind, Seismic, Blast}) + \text{applied forces}) (I_{\max} / I_{\text{tot}})$
 $= (244) (3781/9484)$
 $= 97.24 \text{ Kg}$

Axial Compression, Leg furthest from the Neutral Axis [Sma]:
 $= (W/N_{\text{leg}} + (M_{\text{leg}} / (N_{\text{legm}} \cdot R_n))) / A_{\text{leg}}$
 $= (43057/4 + (3282145 / (2 \cdot 614))) / 6180$
 $= 2.17 \text{ N./mm}^2$

Axial Compression, Leg closest to the Neutral Axis [Sva]:
 $= (W / N_{\text{leg}}) / A_{\text{leg}}$
 $= (4391/4) / 61.8$
 $= 1.74 \text{ N./mm}^2$

Computing Principal Axis and Inertias for Angle:

Leg lengths and thickness:	200.00	200.00	16.000
Distance to geometric centroid:	55.200	55.200	
Arm about YY:	47.200	52.800	
Arm about ZZ:	44.800	47.200	
Leg areas:	32.000	29.440	
Geometric inertia components YY:	719.73	1651.3	
Geometric inertia components ZZ:	1708.9	662.15	
Geometric inertias Iy & Iz:	2371.1	2371.1	
Product of inertia:	1410.3		
Mohrs Radius:	1410.3		
Average Inertia:	2371.1		

QFACT =	1.0000	FBZ	=	0.14893
Principal Axis Inertias (Z&W)			=	960.72 3781.4
Angle to Principal Axis			=	45.000



Distances to extreme fibers CW & CZ = 141.42 63.357
 FOB from Eq 5-5 = 1.9478
 Bending allowables Fby & Fbz = 0.16383 0.14893

Shear Center Coordinates Wo & Zo: 65.814 0.0000

Values for Elastic Flexural-Torsional Buckling Stress:

E, G, J, R0²: 29500. 11346. 1.2670 18.608
 AREA, LENGTH, Kw, Kz: 9.5790 62.992 1.0000 1.0000
 H, Few, Fez, Fej: 0.63919 695.90 176.80 80.651
 Fe computed from C4-1: 77.177

Initial (Kl/r)max, & (Kl/r)equiv = 40.580 61.421
 Final (Kl/r)max, & Cc = 61.421 127.18
 Fa based on Eq 4-1 = 0.11958

Leg Bending and Axial Stresses vs. Allowables: (N./mm²)

	Actual	Allowable
Weak Axis Bending :	7.11	198.53
Strong Axis Bending :	4.03	218.38
Axial Compression :	2.17	159.40

UNITY CHECKS ARE: H1-1 0.000
 H1-2 0.000
 H1-3 0.068

AISC Unity Check : 0.068, should be <= to 1 for Code compliance, [Passed]

WRC 107 Stress Analysis for Leg to Shell Junction, Ope Condition

Rectangular Attachment Parameter C11 282.842 mm.
 Rectangular Attachment Parameter C22 330.000 mm.

Input Echo, WRC107/537 Item 1, Description: LEGS

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Cylindrical	
Internal Corrosion Allowance	Cas	3.0000	mm.
Vessel Diameter	Dv	1075.000	mm.
Vessel Thickness	Tv	12.000	mm.
Design Temperature	T1	200.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Rectangular
Parameter C11	C11	282.84 mm.
Parameter C22	C22	330.00 mm.
Thickness of Reinforcing Pad	Tpad	10.000 mm.
Pad Parameter C11P	C11p	350.000 mm.
Pad Parameter C22P	C22p	350.000 mm.
Design Internal Pressure	Dp	6.400 bars
Include Pressure Thrust		No
Vessel Centerline Direction Cosine	Vx	0.000
Vessel Centerline Direction Cosine	Vy	1.000
Vessel Centerline Direction Cosine	Vz	0.000
Nozzle Centerline Direction Cosine	Nx	1.000
Nozzle Centerline Direction Cosine	Ny	0.000
Nozzle Centerline Direction Cosine	Nz	0.000

Global Force (SUS)	Fx	0.0	Kgf
Global Force (SUS)	Fy	1097.7	Kgf
Global Force (SUS)	Fz	0.0	Kgf
Global Moment (SUS)	Mx	0.0	Kg-m.
Global Moment (SUS)	My	0.0	Kg-m.
Global Moment (SUS)	Mz	71.2	Kg-m.

Internal Pressure (SUS)	P	6.40	bars
Include Pressure Thrust		No	

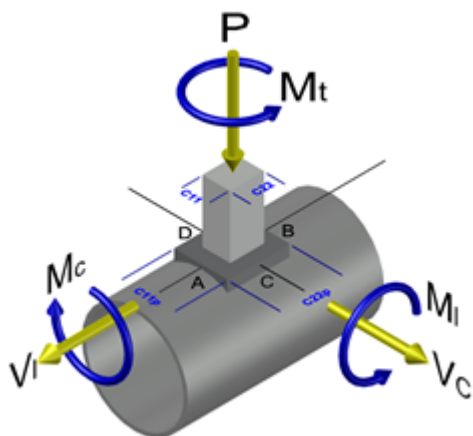
Global Force (OCC)	Fx	97.2	Kgf
Global Force (OCC)	Fy	272.4	Kgf
Global Force (OCC)	Fz	0.0	Kgf
Global Moment (OCC)	Mx	0.0	Kg-m.
Global Moment (OCC)	My	0.0	Kg-m.
Global Moment (OCC)	Mz	95.5	Kg-m.

Occasional Internal Pressure (OCC)	Pvar	0.00	bars
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Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

WRC 107 Stress Calculation for SUSstained loads:

Radial Load	P	0.0	Kgf
Circumferential Shear	VC	-0.0	Kgf
Longitudinal Shear	VL	1097.7	Kgf
Circumferential Moment	MC	-0.0	Kg-m.
Longitudinal Moment	ML	-71.2	Kg-m.
Torsional Moment	MT	-0.0	Kg-m.

Dimensionless Parameters used : Gamma = 28.95

Dimensionless Loads for Cylindrical Shells at Attachment Junction:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.303	4C	3.476	(A,B)
N(PHI) / (P/Rm)	0.303	3C	1.874	(C,D)



M(PHI) / (P)	0.271	2C1	0.026	(A,B)
M(PHI) / (P)	0.271	1C !	0.060	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.271	3A	1.201	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.300	1A	0.073	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.285	3B	2.553	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.289	1B	0.018	(A,B,C,D)
N(x) / (P/Rm)	0.288	3C	1.989	(A,B)
N(x) / (P/Rm)	0.288	4C	3.575	(C,D)
M(x) / (P)	0.291	1C1	0.048	(A,B)
M(x) / (P)	0.291	2C !	0.031	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.271	4A	2.429	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.339	2A	0.029	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.285	4B	1.176	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	0.305	2B	0.028	(A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		1.0	1.0	-1.0	-1.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		1.4	-1.4	-1.4	1.4	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		2.3	-0.4	-2.3	0.4	0.0	0.0	0.0	0.0
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML		0.5	0.5	-0.5	-0.5	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML		2.0	-2.0	-2.0	2.0	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.		2.5	-1.5	-2.5	1.5	0.0	0.0	0.0	0.0
Shear VC		-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL		0.0	0.0	0.0	0.0	-0.9	-0.9	0.9	0.9
Shear MT		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear		0.0	0.0	0.0	0.0	-0.9	-0.9	0.9	0.9
Str. Int.		2.5	1.5	2.5	1.5	1.7	1.7	1.7	1.7

Dimensionless Parameters used : Gamma = 60.56

Dimensionless Loads for Cylindrical Shells at Pad Edge:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.321	4C	5.433	(A,B)
N(PHI) / (P/Rm)	0.321	3C	2.207	(C,D)
M(PHI) / (P)	0.321	2C1	0.010	(A,B)
M(PHI) / (P)	0.321	1C !	0.066	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.321	3A	1.646	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.321	1A	0.058	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.321	3B	3.295	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.321	1B	0.008	(A,B,C,D)
N(x) / (P/Rm)	0.321	3C	2.207	(A,B)

N(x)	/ (P/Rm)	0.321	4C	5.433	(C,D)
M(x)	/ (P)	0.321	1C1	0.026	(A,B)
M(x)	/ (P)	0.321	2C !	0.034	(C,D)
N(x)	/ (MC/(Rm**2 * Beta))	0.321	4A	5.646	(A,B,C,D)
M(x)	/ (MC/(Rm * Beta))	0.321	2A	0.023	(A,B,C,D)
N(x)	/ (ML/(Rm**2 * Beta))	0.321	4B	1.834	(A,B,C,D)
M(x)	/ (ML/(Rm * Beta))	0.321	2B	0.011	(A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		2.7	2.7	-2.7	-2.7	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		2.3	-2.3	-2.3	2.3	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		4.9	0.4	-4.9	-0.4	0.0	0.0	0.0	0.0
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML		1.5	1.5	-1.5	-1.5	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML		3.3	-3.3	-3.3	3.3	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.		4.8	-1.8	-4.8	1.8	0.0	0.0	0.0	0.0
Shear VC		-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL		0.0	0.0	0.0	0.0	-1.7	-1.7	1.7	1.7
Shear MT		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear		0.0	0.0	0.0	0.0	-1.7	-1.7	1.7	1.7
Str. Int.		4.9	2.3	4.9	2.3	3.4	3.4	3.4	3.4

WRC 107 Stress Calculation for OCCasional loads:

Radial Load	P	97.2	Kgf
Circumferential Shear	VC	-0.0	Kgf
Longitudinal Shear	VL	272.4	Kgf
Circumferential Moment	MC	-0.0	Kg-m.
Longitudinal Moment	ML	-95.5	Kg-m.
Torsional Moment	MT	-0.0	Kg-m.

Dimensionless Parameters used : Gamma = 28.95

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.3	-0.3	-0.3	-0.3	-0.2	-0.2	-0.2	-0.2
Circ. Bend. P		-0.4	0.4	-0.4	0.4	-1.0	1.0	-1.0	1.0
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		1.3	1.3	-1.3	-1.3	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		1.8	-1.8	-1.8	1.8	-0.0	-0.0	-0.0	-0.0

Tot. Circ. Str.	2.4	-0.4	-3.8	0.6	-1.1	0.8	-1.1	0.8
Long. Memb. P	-0.2	-0.2	-0.2	-0.2	-0.3	-0.3	-0.3	-0.3
Long. Bend. P	-0.8	0.8	-0.8	0.8	-0.5	0.5	-0.5	0.5
Long. Memb. MC	-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC	-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML	0.7	0.7	-0.7	-0.7	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML	2.6	-2.6	-2.6	2.6	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.	2.3	-1.4	-4.2	2.5	-0.8	0.2	-0.8	0.2
Shear VC	-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL	0.0	0.0	0.0	0.0	-0.2	-0.2	0.2	0.2
Shear MT	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear	0.0	0.0	0.0	0.0	-0.2	-0.2	0.2	0.2
Str. Int.	2.4	1.4	4.2	2.5	1.2	0.8	1.2	0.8

Dimensionless Parameters used : Gamma = 60.56
 Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-1.1	-1.1	-1.1	-1.1	-0.4	-0.4	-0.4	-0.4
Circ. Bend. P		-0.7	0.7	-0.7	0.7	-4.7	4.7	-4.7	4.7
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		3.6	3.6	-3.6	-3.6	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		3.0	-3.0	-3.0	3.0	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		4.9	0.2	-8.4	-0.9	-5.1	4.3	-5.1	4.3
Long. Memb. P		-0.4	-0.4	-0.4	-0.4	-1.1	-1.1	-1.1	-1.1
Long. Bend. P		-1.8	1.8	-1.8	1.8	-2.4	2.4	-2.4	2.4
Long. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML		2.0	2.0	-2.0	-2.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML		4.5	-4.5	-4.5	4.5	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.		4.2	-1.1	-8.7	3.9	-3.5	1.4	-3.5	1.4
Shear VC		-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL		0.0	0.0	0.0	0.0	-0.4	-0.4	0.4	0.4
Shear MT		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear		0.0	0.0	0.0	0.0	-0.4	-0.4	0.4	0.4
Str. Int.		4.9	1.3	8.7	4.8	5.2	4.3	5.2	4.3

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		17.9	18.5	17.9	18.5	17.9	18.5	17.9	18.5
Circ. Pm (OCC)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pm(TOTAL)		17.9	18.5	17.9	18.5	17.9	18.5	17.9	18.5
Circ. Pl (SUS)		1.0	1.0	-1.0	-1.0	0.0	0.0	-0.0	-0.0
Circ. Pl (OCC)		1.0	1.0	-1.6	-1.6	-0.2	-0.2	-0.2	-0.2
Circ. Pl(TOTAL)		2.0	2.0	-2.6	-2.6	-0.2	-0.2	-0.2	-0.2



Circ. Q (SUS)	1.4	-1.4	-1.4	1.4	0.0	0.0	-0.0	0.0
Circ. Q (OCC)	1.4	-1.4	-2.2	2.2	-1.0	1.0	-1.0	1.0
Circ. Q (TOTAL)	2.8	-2.8	-3.6	3.6	-1.0	1.0	-1.0	1.0
Long. Pm (SUS)	8.9	8.9	8.9	8.9	8.9	8.9	8.9	8.9
Long. Pm (OCC)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pm(TOTAL)	8.9	8.9	8.9	8.9	8.9	8.9	8.9	8.9
Long. Pl (SUS)	0.5	0.5	-0.5	-0.5	0.0	0.0	-0.0	-0.0
Long. Pl (OCC)	0.5	0.5	-0.8	-0.8	-0.3	-0.3	-0.3	-0.3
Long. Pl(TOTAL)	1.0	1.0	-1.3	-1.3	-0.3	-0.3	-0.3	-0.3
Long. Q (SUS)	2.0	-2.0	-2.0	2.0	0.0	0.0	-0.0	0.0
Long. Q (OCC)	1.9	-1.9	-3.4	3.4	-0.5	0.5	-0.5	0.5
Long. Q (TOTAL)	3.8	-3.8	-5.3	5.3	-0.5	0.5	-0.5	0.5
Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm (OCC)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm(TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)	0.0	0.0	0.0	0.0	-0.9	-0.9	0.9	0.9
Shear Pl (OCC)	0.0	0.0	0.0	0.0	-0.2	-0.2	0.2	0.2
Shear Pl(TOTAL)	0.0	0.0	0.0	0.0	-1.1	-1.1	1.1	1.1
Shear Q (SUS)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (OCC)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)	17.9	18.5	17.9	18.5	17.9	18.5	17.9	18.5
Pm (SUS+OCC)	17.9	18.5	17.9	18.5	17.9	18.5	17.9	18.5
Pm+Pl (SUS)	18.9	19.5	16.9	17.6	18.0	18.6	18.0	18.6
Pm+Pl (SUS+OCC)	19.8	20.5	15.3	15.9	17.8	18.5	17.8	18.5
Pm+Pl+Q (Total)	22.6	17.7	11.7	19.5	16.9	19.4	16.9	19.4

Vessel Stress Summation Comparison (N/mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	18.53	138.00	Passed
Pm (SUS+OCC)	18.53	165.60	Passed
Pm+Pl (SUS)	19.50	207.00	Passed
Pm+Pl (SUS+OCC)	20.48	248.40	Passed
Pm+Pl+Q (TOTAL)	22.60	450.02	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

WRC 107/537 Stress Summations:**Vessel Stress Summation at Reinforcing Pad Edge (N/mm2)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		38.1	38.8	38.1	38.8	38.1	38.8	38.1	38.8
Circ. Pm (OCC)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pm(TOTAL)		38.1	38.8	38.1	38.8	38.1	38.8	38.1	38.8
Circ. Pl (SUS)		2.7	2.7	-2.7	-2.7	0.0	0.0	-0.0	-0.0
Circ. Pl (OCC)		2.5	2.5	-4.7	-4.7	-0.4	-0.4	-0.4	-0.4
Circ. Pl(TOTAL)		5.2	5.2	-7.3	-7.3	-0.4	-0.4	-0.4	-0.4
Circ. Q (SUS)		2.3	-2.3	-2.3	2.3	0.0	0.0	-0.0	0.0
Circ. Q (OCC)		2.3	-2.3	-3.7	3.7	-4.7	4.7	-4.7	4.7
Circ. Q (TOTAL)		4.6	-4.6	-6.0	6.0	-4.7	4.7	-4.7	4.7



Long. Pm (SUS)	19.1	19.1	19.1	19.1	19.1	19.1	19.1	19.1
Long. Pm (OCC)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pm(TOTAL)	19.1	19.1	19.1	19.1	19.1	19.1	19.1	19.1
Long. P1 (SUS)	1.5	1.5	-1.5	-1.5	0.0	0.0	-0.0	-0.0
Long. P1 (OCC)	1.6	1.6	-2.4	-2.4	-1.1	-1.1	-1.1	-1.1
Long. P1(TOTAL)	3.1	3.1	-3.9	-3.9	-1.1	-1.1	-1.1	-1.1
Long. Q (SUS)	3.3	-3.3	-3.3	3.3	0.0	0.0	-0.0	0.0
Long. Q (OCC)	2.6	-2.6	-6.3	6.3	-2.4	2.4	-2.4	2.4
Long. Q (TOTAL)	6.0	-6.0	-9.7	9.7	-2.4	2.4	-2.4	2.4
Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm (OCC)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm(TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear P1 (SUS)	0.0	0.0	0.0	0.0	-1.7	-1.7	1.7	1.7
Shear P1 (OCC)	0.0	0.0	0.0	0.0	-0.4	-0.4	0.4	0.4
Shear P1(TOTAL)	0.0	0.0	0.0	0.0	-2.1	-2.1	2.1	2.1
Shear Q (SUS)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (OCC)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)	38.1	38.8	38.1	38.8	38.1	38.8	38.1	38.8
Pm (SUS+OCC)	38.1	38.8	38.1	38.8	38.1	38.8	38.1	38.8
Pm+P1 (SUS)	40.8	41.4	35.4	36.1	38.3	38.9	38.3	38.9
Pm+P1 (SUS+OCC)	43.3	44.0	30.8	31.4	37.9	38.6	37.9	38.6
Pm+P1+Q (Total)	47.9	39.4	24.8	37.4	33.3	43.2	33.3	43.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	38.76	138.00	Passed
Pm (SUS+OCC)	38.76	165.60	Passed
Pm+P1 (SUS)	41.44	207.00	Passed
Pm+P1 (SUS+OCC)	43.98	248.40	Passed
Pm+P1+Q (TOTAL)	47.93	450.02	Passed

The Pm+P1+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+P1+Q allowable was based on a temperature range cycling from ambient to design temperature.

Bolting Size Requirement for Leg Baseplates :

Baseplate Material	SA-516 70
Baseplate Allowable Stress	SBA 137.90 N./mm ²
Baseplate Length	B 350.0000 mm.
Baseplate Width	D 350.0000 mm.
Baseplate Thickness	BTHK 20.0000 mm.
Leg Dimension Along Baseplate Length	d 199.9996 mm.
Leg Dimension Along Baseplate Width	b 199.9996 mm.
Bolt Circle Diameter	BCD 1045.0000 mm.
Bolt Material	SA-307 A
Bolt Allowable Tensile Stress	STBA 138.00 N./mm ²
Bolt Allowable Shear Stress	SBShear 82.80 N./mm ²
Anchor Bolt Nominal Diameter	BOD 36.0000 mm.
Number of Anchor Bolts in Tension per Leg	NB 1
Total Number of Anchors Bolt per Leg	NBT 1
Bolt Corrosion Allowance	bca 0.0000 mm.
Ultimate 28-day Concrete Strength	FCPRIME 20.685 N./mm ²

Leg Baseplate and Bolting Analysis, including Moments

Shear Stress in a Single Bolt [rb]:

= Shear Force/(Bolt Area • Number of Bolts)

= 244/(7.38 • 4)



= 0.8 N./mm². Must be less than 82.8 N./mm².

Plate thickness required (3/4"; 19mm) per D. Moss

Since there is only 1 bolt, the standard computation for this geometry is not possible. A reasonable bolt size should be selected. Per Megyesy, use a 1 inch (25.4 mm) bolt.

Computing the required bolt area based on a standard method not including the baseplate's interaction with the foundation.

Calculation of Load per Bolt, Wind Empty [W,wind]:
= ((4 • Moment/BCD) - Weight)/Num Bolts
= ((4 • 307/1045) - 2880)/4
= -426.615 Kgf

Calculation of Load per Bolt, Seismic Operating [W,seismic]:
= ((4 • Moment/BCD) - Weight)/Num Bolts
= ((4 • 335/1045) - 4707)/4
= -856.433 Kgf

There was a vertical load reducing the weight on the support.

Note: The bolts are not loaded in tension.

Review notes about nozzle loadings and supports in the Warnings report.

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RESULTS FOR LEGS : HydroTest Case Description: LEGS

Legs attached to: SHELL-1

Section Properties : Single Angle L200X200X16

European Structural Steel Data

Overall Leg Length		1830.000	mm.
Effective Leg Length	Leglen	1600.000	mm.
Distance Leg Up Side of Vessel		330.000	mm.
Number of Legs	Nleg	4	
Cross Sectional Area for L200X200X16	Aleg	61.800	cm ²
Section Inertia (strong axis)		2339.988	cm ⁴
Section Inertia (weak axis)		2339.988	cm ⁴
Section Modulus (strong axis)		162000.672	mm. ³
Section Modulus (weak axis)		162000.672	mm. ³
Radius of Gyration (strong axis)		61.600	mm.
Radius of Gyration (weak axis)		61.600	mm.

Leg Orientation - Diagonal

Overturning Moment at top of Legs		153.3	Kg-m.
Total Weight Load at top of Legs	W	5277.3	Kgf
Total Shear force at top of Legs		121.9	Kgf
Additional force in Leg due to Bracing	Fadd	0.0	Kgf
Occasional Load Factor	Occfac	1.333	
Effective Leg End Condition Factor	k	1.000	

Note: The Legs are Not Cross Braced

The Leg Shear Force includes Wind, Seismic, and Applied Load Effects.

Pad Width along Circumference	C11P	350.000	mm.
Pad Length along Vessel Axis	C22P	350.000	mm.
Pad Thickness	Tpad	10.000	mm.

Maximum Shear at top of one Leg [Vleg]:
 $= (\max(\text{Wind, Seismic, Blast}) + \text{applied forces}) (I_{\max} / I_{\text{tot}})$
 $= (122) (3781/9484)$
 $= 48.62 \text{ Kg}$

Axial Compression, Leg furthest from the Neutral Axis [Sma]:
 $= (W/N_{\text{leg}} + (M_{\text{leg}} / (N_{\text{legm}} \cdot R_n))) / A_{\text{leg}}$
 $= (51752/4 + (1503669 / (2 \cdot 614))) / 6180$
 $= 2.29 \text{ N./mm}^2$

Axial Compression, Leg closest to the Neutral Axis [Sva]:
 $= (W / N_{\text{leg}}) / A_{\text{leg}}$
 $= (5277/4) / 61.8$
 $= 2.09 \text{ N./mm}^2$

Computing Principal Axis and Inertias for Angle:

Leg lengths and thickness:	200.00	200.00	16.000
Distance to geometric centroid:	55.200	55.200	
Arm about YY:	47.200	52.800	
Arm about ZZ:	44.800	47.200	
Leg areas:	32.000	29.440	
Geometric inertia components YY:	719.73	1651.3	
Geometric inertia components ZZ:	1708.9	662.15	
Geometric inertias Iy & Iz:	2371.1	2371.1	
Product of inertia:	1410.3		
Mohrs Radius:	1410.3		
Average Inertia:	2371.1		

QFACT =	1.0000	FBZ	=	0.14893
Principal Axis Inertias (Z&W)			=	960.72 3781.4
Angle to Principal Axis			=	45.000

Distances to extreme fibers CW & CZ = 141.42 63.357
 FOB from Eq 5-5 = 1.9478
 Bending allowables Fby & Fbz = 0.16383 0.14893

Shear Center Coordinates Wo & Zo: 65.814 0.0000

Values for Elastic Flexural-Torsional Buckling Stress:

E, G, J, R0²: 29500. 11346. 1.2670 18.608
 AREA, LENGTH, Kw, Kz: 9.5790 62.992 1.0000 1.0000
 H, Few, Fez, Fej: 0.63919 695.90 176.80 80.651
 Fe computed from C4-1: 77.177

Initial (Kl/r)max, & (Kl/r)equiv = 40.580 61.421
 Final (Kl/r)max, & Cc = 61.421 127.18
 Fa based on Eq 4-1 = 0.11958

Leg Bending and Axial Stresses vs. Allowables: (N./mm²)

	Actual	Allowable
Weak Axis Bending :	3.56	198.53
Strong Axis Bending :	2.02	218.38
Axial Compression :	2.29	159.40

UNITY CHECKS ARE: H1-1 0.000
 H1-2 0.000
 H1-3 0.042

AISC Unity Check : 0.042, should be <= to 1 for Code compliance, [Passed]

WRC 107 Stress Analysis for Leg to Shell Junction, Test Condition

Rectangular Attachment Parameter C11 282.842 mm.
 Rectangular Attachment Parameter C22 330.000 mm.

Input Echo, WRC107/537 Item 1, Description: LEGS

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Cylindrical	
Internal Corrosion Allowance	Cas	3.0000	mm.
Vessel Diameter	Dv	1075.000	mm.
Vessel Thickness	Tv	12.000	mm.
Design Temperature	T1	200.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	234.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	234.00	N./mm ²
Attachment Type	Type	Rectangular	
Parameter C11	C11	282.84	mm.
Parameter C22	C22	330.00	mm.
Thickness of Reinforcing Pad	Tpad	10.000	mm.
Pad Parameter C11P	C11p	350.000	mm.
Pad Parameter C22P	C22p	350.000	mm.
Design Internal Pressure	Dp	6.400	bars
Include Pressure Thrust		No	
Vessel Centerline Direction Cosine	Vx	0.000	
Vessel Centerline Direction Cosine	Vy	1.000	
Vessel Centerline Direction Cosine	Vz	0.000	
Nozzle Centerline Direction Cosine	Nx	1.000	
Nozzle Centerline Direction Cosine	Ny	0.000	
Nozzle Centerline Direction Cosine	Nz	0.000	
Global Force (SUS)	Fx	0.0	Kgf
Global Force (SUS)	Fy	1319.3	Kgf

Global Force (SUS)	Fz	0.0	Kgf
Global Moment (SUS)	Mx	0.0	Kg-m.
Global Moment (SUS)	My	0.0	Kg-m.
Global Moment (SUS)	Mz	85.6	Kg-m.

Internal Pressure (SUS)	P	6.40	bars
Include Pressure Thrust		No	

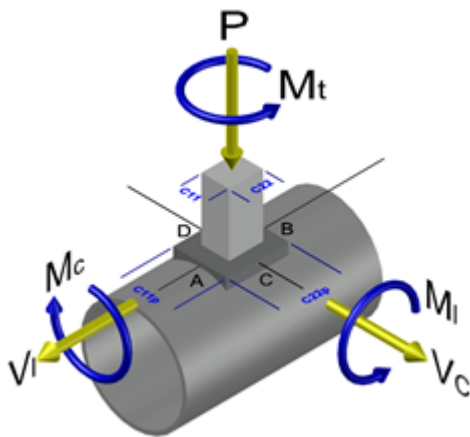
Global Force (OCC)	Fx	48.6	Kgf
Global Force (OCC)	Fy	124.8	Kgf
Global Force (OCC)	Fz	0.0	Kgf
Global Moment (OCC)	Mx	0.0	Kg-m.
Global Moment (OCC)	My	0.0	Kg-m.
Global Moment (OCC)	Mz	47.0	Kg-m.

Occasional Internal Pressure (OCC)	Pvar	2.23	bars
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Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

WRC 107 Stress Calculation for SUStained loads:

Radial Load	P	0.0	Kgf
Circumferential Shear	VC	-0.0	Kgf
Longitudinal Shear	VL	1319.3	Kgf
Circumferential Moment	MC	-0.0	Kg-m.
Longitudinal Moment	ML	-85.6	Kg-m.
Torsional Moment	MT	-0.0	Kg-m.

Dimensionless Parameters used : Gamma = 28.95

Dimensionless Loads for Cylindrical Shells at Attachment Junction:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.303	4C	3.476	(A,B)
N(PHI) / (P/Rm)	0.303	3C	1.874	(C,D)
M(PHI) / (P)	0.271	2C1	0.026	(A,B)
M(PHI) / (P)	0.271	1C !	0.060	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.271	3A	1.201	(A,B,C,D)



M(PHI) / (MC/(Rm * Beta))	0.300	1A	0.073	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.285	3B	2.553	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.289	1B	0.018	(A,B,C,D)
N(x) / (P/Rm)	0.288	3C	1.989	(A,B)
N(x) / (P/Rm)	0.288	4C	3.575	(C,D)
M(x) / (P)	0.291	1C1	0.048	(A,B)
M(x) / (P)	0.291	2C !	0.031	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.271	4A	2.429	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.339	2A	0.029	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.285	4B	1.176	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	0.305	2B	0.028	(A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		1.2	1.2	-1.2	-1.2	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		1.6	-1.6	-1.6	1.6	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		2.8	-0.5	-2.8	0.5	0.0	0.0	0.0	0.0
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML		0.6	0.6	-0.6	-0.6	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML		2.4	-2.4	-2.4	2.4	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.		2.9	-1.8	-2.9	1.8	0.0	0.0	0.0	0.0
Shear VC		-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL		0.0	0.0	0.0	0.0	-1.0	-1.0	1.0	1.0
Shear MT		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear		0.0	0.0	0.0	0.0	-1.0	-1.0	1.0	1.0
Str. Int.		2.9	1.8	2.9	1.8	2.1	2.1	2.1	2.1

Dimensionless Parameters used : Gamma = 60.56

Dimensionless Loads for Cylindrical Shells at Pad Edge:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.321	4C	5.433	(A,B)
N(PHI) / (P/Rm)	0.321	3C	2.207	(C,D)
M(PHI) / (P)	0.321	2C1	0.010	(A,B)
M(PHI) / (P)	0.321	1C !	0.066	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.321	3A	1.646	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.321	1A	0.058	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.321	3B	3.295	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.321	1B	0.008	(A,B,C,D)
N(x) / (P/Rm)	0.321	3C	2.207	(A,B)
N(x) / (P/Rm)	0.321	4C	5.433	(C,D)
M(x) / (P)	0.321	1C1	0.026	(A,B)
M(x) / (P)	0.321	2C !	0.034	(C,D)

N(x) / (MC/(Rm**2 * Beta)) 0.321 4A 5.646 (A,B,C,D)
 M(x) / (MC/(Rm * Beta)) 0.321 2A 0.023 (A,B,C,D)
 N(x) / (ML/(Rm**2 * Beta)) 0.321 4B 1.834 (A,B,C,D)
 M(x) / (ML/(Rm * Beta)) 0.321 2B 0.011 (A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		3.2	3.2	-3.2	-3.2	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		2.7	-2.7	-2.7	2.7	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		5.9	0.5	-5.9	-0.5	0.0	0.0	0.0	0.0
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML		1.8	1.8	-1.8	-1.8	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML		4.0	-4.0	-4.0	4.0	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.		5.8	-2.2	-5.8	2.2	0.0	0.0	0.0	0.0
Shear VC		-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL		0.0	0.0	0.0	0.0	-2.1	-2.1	2.1	2.1
Shear MT		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear		0.0	0.0	0.0	0.0	-2.1	-2.1	2.1	2.1
Str. Int.		5.9	2.7	5.9	2.7	4.1	4.1	4.1	4.1

WRC 107 Stress Calculation for OCCasional loads:

Radial Load	P	48.6	Kgf
Circumferential Shear	VC	-0.0	Kgf
Longitudinal Shear	VL	124.8	Kgf
Circumferential Moment	MC	-0.0	Kg-m.
Longitudinal Moment	ML	-47.0	Kg-m.
Torsional Moment	MT	-0.0	Kg-m.

Dimensionless Parameters used : Gamma = 28.95

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.2	-0.2	-0.2	-0.2	-0.1	-0.1	-0.1	-0.1
Circ. Bend. P		-0.2	0.2	-0.2	0.2	-0.5	0.5	-0.5	0.5
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		0.6	0.6	-0.6	-0.6	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		0.9	-0.9	-0.9	0.9	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		1.2	-0.2	-1.9	0.3	-0.6	0.4	-0.6	0.4
Long. Memb. P		-0.1	-0.1	-0.1	-0.1	-0.2	-0.2	-0.2	-0.2

Long. Bend. P	-0.4	0.4	-0.4	0.4	-0.2	0.2	-0.2	0.2
Long. Memb. MC	-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC	-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML	0.3	0.3	-0.3	-0.3	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML	1.3	-1.3	-1.3	1.3	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.	1.1	-0.7	-2.1	1.3	-0.4	0.1	-0.4	0.1
Shear VC	-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL	0.0	0.0	0.0	0.0	-0.1	-0.1	0.1	0.1
Shear MT	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear	0.0	0.0	0.0	0.0	-0.1	-0.1	0.1	0.1
Str. Int.	1.2	0.7	2.1	1.3	0.6	0.4	0.6	0.4

Dimensionless Parameters used : Gamma = 60.56
 Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress Load		Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.5	-0.5	-0.5	-0.5	-0.2	-0.2	-0.2	-0.2
Circ. Bend. P		-0.4	0.4	-0.4	0.4	-2.3	2.3	-2.3	2.3
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		1.8	1.8	-1.8	-1.8	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		1.5	-1.5	-1.5	1.5	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		2.4	0.1	-4.1	-0.5	-2.6	2.1	-2.6	2.1
Long. Memb. P		-0.2	-0.2	-0.2	-0.2	-0.5	-0.5	-0.5	-0.5
Long. Bend. P		-0.9	0.9	-0.9	0.9	-1.2	1.2	-1.2	1.2
Long. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML		1.0	1.0	-1.0	-1.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML		2.2	-2.2	-2.2	2.2	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.		2.1	-0.5	-4.3	1.9	-1.7	0.7	-1.7	0.7
Shear VC		-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL		0.0	0.0	0.0	0.0	-0.2	-0.2	0.2	0.2
Shear MT		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear		0.0	0.0	0.0	0.0	-0.2	-0.2	0.2	0.2
Str. Int.		2.4	0.6	4.3	2.4	2.6	2.2	2.6	2.2

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		17.9	18.5	17.9	18.5	17.9	18.5	17.9	18.5
Circ. Pm (OCC)		6.2	6.5	6.2	6.5	6.2	6.5	6.2	6.5
Circ. Pm(TOTAL)		24.1	25.0	24.1	25.0	24.1	25.0	24.1	25.0
Circ. Pl (SUS)		1.2	1.2	-1.2	-1.2	0.0	0.0	-0.0	-0.0
Circ. Pl (OCC)		0.5	0.5	-0.8	-0.8	-0.1	-0.1	-0.1	-0.1
Circ. Pl(TOTAL)		1.6	1.6	-2.0	-2.0	-0.1	-0.1	-0.1	-0.1
Circ. Q (SUS)		1.6	-1.6	-1.6	1.6	0.0	0.0	-0.0	0.0
Circ. Q (OCC)		0.7	-0.7	-1.1	1.1	-0.5	0.5	-0.5	0.5
Circ. Q (TOTAL)		2.3	-2.3	-2.7	2.7	-0.5	0.5	-0.5	0.5

Long. Pm (SUS)	8.9	8.9	8.9	8.9	8.9	8.9	8.9	8.9
Long. Pm (OCC)	3.1	3.1	3.1	3.1	3.1	3.1	3.1	3.1
Long. Pm(TOTAL)	12.1	12.1	12.1	12.1	12.1	12.1	12.1	12.1
Long. Pl (SUS)	0.6	0.6	-0.6	-0.6	0.0	0.0	-0.0	-0.0
Long. Pl (OCC)	0.2	0.2	-0.4	-0.4	-0.2	-0.2	-0.2	-0.2
Long. Pl(TOTAL)	0.8	0.8	-1.0	-1.0	-0.2	-0.2	-0.2	-0.2
Long. Q (SUS)	2.4	-2.4	-2.4	2.4	0.0	0.0	-0.0	0.0
Long. Q (OCC)	0.9	-0.9	-1.7	1.7	-0.2	0.2	-0.2	0.2
Long. Q (TOTAL)	3.3	-3.3	-4.0	4.0	-0.2	0.2	-0.2	0.2
Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm (OCC)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm(TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)	0.0	0.0	0.0	0.0	-1.0	-1.0	1.0	1.0
Shear Pl (OCC)	0.0	0.0	0.0	0.0	-0.1	-0.1	0.1	0.1
Shear Pl(TOTAL)	0.0	0.0	0.0	0.0	-1.1	-1.1	1.1	1.1
Shear Q (SUS)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (OCC)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)	17.9	18.5	17.9	18.5	17.9	18.5	17.9	18.5
Pm (SUS+OCC)	24.1	25.0	24.1	25.0	24.1	25.0	24.1	25.0
Pm+Pl (SUS)	19.1	19.7	16.7	17.4	18.0	18.6	18.0	18.6
Pm+Pl (SUS+OCC)	25.8	26.6	22.2	23.0	24.1	25.0	24.1	25.0
Pm+Pl+Q (Total)	28.1	24.3	19.4	25.7	23.7	25.5	23.7	25.5

Vessel Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	18.53	234.00	Passed
Pm (SUS+OCC)	24.99	280.79	Passed
Pm+P1 (SUS)	19.70	350.99	Passed
Pm+P1 (SUS+OCC)	26.63	421.19	Passed
Pm+P1+0 (TOTAL)	28.08	701.99	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The P_m+P_I+Q allowable was based on a temperature range cycling from ambient to design temperature.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²)

[illegible]

Long. P1 (SUS)	1.8	1.8	-1.8	-1.8	0.0	0.0	-0.0	-0.0
Long. P1 (OCC)	0.8	0.8	-1.2	-1.2	-0.5	-0.5	-0.5	-0.5
Long. P1(TOTAL)	2.6	2.6	-3.0	-3.0	-0.5	-0.5	-0.5	-0.5
Long. Q (SUS)	4.0	-4.0	-4.0	4.0	0.0	0.0	-0.0	0.0
Long. Q (OCC)	1.3	-1.3	-3.1	3.1	-1.2	1.2	-1.2	1.2
Long. Q (TOTAL)	5.3	-5.3	-7.1	7.1	-1.2	1.2	-1.2	1.2
Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm (OCC)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm(TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear P1 (SUS)	0.0	0.0	0.0	0.0	-2.1	-2.1	2.1	2.1
Shear P1 (OCC)	0.0	0.0	0.0	0.0	-0.2	-0.2	0.2	0.2
Shear P1(TOTAL)	0.0	0.0	0.0	0.0	-2.2	-2.2	2.2	2.2
Shear Q (SUS)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (OCC)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)	38.1	38.8	38.1	38.8	38.1	38.8	38.1	38.8
Pm (SUS+OCC)	51.4	52.3	51.4	52.3	51.4	52.3	51.4	52.3
Pm+P1 (SUS)	41.3	42.0	34.9	35.5	38.3	39.0	38.3	39.0
Pm+P1 (SUS+OCC)	55.9	56.7	45.9	46.7	51.4	52.2	51.4	52.2
Pm+P1+Q (Total)	59.7	52.9	41.3	51.3	49.0	54.6	49.0	54.6

Vessel Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	38.76	234.00	Passed
Pm (SUS+OCC)	52.26	280.79	Passed
Pm+P1 (SUS)	41.98	350.99	Passed
Pm+P1 (SUS+OCC)	56.73	421.19	Passed
Pm+P1+Q (TOTAL)	59.73	701.99	Passed

The Pm+P1+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+P1+Q allowable was based on a temperature range cycling from ambient to design temperature.

Bolting Size Requirement for Leg Baseplates :

Baseplate Material	SA-516 70
Baseplate Allowable Stress	SBA 137.90 N./mm2
Baseplate Length	B 350.0000 mm.
Baseplate Width	D 350.0000 mm.
Baseplate Thickness	BTHK 20.0000 mm.
Leg Dimension Along Baseplate Length	d 199.9996 mm.
Leg Dimension Along Baseplate Width	b 199.9996 mm.
Bolt Circle Diameter	BCD 1045.0000 mm.
Bolt Material	SA-307 A
Bolt Allowable Tensile Stress	STBA 138.00 N./mm2
Bolt Allowable Shear Stress	SBShear 82.80 N./mm2
Anchor Bolt Nominal Diameter	BOD 36.0000 mm.
Number of Anchor Bolts in Tension per Leg	NB 1
Total Number of Anchors Bolt per Leg	NBT 1
Bolt Corrosion Allowance	bca 0.0000 mm.
Ultimate 28-day Concrete Strength	FCPRIME 20.685 N./mm2

Leg Baseplate and Bolting Analysis, including Moments

Shear Stress in a Single Bolt [rb]:
 = Shear Force/(Bolt Area • Number of Bolts)
 = 244/(7.38 • 4)
 = 0.8 N./mm². Must be less than 82.8 N./mm².

Plate thickness required (3/4"; 19mm) per D. Moss



Since there is only 1 bolt, the standard computation for this geometry is not possible. A reasonable bolt size should be selected. Per Megyesy, use a 1 inch (25.4 mm) bolt.

Review notes about nozzle loadings and supports in the Warnings report.

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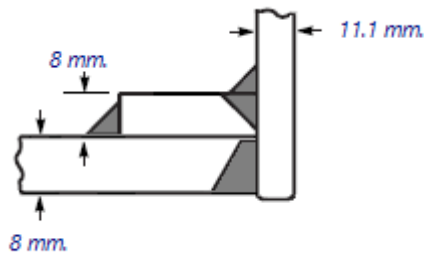
Input, Nozzle Desc: N3-4"
From: 10

Pressure for Reinforcement Calculations	P	6.624	bars
Temperature for Internal Pressure	Temp	200	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	1075.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	8.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm2
Allowable Stress At Ambient	Sna	118.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		120	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	160.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	8.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	220.0000	mm.
Thickness of Pad	te	8.0000	mm.
Weld leg size between Pad and Shell	Wp	6.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	8.0000	mm.
Reinforcing Pad Width		52.8500	mm.
Flange Class		300	
Flange Grade		GR 1.1	
Piping Exit Angle [North Clockwise reference]		0.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		790.0	mm.
Centerline Distance to Tangent		630.0	mm.

The Pressure Design option was Overall MAWP.

**Reinforcement CALCULATION, Description: N3-4"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	4.500 in.
Actual Thickness Used in Calculation	0.438 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 6.62 \cdot 1081 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 6.62)$
 $= 2.3240$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 6.62 \cdot 57.1 / (118 \cdot 1 + 0.4 \cdot 6.62)$
 $= 0.3201$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.4269 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	196.0992 mm.
Parallel to Vessel Wall, opening length	d	98.0496 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		12.5000 mm.

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118/138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118/138)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, Sp/Sv)$
 $= \min(1, 138/138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$

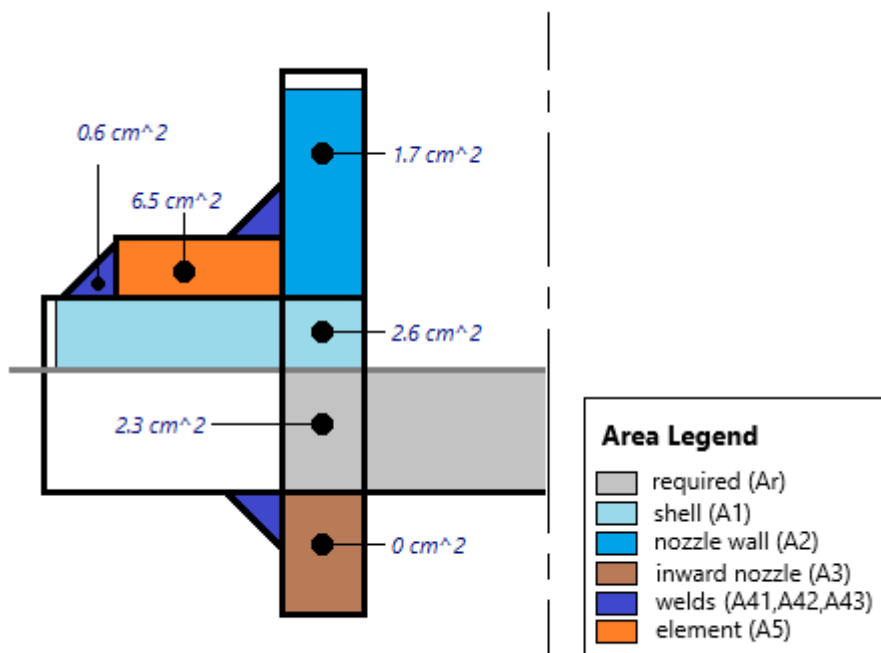


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	2.333	1.441	NA
Area in Shell	A1	2.561	2.038	NA
Area in Nozzle Wall	A2	1.668	1.646	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.596	0.596	NA
Area in Element	A5	6.544	6.544	NA
TOTAL AREA AVAILABLE	Atot	11.370	10.824	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (98 \cdot 2.32 \cdot 1 + 2 \cdot 8.13 \cdot 2.32 \cdot 1 \cdot (1 - 0.86)) \\
 &= 2.333 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 98(1 \cdot 5 - 1 \cdot 2.32) - 2 \cdot 8.13 \\
 &\quad (1 \cdot 5 - 1 \cdot 2.32) \cdot (1 - 0.86) \\
 &= 2.561 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\
 &= (2 \cdot 12.5) \cdot (8.13 - 0.32) \cdot 0.86 \\
 &= 1.668 \text{ cm}^2
 \end{aligned}$$

Area Available in Welds [A41 + A42 + A43]:

$$\begin{aligned}
 &= (Wo^2 - Ar \text{ Lost}) \cdot Fr3 + ((Wi - can/0.707)^2 - Ar \text{ Lost}) \cdot fr2 + Wp^2 \cdot fr4 \\
 &= (0.7) \cdot 0.86 + (0) \cdot 0.86 + 0^2 \cdot 1
 \end{aligned}$$

= 0.596 cm²

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lp}, t_e)) \cdot f_r$$

$$= (196 - 114) \cdot 8 \cdot 1$$

$$= 6.544 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.4269 mm.
Wall Thickness per UG16(b),	tr16b = 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 5.5768 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 5.5768 mm.
Wall Thickness, shell/head, external pressure	trb2 = 3.4021 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm.
Wall Thickness per table UG-45	tb3 = 8.2578 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[8.26, \max(5.58, 4.5)]$$

$$= 5.5768 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.43, 5.58)$$

$$= 5.5768 \text{ mm.}$$

Available Nozzle Neck Thickness = 0.875 * 11.125 = 9.735 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 34.1,	Allowable	: 118.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 260.9 N./mm ²	Passed
Occasional	: 1.8,	Allowable	: 156.9 N./mm ²	Passed
Shear	: 21.9,	Allowable	: 82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 9.73, tr = 0.32, c = 3 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.048, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 8, c = 3 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.46, Temp. Reduction = 37 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 8, c = 3 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.46, Temp. Reduction = 37 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 8, tr = 2.32, c = 3 mm., E* = 1

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.46$, Temp. Reduction = 37 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $tg = 8$, $tr = 2.32$, $c = 3$ mm., $E^* = 1$
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.46$, Temp. Reduction = 37 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Gov. MDMT of the Nozzle : -48 °C
 Gov. MDMT of the Reinforcement Pad : -48 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $6.62/51.00 = 0.130$

Weld Size Calculations, Description: N3-4"

Intermediate Calc. for nozzle/shell Welds T_{min} 8.0000 mm.
 Intermediate Calc. for pad/shell Welds T_{minPad} 7.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.6000 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm.
Pad Weld	$3.5000 = 0.5 \cdot T_{minPad}$	$4.2420 = 0.7 \cdot W_p$ mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (2.33 - 2.56 + 2 \cdot 8.13 \cdot 0.86 \cdot (1 \cdot 5 - 2.32)) \cdot 138)$
 $= 203.17$ Kg

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (Wi - Can / .707)^2 \cdot fr2) \cdot Sv$
 $= (1.67 + 6.54 + 0.6 - 0 \cdot 0.86) \cdot 138$
 $= 12395.40$ Kg

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (1.67 + 0 + 0.86 + (0.69)) \cdot 138$
 $= 4528.67$ Kg

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (1.67 + 0 + 0.6 + 6.54 + (0.69)) \cdot 138$
 $= 13373.04$ Kg

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_o \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 114 \cdot 10 \cdot 0.49 \cdot 118$
 $= 10585$ Kg

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.14/2.0) \cdot 220 \cdot 6 \cdot 0.49 \cdot 138$



= 14297. Kgf

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 53.1) \cdot (11.1 - 3) \cdot 0.7 \cdot 118$$

$$= 11414. Kgf$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.14/2) \cdot 114 \cdot 8 \cdot 0.74 \cdot 138$$

$$= 14957. Kgf$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 114 \cdot (8 - 3) \cdot 0.74 \cdot 138$$

$$= 9348. Kgf$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (14297 + 11414) = 25710 Kgf$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (10585 + 14957 + 9348 + 0) = 34890 Kgf$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (14297 + 9348 + 0) = 23644 Kgf$$

Summary of Failure Path Calculations:

Path 1-1 = 25710 Kgf, must exceed W = 203 Kgf	or W1 = 12395 Kgf
Path 2-2 = 34889 Kgf, must exceed W = 203 Kgf	or W2 = 4528 Kgf
Path 3-3 = 23644 Kgf, must exceed W = 203 Kgf	or W3 = 13373 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6.4 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1.03 bars

The Drop for this Nozzle is : 1.6783 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 169.6783 mm.

Input Echo, WRC107/537 Item 1, Description: N3-4" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm.
Vessel Diameter	Dv	1935.000	mm.
Vessel Thickness	Tv	8.000	mm.
Design Temperature	T1	200.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm2
Vessel Hot S.I. Allowable	Smh	138.00	N./mm2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round
WRC107 Attachment Classification	Holsol	Hollow

Diameter Basis for Nozzle	Nbasis	OD
Corrosion Allowance for Nozzle	Can	3.0000 mm.
Nozzle Diameter	Dn	114.300 mm.



Nozzle Thickness	Tn	11.125	mm.
Nozzle Material		SA-106 B	
Nozzle UNS Number		K03006	
Nozzle Cold S.I. Allowable	SNmc	118.00	N./mm2
Nozzle Hot S.I. Allowable	SNmh	118.00	N./mm2
Thickness of Reinforcing Pad	Tpad	8.000	mm.
Diameter of Reinforcing Pad	Dpad	220.000	mm.
Design Internal Pressure	Dp	6.624	bars
Include Pressure Thrust		No	

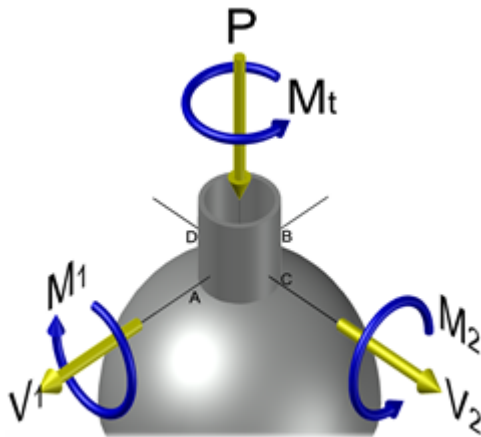
External Forces and Moments in WRC 107/537 Convention:

Radial Load	(SUS)	P	764.5	Kgf
Longitudinal Shear	(SUS)	(V1) V1	509.7	Kgf
Circumferential Shear	(SUS)	(Vc) V2	509.7	Kgf
Circumferential Moment	(SUS)	(Mc) M1	142.7	Kg-m.
Longitudinal Moment	(SUS)	(M1) M2	142.7	Kg-m.
Torsional Moment	(SUS)	Mt	203.9	Kg-m.

Use Interactive Control No
WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No
Compute Pressure Stress per WRC-368 No
Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))} \\
 &= 114.3 + 2 \cdot 1.65 \cdot \sqrt{(973.0 (8.0 - 3.0))} \\
 &= 344.473 \text{ mm.}
 \end{aligned}$$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load		P	764.5	Kgf
Circumferential Shear	(VC)	V2	509.7	Kgf
Longitudinal Shear	(VL)	V1	509.7	Kgf
Circumferential Moment	(MC)	M1	142.7	Kg-m.
Longitudinal Moment	(ML)	M2	142.7	Kg-m.
Torsional Moment		MT	203.9	Kg-m.

Dimensionless Parameters: U = 0.51 TAU = 6.53 RHO = 1.60

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SP 2	0.07178	(A,B,C,D)
M(x) / P	SP 2	0.06457	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SM 2	0.15243	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SM 2	0.16227	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SM 2	0.15243	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SM 2	0.16227	(A,B,C,D)
N(y) * T / P	SP 2	0.17437	(A,B,C,D)
M(y) / P	SP 2	0.04159	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SM 2	0.14421	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SM 2	0.14050	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SM 2	0.14421	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SM 2	0.14050	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		-3.2	-3.2	-3.2	-3.2	-3.2	-3.2	-3.2	-3.2
Rad. Bend. P		-17.2	17.2	-17.2	17.2	-17.2	17.2	-17.2	17.2
Rad. Memb. MC		0.0	0.0	0.0	0.0	-11.2	-11.2	11.2	11.2
Rad. Bend. MC		0.0	0.0	0.0	0.0	-71.5	71.5	71.5	-71.5
Rad. Memb. ML		-11.2	-11.2	11.2	11.2	0.0	0.0	0.0	0.0
Rad. Bend. ML		-71.5	71.5	71.5	-71.5	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-103.1	74.3	62.4	-46.3	-103.1	74.3	62.4	-46.3
Tang. Memb. P		-7.7	-7.7	-7.7	-7.7	-7.7	-7.7	-7.7	-7.7
Tang. Bend. P		-11.1	11.1	-11.1	11.1	-11.1	11.1	-11.1	11.1
Tang. Memb. MC		0.0	0.0	0.0	0.0	-10.6	-10.6	10.6	10.6
Tang. Bend. MC		0.0	0.0	0.0	0.0	-61.9	61.9	61.9	-61.9
Tang. Memb. ML		-10.6	-10.6	10.6	10.6	0.0	0.0	0.0	0.0
Tang. Bend. ML		-61.9	61.9	61.9	-61.9	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-91.4	54.7	53.7	-48.0	-91.4	54.7	53.7	-48.0
Shear VC		2.1	2.1	-2.1	-2.1	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-2.1	-2.1	2.1	2.1
Shear MT		7.5	7.5	7.5	7.5	7.5	7.5	7.5	7.5
Tot. Shear		9.6	9.6	5.4	5.4	5.4	5.4	9.6	9.6
Str. Int.		108.5	78.3	64.9	52.6	105.2	75.7	68.6	56.9

Unitless Prm: U = 1.58 TAU = 0.00 (13.04) RHO = 0.00 (0.62)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SR 2	0.03549	(A,B,C,D)
M(x) / P	SR 2	0.01725	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SR 3	0.03476	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SR 3	0.02507	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SR 3	0.03476	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SR 3	0.02507	(A,B,C,D)
N(y) * T / P	SR 2	0.01059	(A,B,C,D)
M(y) / P	SR 2	0.00517	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SR 3	0.01039	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SR 3	0.00743	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SR 3	0.01039	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SR 3	0.00743	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		-10.6	-10.6	-10.6	-10.6	-10.6	-10.6	-10.6	-10.6
Rad. Bend. P		-31.0	31.0	-31.0	31.0	-31.0	31.0	-31.0	31.0
Rad. Memb. MC		0.0	0.0	0.0	0.0	-27.9	-27.9	27.9	27.9
Rad. Bend. MC		0.0	0.0	0.0	0.0	-120.7	120.7	120.7	-120.7
Rad. Memb. ML		-27.9	-27.9	27.9	27.9	0.0	0.0	0.0	0.0
Rad. Bend. ML		-120.7	120.7	120.7	-120.7	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-190.3	113.2	106.9	-72.4	-190.3	113.2	106.9	-72.4
Tang. Memb. P		-3.2	-3.2	-3.2	-3.2	-3.2	-3.2	-3.2	-3.2
Tang. Bend. P		-9.3	9.3	-9.3	9.3	-9.3	9.3	-9.3	9.3
Tang. Memb. MC		0.0	0.0	0.0	0.0	-8.3	-8.3	8.3	8.3
Tang. Bend. MC		0.0	0.0	0.0	0.0	-35.8	35.8	35.8	-35.8
Tang. Memb. ML		-8.3	-8.3	8.3	8.3	0.0	0.0	0.0	0.0
Tang. Bend. ML		-35.8	35.8	35.8	-35.8	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-56.6	33.6	31.6	-21.3	-56.6	33.6	31.6	-21.3
Shear VC		2.9	2.9	-2.9	-2.9	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-2.9	-2.9	2.9	2.9
Shear MT		5.3	5.3	5.3	5.3	5.3	5.3	5.3	5.3
Tot. Shear		8.2	8.2	2.4	2.4	2.4	2.4	8.2	8.2
Str. Int.		190.8	114.0	107.0	72.5	190.3	113.3	107.8	73.7

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		24.7	24.7	24.7	24.7	24.7	24.7	24.7	24.7
Rad. Pl (SUS)		-14.4	-14.4	8.0	8.0	-14.4	-14.4	8.0	8.0
Rad. Q (SUS)		-88.7	88.7	54.4	-54.4	-88.7	88.7	54.4	-54.4
Long. Pm (SUS)		24.7	24.7	24.7	24.7	24.7	24.7	24.7	24.7
Long. Pl (SUS)		-18.3	-18.3	2.9	2.9	-18.3	-18.3	2.9	2.9
Long. Q (SUS)		-73.0	73.0	50.9	-50.9	-73.0	73.0	50.9	-50.9
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		2.1	2.1	-2.1	-2.1	-2.1	-2.1	2.1	2.1
Shear Q (SUS)		7.5	7.5	7.5	7.5	7.5	7.5	7.5	7.5
Pm (SUS)		24.7	24.7	24.7	24.7	24.7	24.7	24.7	24.7
Pm+Pl (SUS)		11.3	11.3	33.5	33.5	11.3	11.3	33.5	33.5
Pm+Pl+Q (Total)		83.8	103.0	89.7	27.9	80.5	100.4	93.3	32.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	24.73	138.00	Passed
Pm+Pl (SUS)	33.52	207.00	Passed
Pm+Pl+Q (TOTAL)	103.01	413.99	Passed



Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	64.3	64.3	64.3	64.3	64.3	64.3	64.3	64.3
Rad.	Pl (SUS)	-38.5	-38.5	17.3	17.3	-38.5	-38.5	17.3	17.3
Rad.	Q (SUS)	-151.8	151.8	89.7	-89.7	-151.8	151.8	89.7	-89.7
Long.	Pm (SUS)	64.3	64.3	64.3	64.3	64.3	64.3	64.3	64.3
Long.	Pl (SUS)	-11.5	-11.5	5.2	5.2	-11.5	-11.5	5.2	5.2
Long.	Q (SUS)	-45.1	45.1	26.5	-26.5	-45.1	45.1	26.5	-26.5
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	2.9	2.9	-2.9	-2.9	-2.9	-2.9	2.9	2.9
Shear	Q (SUS)	5.3	5.3	5.3	5.3	5.3	5.3	5.3	5.3
Pm (SUS)		64.3	64.3	64.3	64.3	64.3	64.3	64.3	64.3
Pm+Pl (SUS)		53.1	53.1	82.2	82.2	53.1	53.1	82.2	82.2
Pm+Pl+Q (Total)		134.7	178.3	171.3	51.3	133.8	177.6	172.1	53.6

Vessel Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	64.29	138.00	Passed
Pm+Pl (SUS)	82.20	207.00	Passed
Pm+Pl+Q (TOTAL)	178.33	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N1-12"

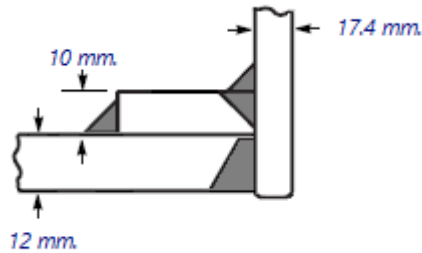
From: 20

Pressure for Reinforcement Calculations	P	6.624	bars
Temperature for Internal Pressure	Temp	200	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	1075.00	mm.
Design Length of Section	L	3454.1667	mm.
Shell Finished (Minimum) Thickness	t	12.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		2050.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm2
Allowable Stress At Ambient	Sna	118.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		12.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	330.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	8.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	750.0000	mm.
Thickness of Pad	te	10.0000	mm.
Weld leg size between Pad and Shell	Wp	8.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	10.0000	mm.
Reinforcing Pad Width		213.0750	mm.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Overall MAWP.

**Reinforcement CALCULATION, Description: N1-12"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	12.750 in.
Actual Thickness Used in Calculation	0.687 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.62 \cdot 540 / (138 \cdot 1 - 0.6 \cdot 6.62)$
 $= 2.6021$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 6.62 \cdot 162 / (118 \cdot 1 + 0.4 \cdot 6.62)$
 $= 0.9070$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 1.0628 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	589.9008 mm.
Parallel to Vessel Wall, opening length	d	294.9504 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		22.5000 mm.

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$

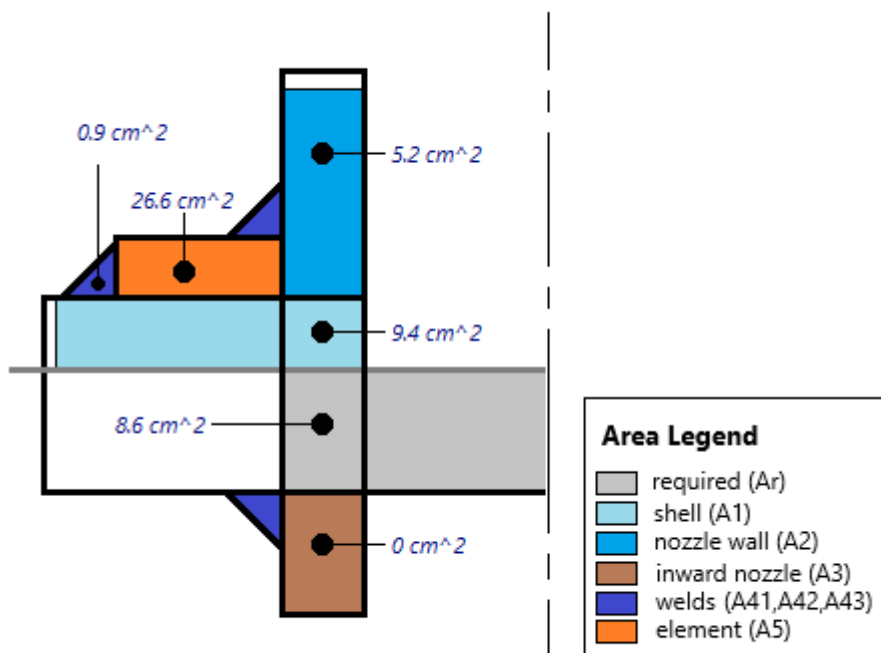


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	7.784	8.601	NA
Area in Shell	A1	18.603	9.448	NA
Area in Nozzle Wall	A2	5.211	5.151	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.855	0.855	NA
Area in Element	A5	26.605	26.605	NA
TOTAL AREA AVAILABLE	Atot	51.274	42.059	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1-fr1)) \text{ per UG-37(d)}$$

$$= 0.5(295 \cdot 5.75 \cdot 1 + 2 \cdot 14.4 \cdot 5.75 \cdot 1(1-0.86))$$

$$= 8.601 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 295(1 \cdot 9 - 1 \cdot 5.75) - 2 \cdot 14.4$$

$$(1 \cdot 9 - 1 \cdot 5.75) \cdot (1 - 0.86)$$

$$= 9.448 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 22.5) \cdot (14.4 - 1.06) \cdot 0.86$$

$$= 5.151 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= Wo^2 \cdot fr3 + (Wi-can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$$

$$= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86 + 0^2 \cdot 1$$

= 0.855 cm²

Area Available in Element [A5]:

$$= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr4$$

$$= (590 - 324) \cdot 10 \cdot 1$$

$$= 26.605 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.0628 mm.
Wall Thickness per UG16(b),	tr16b = 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 5.6021 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 5.6021 mm.
Wall Thickness, shell/head, external pressure	trb2 = 3.4053 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm.
Wall Thickness per table UG-45	tb3 = 11.3312 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.3, \max(5.6, 4.5)]$$

$$= 5.6021 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(4.06, 5.6)$$

$$= 5.6021 \text{ mm.}$$

Available Nozzle Neck Thickness = 0.875 * 17.450 = 15.269 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 21.1,	Allowable	: 118.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 273.9 N./mm ²	Passed
Occasional	: 3.2,	Allowable	: 156.9 N./mm ²	Passed
Shear	: 12.3,	Allowable	: 82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 15.3, tr = 0.91, c = 3 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.074, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	4 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 10, c = 3 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.29, Temp. Reduction = 78 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 10, c = 3 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.29, Temp. Reduction = 78 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 10, c = 3 mm., E* = 1

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.29$, Temp. Reduction = 78 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $tg = 12$, $tr = 2.6$, $c = 3$ mm., $E^* = 1$
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.29$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -104 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $6.62/102.00 = 0.065$

Weld Size Calculations, Description: N1-12"

Intermediate Calc. for nozzle/shell Welds $T_{min} = 10.0000$ mm.
 Intermediate Calc. for pad/shell Welds $T_{minPad} = 9.0000$ mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o$ mm.
Pad Weld	$4.5000 = 0.5 \cdot T_{minPad}$	$5.6560 = 0.7 \cdot W_p$ mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (8.6 - 9.45 + 2 \cdot 14.4 \cdot 0.86 \cdot (1 \cdot 9 - 5.75)) \cdot 138)$
 $= \max(0, -61.92) \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (W_i - Can / .707)^2 \cdot f_{r2}) \cdot S_v$
 $= (5.15 + 26.6 + 0.86 - 0 \cdot 0.86) \cdot 138$
 $= 45889.13 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (5.15 + 0 + 0.86 + (2.22)) \cdot 138$
 $= 11581.14 \text{ Kgf}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (5.15 + 0 + 0.86 + 26.6 + (2.22)) \cdot 138$
 $= 49018.67 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 324 \cdot 10 \cdot 0.49 \cdot 118$
 $= 29992. \text{ Kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$



$$= (3.14/2.0) \cdot 750 \cdot 8 \cdot 0.49 \cdot 138$$

$$= 64984. \text{ Kg}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 155) \cdot (17.4 - 3) \cdot 0.7 \cdot 118$$

$$= 59149. \text{ Kg}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.14/2) \cdot 324 \cdot 10 \cdot 0.74 \cdot 138$$

$$= 52971. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 324 \cdot (8 - 0) \cdot 0.74 \cdot 138$$

$$= 42377. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (64984 + 59149) = 124133 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (29992 + 52971 + 42377 + 0) = 125340 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (64984 + 42377 + 0) = 107361 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 124133 Kg, must exceed W = 0 Kg	or W1 = 45889 Kg
Path 2-2 = 125339 Kg, must exceed W = 0 Kg	or W2 = 11581 Kg
Path 3-3 = 107361 Kg, must exceed W = 0 Kg	or W3 = 49018 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6.4 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1.03 bars

The Drop for this Nozzle is : 24.9704 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 366.9704 mm.

Input Echo, WRC297 Item 1, Description: N1-12" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	1075.000	mm.
Shell Thickness	12.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	323.850	mm.
Nozzle Thickness	17.4498	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00 N./mm ²



Nozzle Hot S.I. Allowable SNmh 118.00 N./mm²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter of Reinforcing Pad 750.000 mm.
 Thickness of Reinforcing Pad 10.0000 mm.

Note:
External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSTAINED loads.

Design Internal Pressure	Dp	6.62	bars
Radial Load	P	2294.51	Kgf
Circumferential Shear	Vc	1529.68	Kgf
Longitudinal Shear	Vl	1529.68	Kgf
Circumferential Moment	Mc	1223.27	Kg-m.
Longitudinal Moment	Ml	1223.27	Kg-m.
Torsional Moment	Mt	1834.90	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 323.85 + 2 \cdot 1.65 \cdot \sqrt{(550.0(12.0 - 3.0))}$
 $= 556.026 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1100.000 mm.
Nozzle Outside Diameter	(d) =	323.850 mm.
Vessel Thickness used	(T) =	19.000 mm.
Nozzle Thickness used	(t) =	14.450 mm.
T / t	=	1.315
d / t	=	22.412

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.240$$

Nr/P	=	0.060
Mr/P	=	0.103
M0/P	=	0.041
N0/P	=	0.124
MrD/Mc	=	0.230
NrDL/Mc	=	0.104
M0d/Mc	=	0.102
N0DL/Mc	=	0.124
MrD/Ml	=	0.115
NrDL/Ml	=	0.059
M0D/Ml	=	0.051
N0DL/Ml	=	0.194

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-7	-7	-7	-7
Outplane Bending (P)	-15	15	-15	15
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-19	-19	19	19
Outplane Bending (Ml)	-31	31	31	-31
Normal Pressure Stress	18	19	18	19
Outplane Stress Summary	-55	38	46	15

**Vessel Stresses (N./mm²):**

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	-3	-3	-3	-3
Inplane Bending (P)	-38	38	-38	38
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-6	-6	6	6
Inplane Bending (ML)	-71	71	71	-71
Inplane Pressure Stress	9	9	9	9
Inplane Stress Summary	-109	108	44	-21

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	5	5	5	5
Shear Stress Summary	7	7	4	4

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	110	109	49	37

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	D1
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	-7	-7	-7	-7
Outplane Bending (P)	-15	15	-15	15
Outplane Membrane (Mc)	-12	-12	12	12
Outplane Bending (Mc)	-63	63	63	-63
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	9	9	9	9
Outplane Stress Summary	-89	67	62	-33

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	D1
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	-3	-3	-3	-3
Inplane Bending (P)	-38	38	-38	38
Inplane Membrane (Mc)	-10	-10	10	10
Inplane Bending (Mc)	-141	141	141	-141
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	18	19	18	19
Inplane Stress Summary	-175	184	128	-76

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	D1
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0



Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	5	5	5	5

Shear Stress Summary	4	4	7	7
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Two * Max Shear Stress	175	184	129	78
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Hoop Membrane (P)	-7	-7	-7	-7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-19	-19	19	19
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	7	8	7	8

Hoop Stress Summary	-19	-19	19	20
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-47	47	-47	47
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-11	-11	11	11
Axial Bending (ML)	-91	91	91	-91
Axial Pressure Stress	3	3	3	3

Axial Stress Summary	-147	129	58	-30
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
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Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	7	7	7	7

Shear Stress Summary	9	9	5	5
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
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Two * Max Shear Stress	148	149	58	52
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	-7	-7	-7	-7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-12	-12	12	12
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	7	8	7	8



Hoop Stress Summary	-12	-11	12	13
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-47	47	-47	47
Axial Membrane (Mc)	-11	-11	11	11
Axial Bending (Mc)	-189	189	189	-189
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3

Axial Stress Summary	-245	226	155	-128
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
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Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	7	7	7	7

Shear Stress Summary	5	5	9	9
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
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Two * Max Shear Stress	245	238	156	142
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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 323.85 + 2 \cdot 1.65 \cdot \sqrt{545.0(12.0 - 3.0)}$$

$$= 554.968 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1090.000 mm.

Nozzle Outside Diameter (d) = 750.000 mm.

Vessel Thickness used (T) = 9.000 mm.

Nozzle Thickness used (t) = 227.525 mm.

T / t = 0.040

d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 7.572$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.046

Mr/P = 0.071

M0/P = 0.019

N0/P = 0.021

MrD/Mc = 0.237

NrDL/Mc = 0.092

M0d/Mc = 0.066

N0DL/Mc = 0.039

MrD/Ml = 0.034

NrDL/Ml = 0.035

M0D/Ml = 0.011

N0DL/Ml = 0.016

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
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(Stresses Normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	-5	-5	-5	-5
Outplane Bending (P)	-30	30	-30	30
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-3	-3	3	3
Outplane Bending (ML)	-13	13	13	-13
Normal Pressure Stress	39	40	39	40
Outplane Stress Summary	-13	75	19	55

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	-12	-12	-12	-12
Inplane Bending (P)	-118	118	-118	118
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-6	-6	6	6
Inplane Bending (ML)	-40	40	40	-40
Inplane Pressure Stress	19	19	19	19
Inplane Stress Summary	-159	159	-64	92

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	159	159	84	92

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	-5	-5	-5	-5
Outplane Bending (P)	-30	30	-30	30
Outplane Membrane (Mc)	-7	-7	7	7
Outplane Bending (Mc)	-77	77	77	-77
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	19	19	19	19
Outplane Stress Summary	-102	114	68	-25

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	-12	-12	-12	-12
Inplane Bending (P)	-118	118	-118	118
Inplane Membrane (Mc)	-18	-18	18	18
Inplane Bending (Mc)	-280	280	280	-280



Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	39	40	39	40
Inplane Stress Summary	-391	408	206	-116

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	2	2	2	2
Shear Stress Summary	0	0	3	3

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	391	408	206	116

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	18	19	18	19	18	19	18	19
Circ. Pl (SUS)	-26	-26	12	12	-13	-13	7	7
Circ. Q (SUS)	-46	46	16	-16	-179	179	103	-103
Long. Pm (SUS)	9	9	9	9	9	9	9	9
Long. Pl (SUS)	-9	-9	3	3	-19	-19	5	5
Long. Q (SUS)	-109	109	33	-33	-78	78	48	-48
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	18.0	19.0	18.0	19.0	18.0	19.0	18.0	19.0
Pm+Pl (SUS)	8.2	7.3	30.1	31.1	15.1	16.1	25.1	26.1
Pm+Pl+Q (Total)	109.6	109.5	49.5	36.9	174.2	185.1	128.5	77.8

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	19.00	138.00	Passed
Pm+Pl (SUS)	31.05	207.00	Passed
Pm+Pl+Q (TOTAL)	185.14	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl



Circ. Pm (SUS)	39	40	39	40	39	40	39	40
Circ. Pl (SUS)	-8	-8	-2	-2	-30	-30	6	6
Circ. Q (SUS)	-43	43	-17	17	-398	398	162	-162
Long. Pm (SUS)	19	19	19	19	19	19	19	19
Long. Pl (SUS)	-18	-18	-6	-6	-12	-12	2	2
Long. Q (SUS)	-158	158	-78	78	-107	107	47	-47
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	39.0	40.0	39.0	40.0	39.0	40.0	39.0	40.0
Pm+Pl (SUS)	31.0	32.0	37.0	38.0	9.4	10.3	45.0	46.0
Pm+Pl+Q (Total)	157.1	159.1	85.0	91.0	389.0	408.0	207.1	116.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	40.00	138.00	Passed
Pm+Pl (SUS)	46.04	207.00	Passed
Pm+Pl+Q (TOTAL)	408.00	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	7	8	7	8	7	8	7	8
Circ. Pl (SUS)	-26	-26	12	12	-19	-19	5	5
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. Pl (SUS)	-12	-12	10	10	-12	-12	10	10
Long. Q (SUS)	-138	138	44	-44	-236	236	142	-142
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	7	7	7	7	7	7	7	7
Pm (SUS)	7.0	8.0	7.0	8.0	7.0	8.0	7.0	8.0
Pm+Pl (SUS)	19.4	18.4	19.6	20.5	13.0	12.2	14.6	15.0
Pm+Pl+Q (Total)	147.6	148.1	57.6	52.0	245.1	238.2	155.6	143.1

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	8.00	118.00	Passed
Pm+Pl (SUS)	20.53	177.00	Passed
Pm+Pl+Q (TOTAL)	245.11	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: K1-2"

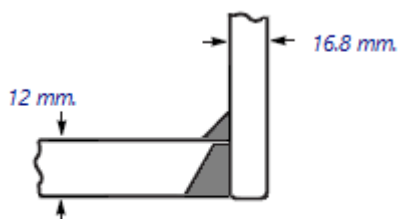
From: 20

Pressure for Reinforcement Calculations	P	6.624	bars
Temperature for Internal Pressure	Temp	200	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	1075.00	mm.
Design Length of Section	L	3454.1667	mm.
Shell Finished (Minimum) Thickness	t	12.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		2300.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	240.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		8.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Overall MAWP.



Reinforcement CALCULATION, Description: K1-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $6.62 \cdot 540 / (138 \cdot 1 - 0.6 \cdot 6.62)$
= 2.6021 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $6.62 \cdot 28.4 / (138 \cdot 1 - 0.6 \cdot 6.62)$
= 0.1367 mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.4247 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000 mm.
Parallel to Vessel Wall, opening length	d	56.8000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000 mm.

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n/S_v)$
= $\min(1, 138/138)$
= 1.000

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n/S_v)$
= $\min(1, 138/138)$
= 1.000

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(1, 1)$
= 1.000

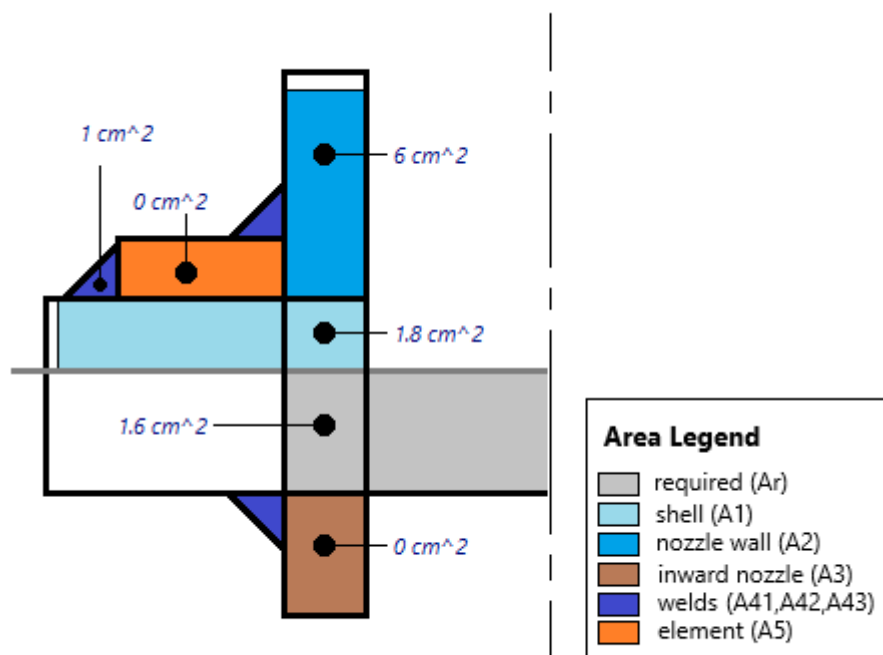


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5	MAWP	External	Mapnc
--------------------------	------	----------	-------



Area Required	Ar	1.478	1.633	NA
Area in Shell	A1	3.634	1.846	NA
Area in Nozzle Wall	A2	6.132	6.003	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	10.766	8.848	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5 (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1))$ per UG-37(d)
 $= 0.5(56.8 \cdot 5.75 \cdot 1 + 2 \cdot 13.8 \cdot 5.75 \cdot 1(1 - 1))$
 $= 1.633 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 56.8 (1 \cdot 9 - 1 \cdot 5.75) - 2 \cdot 13.8$
 $(1 \cdot 9 - 1 \cdot 5.75) \cdot (1 - 1)$
 $= 1.846 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp) (tn - trn) fr2$
 $= (2 \cdot 22.5) (13.8 - 0.42) 1$
 $= 6.003 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 1 + (0)^2 \cdot 1$
 $= 1.000 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 3.4247 \text{ mm.}$
 Wall Thickness per UG16(b), $tr16b = 4.5000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 5.6021 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 5.6021 \text{ mm.}$
 Wall Thickness, shell/head, external pressure $trb2 = 3.4053 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 7.8000 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(5.6, 4.5)]$
 $= 5.6021 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.42, 5.6)$
 $= 5.6021 \text{ mm.}$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:B)

Govrn. thk, $tg = 12$, $tr = 2.6$, $c = 3 \text{ mm.}$, $E* = 1$
 Thickness Ratio $= tr \cdot E* / (tg - c) = 0.29$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A -3 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = 6.62/51.00 = 0.130

Weld Size Calculations, Description: K1-2"

Intermediate Calc. for nozzle/shell Welds Tmin 9.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (1.63 - 1.85 + 2 \cdot 13.8 \cdot 1 \cdot (1 \cdot 9 - 5.75)) \cdot 138)$$

$$= 959.67 \text{ Kgf}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (6 + 0 + 1 - 0 \cdot 1) \cdot 138$$

$$= 9853.85 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (6 + 0 + 1 + (2.48)) \cdot 138$$

$$= 13340.11 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (6 + 0 + 1 + 0 + (2.48)) \cdot 138$$

$$= 13340.11 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 84.3 \cdot 10 \cdot 0.49 \cdot 138$$

$$= 9133. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 35.3) \cdot (16.8 - 3) \cdot 0.7 \cdot 138$$

$$= 15028. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 84.3 \cdot (8 - 0) \cdot 0.74 \cdot 138$$

$$= 11035. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (9133 + 15028) = 24161 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (9133 + 0 + 11035 + 0) = 20168 \text{ Kgf}$$



Failure Path [Path 3-3]:

= (Sonw + Tngw + Sinw)

= (9133 + 11035 + 0) = 20168 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 24160 Kgf, must exceed W = 959 Kgf or W1 = 9853 Kgf

Path 2-2 = 20167 Kgf, must exceed W = 959 Kgf or W2 = 13340 Kgf

Path 3-3 = 20167 Kgf, must exceed W = 959 Kgf or W3 = 13340 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6.4 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1.03 bars

The Drop for this Nozzle is : 1.6563 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 253.6563 mm.

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Input, Nozzle Desc: K2-2"

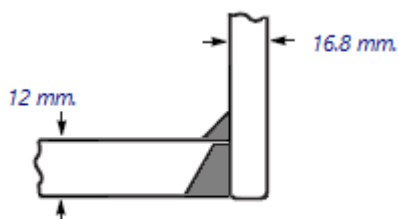
From: 20

Pressure for Reinforcement Calculations	P	6.624	bars
Temperature for Internal Pressure	Temp	200	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	1075.00	mm.
Design Length of Section	L	3454.1667	mm.
Shell Finished (Minimum) Thickness	t	12.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		175.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	240.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		8.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Overall MAWP.



Reinforcement CALCULATION, Description: K2-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $6.62 \cdot 540 / (138 \cdot 1 - 0.6 \cdot 6.62)$
= 2.6021 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $6.62 \cdot 28.4 / (138 \cdot 1 - 0.6 \cdot 6.62)$
= 0.1367 mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.4247 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000 mm.
Parallel to Vessel Wall, opening length	d	56.8000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000 mm.

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n/S_v)$
= $\min(1, 138/138)$
= 1.000

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n/S_v)$
= $\min(1, 138/138)$
= 1.000

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(1, 1)$
= 1.000

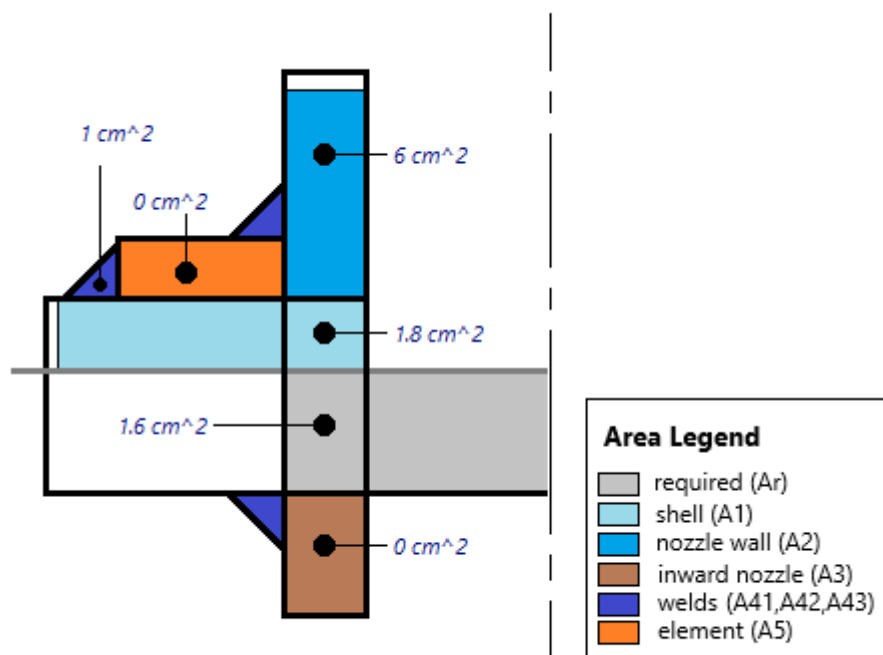


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5	MAWP	External	Mapnc
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Area Required	Ar	1.478	1.633	NA
Area in Shell	A1	3.634	1.846	NA
Area in Nozzle Wall	A2	6.132	6.003	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	10.766	8.848	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1)) \text{ per UG-37(d)}$$

$$= 0.5(56.8 \cdot 5.75 \cdot 1 + 2 \cdot 13.8 \cdot 5.75 \cdot 1(1 - 1))$$

$$= 1.633 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 56.8(1 \cdot 9 - 1 \cdot 5.75) - 2 \cdot 13.8$$

$$(1 \cdot 9 - 1 \cdot 5.75) \cdot (1 - 1)$$

$$= 1.846 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 22.5)(13.8 - 0.42)1$$

$$= 6.003 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

$$\text{Wall Thickness for Internal/External pressures } ta = 3.4247 \text{ mm.}$$

$$\text{Wall Thickness per UG16(b), } tr16b = 4.5000 \text{ mm.}$$

$$\text{Wall Thickness, shell/head, internal pressure } trb1 = 5.6021 \text{ mm.}$$

$$\text{Wall Thickness } tb1 = \max(trb1, tr16b) = 5.6021 \text{ mm.}$$

$$\text{Wall Thickness, shell/head, external pressure } trb2 = 3.4053 \text{ mm.}$$

$$\text{Wall Thickness } tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm.}$$

$$\text{Wall Thickness per table UG-45 } tb3 = 7.8000 \text{ mm.}$$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[7.8, \max(5.6, 4.5)]$$

$$= 5.6021 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.42, 5.6)$$

$$= 5.6021 \text{ mm.}$$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:B)

Govrn. thk, $tg = 12$, $tr = 2.6$, $c = 3 \text{ mm.}$, $E* = 1$

Thickness Ratio = $tr \cdot E*/(tg - c) = 0.29$, Temp. Reduction = $78 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A $-3 \text{ }^\circ\text{C}$

Min Metal Temp. at Required thickness (UCS 66.1) $-104 \text{ }^\circ\text{C}$



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $6.62/51.00 = 0.130$

Weld Size Calculations, Description: K2-2"

Intermediate Calc. for nozzle/shell Welds Tmin 9.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (1.63 - 1.85 + 2 \cdot 13.8 \cdot 1 \cdot (1 \cdot 9 - 5.75)) \cdot 138)$$

$$= 959.67 \text{ Kg}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(W_i - \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (6 + 0 + 1 - 0 \cdot 1) \cdot 138$$

$$= 9853.85 \text{ Kg}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (6 + 0 + 1 + (2.48)) \cdot 138$$

$$= 13340.11 \text{ Kg}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (6 + 0 + 1 + 0 + (2.48)) \cdot 138$$

$$= 13340.11 \text{ Kg}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 84.3 \cdot 10 \cdot 0.49 \cdot 138$$

$$= 9133. \text{ Kg}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 35.3) \cdot (16.8 - 3) \cdot 0.7 \cdot 138$$

$$= 15028. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 84.3 \cdot (8 - 0) \cdot 0.74 \cdot 138$$

$$= 11035. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (9133 + 15028) = 24161 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (9133 + 0 + 11035 + 0) = 20168 \text{ Kg}$$



Failure Path [Path 3-3]:

= (Sonw + Tngw + Sinw)

= (9133 + 11035 + 0) = 20168 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 24160 Kgf, must exceed W = 959 Kgf or W1 = 9853 Kgf

Path 2-2 = 20167 Kgf, must exceed W = 959 Kgf or W2 = 13340 Kgf

Path 3-3 = 20167 Kgf, must exceed W = 959 Kgf or W3 = 13340 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6.4 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1.03 bars

The Drop for this Nozzle is : 1.6563 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 253.6563 mm.

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Input, Nozzle Desc: K4-2"

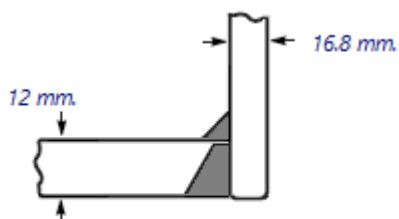
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Pressure for Reinforcement Calculations	P	6.624	bars
Temperature for Internal Pressure	Temp	200	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	1075.00	mm.
Design Length of Section	L	3454.1667	mm.
Shell Finished (Minimum) Thickness	t	12.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1850.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	240.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		8.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Overall MAWP.



Reinforcement CALCULATION, Description: K4-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $6.62 \cdot 540 / (138 \cdot 1 - 0.6 \cdot 6.62)$
= 2.6021 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $6.62 \cdot 28.4 / (138 \cdot 1 - 0.6 \cdot 6.62)$
= 0.1367 mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.4247 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000 mm.
Parallel to Vessel Wall, opening length	d	56.8000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000 mm.

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n/S_v)$
= $\min(1, 138/138)$
= 1.000

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n/S_v)$
= $\min(1, 138/138)$
= 1.000

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(1, 1)$
= 1.000

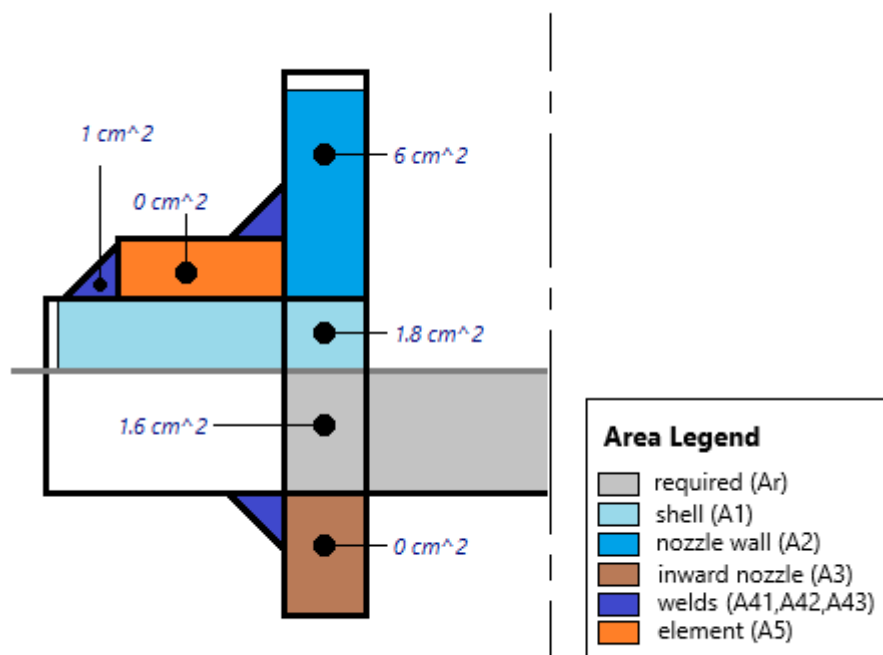


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5	MAWP	External	Mapnc
--------------------------	------	----------	-------



Area Required	Ar	1.478	1.633	NA
Area in Shell	A1	3.634	1.846	NA
Area in Nozzle Wall	A2	6.132	6.003	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	10.766	8.848	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1)) \text{ per UG-37(d)}$$

$$= 0.5(56.8 \cdot 5.75 \cdot 1 + 2 \cdot 13.8 \cdot 5.75 \cdot 1(1 - 1))$$

$$= 1.633 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 56.8(1 \cdot 9 - 1 \cdot 5.75) - 2 \cdot 13.8$$

$$(1 \cdot 9 - 1 \cdot 5.75) \cdot (1 - 1)$$

$$= 1.846 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 22.5)(13.8 - 0.42)1$$

$$= 6.003 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

$$\text{Wall Thickness for Internal/External pressures } ta = 3.4247 \text{ mm.}$$

$$\text{Wall Thickness per UG16(b), } tr16b = 4.5000 \text{ mm.}$$

$$\text{Wall Thickness, shell/head, internal pressure } trb1 = 5.6021 \text{ mm.}$$

$$\text{Wall Thickness } tb1 = \max(trb1, tr16b) = 5.6021 \text{ mm.}$$

$$\text{Wall Thickness, shell/head, external pressure } trb2 = 3.4053 \text{ mm.}$$

$$\text{Wall Thickness } tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm.}$$

$$\text{Wall Thickness per table UG-45 } tb3 = 7.8000 \text{ mm.}$$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[7.8, \max(5.6, 4.5)]$$

$$= 5.6021 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.42, 5.6)$$

$$= 5.6021 \text{ mm.}$$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:B)

Govrn. thk, $tg = 12$, $tr = 2.6$, $c = 3 \text{ mm.}$, $E* = 1$

Thickness Ratio = $tr \cdot E*/(tg - c) = 0.29$, Temp. Reduction = $78 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A $-3 \text{ }^\circ\text{C}$

Min Metal Temp. at Required thickness (UCS 66.1) $-104 \text{ }^\circ\text{C}$



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 6.62/51.00 = 0.130

Weld Size Calculations, Description: K4-2"

Intermediate Calc. for nozzle/shell Welds Tmin 9.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (1.63 - 1.85 + 2 \cdot 13.8 \cdot 1 \cdot (1 \cdot 9 - 5.75)) \cdot 138)$
 $= 959.67 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i - \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (6 + 0 + 1 - 0 \cdot 1) \cdot 138$
 $= 9853.85 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (6 + 0 + 1 + (2.48)) \cdot 138$
 $= 13340.11 \text{ Kg}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (6 + 0 + 1 + 0 + (2.48)) \cdot 138$
 $= 13340.11 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 84.3 \cdot 10 \cdot 0.49 \cdot 138$
 $= 9133. \text{ Kg}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 35.3) \cdot (16.8 - 3) \cdot 0.7 \cdot 138$
 $= 15028. \text{ Kg}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 84.3 \cdot (8 - 0) \cdot 0.74 \cdot 138$
 $= 11035. \text{ Kg}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (9133 + 15028) = 24161 \text{ Kg}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (9133 + 0 + 11035 + 0) = 20168 \text{ Kg}$



Failure Path [Path 3-3]:

= (Sonw + Tngw + Sinw)

= (9133 + 11035 + 0) = 20168 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 24160 Kgf, must exceed W = 959 Kgf or W1 = 9853 Kgf

Path 2-2 = 20167 Kgf, must exceed W = 959 Kgf or W2 = 13340 Kgf

Path 3-3 = 20167 Kgf, must exceed W = 959 Kgf or W3 = 13340 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6.4 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1.03 bars

The Drop for this Nozzle is : 1.6563 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 253.6563 mm.

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Input, Nozzle Desc: K5-2"

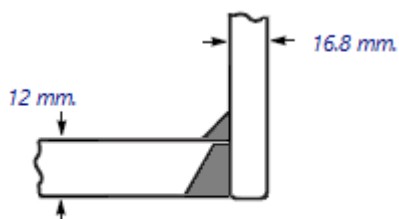
From: 20

Pressure for Reinforcement Calculations	P	6.624	bars
Temperature for Internal Pressure	Temp	200	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	1075.00	mm.
Design Length of Section	L	3454.1667	mm.
Shell Finished (Minimum) Thickness	t	12.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		150.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	240.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		8.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Overall MAWP.



Reinforcement CALCULATION, Description: K5-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $6.62 \cdot 540 / (138 \cdot 1 - 0.6 \cdot 6.62)$
= 2.6021 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $6.62 \cdot 28.4 / (138 \cdot 1 - 0.6 \cdot 6.62)$
= 0.1367 mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.4247 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000 mm.
Parallel to Vessel Wall, opening length	d	56.8000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000 mm.

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n/S_v)$
= $\min(1, 138/138)$
= 1.000

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n/S_v)$
= $\min(1, 138/138)$
= 1.000

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(1, 1)$
= 1.000

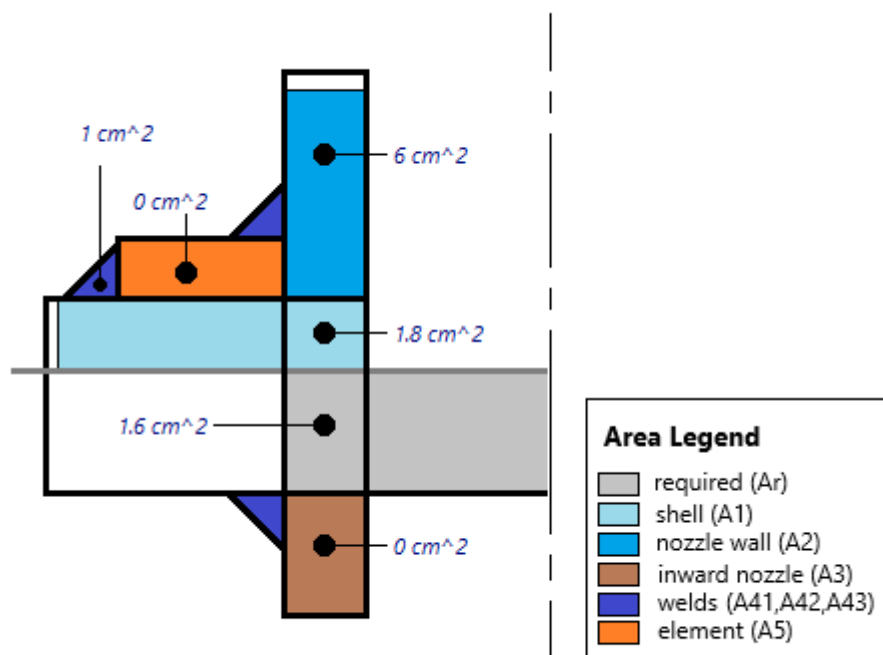


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5	MAWP	External	Mapnc
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Area Required	Ar	1.478	1.633	NA
Area in Shell	A1	3.634	1.846	NA
Area in Nozzle Wall	A2	6.132	6.003	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	10.766	8.848	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1)) \text{ per UG-37(d)}$$

$$= 0.5(56.8 \cdot 5.75 \cdot 1 + 2 \cdot 13.8 \cdot 5.75 \cdot 1(1 - 1))$$

$$= 1.633 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 56.8(1 \cdot 9 - 1 \cdot 5.75) - 2 \cdot 13.8$$

$$(1 \cdot 9 - 1 \cdot 5.75) \cdot (1 - 1)$$

$$= 1.846 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 22.5)(13.8 - 0.42)1$$

$$= 6.003 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

$$\text{Wall Thickness for Internal/External pressures } ta = 3.4247 \text{ mm.}$$

$$\text{Wall Thickness per UG16(b), } tr16b = 4.5000 \text{ mm.}$$

$$\text{Wall Thickness, shell/head, internal pressure } trb1 = 5.6021 \text{ mm.}$$

$$\text{Wall Thickness } tb1 = \max(trb1, tr16b) = 5.6021 \text{ mm.}$$

$$\text{Wall Thickness, shell/head, external pressure } trb2 = 3.4053 \text{ mm.}$$

$$\text{Wall Thickness } tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm.}$$

$$\text{Wall Thickness per table UG-45 } tb3 = 7.8000 \text{ mm.}$$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[7.8, \max(5.6, 4.5)]$$

$$= 5.6021 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.42, 5.6)$$

$$= 5.6021 \text{ mm.}$$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:B)

Govrn. thk, $tg = 12$, $tr = 2.6$, $c = 3 \text{ mm.}$, $E* = 1$

Thickness Ratio = $tr \cdot E*/(tg - c) = 0.29$, Temp. Reduction = $78 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A $-3 \text{ }^\circ\text{C}$

Min Metal Temp. at Required thickness (UCS 66.1) $-104 \text{ }^\circ\text{C}$

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 6.62/51.00 = 0.130

Weld Size Calculations, Description: K5-2"

Intermediate Calc. for nozzle/shell Welds Tmin 9.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (1.63 - 1.85 + 2 \cdot 13.8 \cdot 1 \cdot (1 \cdot 9 - 5.75)) \cdot 138)$
 $= 959.67 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i - \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (6 + 0 + 1 - 0 \cdot 1) \cdot 138$
 $= 9853.85 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (6 + 0 + 1 + (2.48)) \cdot 138$
 $= 13340.11 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (6 + 0 + 1 + 0 + (2.48)) \cdot 138$
 $= 13340.11 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 84.3 \cdot 10 \cdot 0.49 \cdot 138$
 $= 9133. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 35.3) \cdot (16.8 - 3) \cdot 0.7 \cdot 138$
 $= 15028. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 84.3 \cdot (8 - 0) \cdot 0.74 \cdot 138$
 $= 11035. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (9133 + 15028) = 24161 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (9133 + 0 + 11035 + 0) = 20168 \text{ Kgf}$



Failure Path [Path 3-3]:

= (Sonw + Tngw + Sinw)

= (9133 + 11035 + 0) = 20168 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 24160 Kgf, must exceed W = 959 Kgf or W1 = 9853 Kgf

Path 2-2 = 20167 Kgf, must exceed W = 959 Kgf or W2 = 13340 Kgf

Path 3-3 = 20167 Kgf, must exceed W = 959 Kgf or W3 = 13340 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6.4 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1.03 bars

The Drop for this Nozzle is : 1.6563 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 253.6563 mm.

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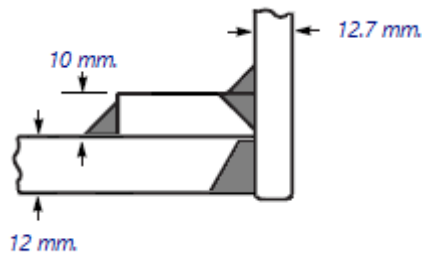
Input, Nozzle Desc: K3-8"
From: 30

Pressure for Reinforcement Calculations	P	6.624	bars
Temperature for Internal Pressure	Temp	200	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	1075.00	mm.
Design Length of Section	L	3454.1667	mm.
Shell Finished (Minimum) Thickness	t	12.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		2850.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm2
Allowable Stress At Ambient	Sna	118.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		8.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		XS	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	330.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	8.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	400.0000	mm.
Thickness of Pad	te	10.0000	mm.
Weld leg size between Pad and Shell	Wp	8.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	10.0000	mm.
Reinforcing Pad Width		90.4625	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Overall MAWP.

**Reinforcement CALCULATION, Description: K3-8"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	8.625 in.
Actual Thickness Used in Calculation	0.500 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.62 \cdot 540 / (138 \cdot 1 - 0.6 \cdot 6.62)$
 $= 2.6021 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 6.62 \cdot 110 / (118 \cdot 1 + 0.4 \cdot 6.62)$
 $= 0.6136 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.8443 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	399.3500 mm.
Parallel to Vessel Wall, opening length	d	199.6750 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		22.5000 mm.

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$

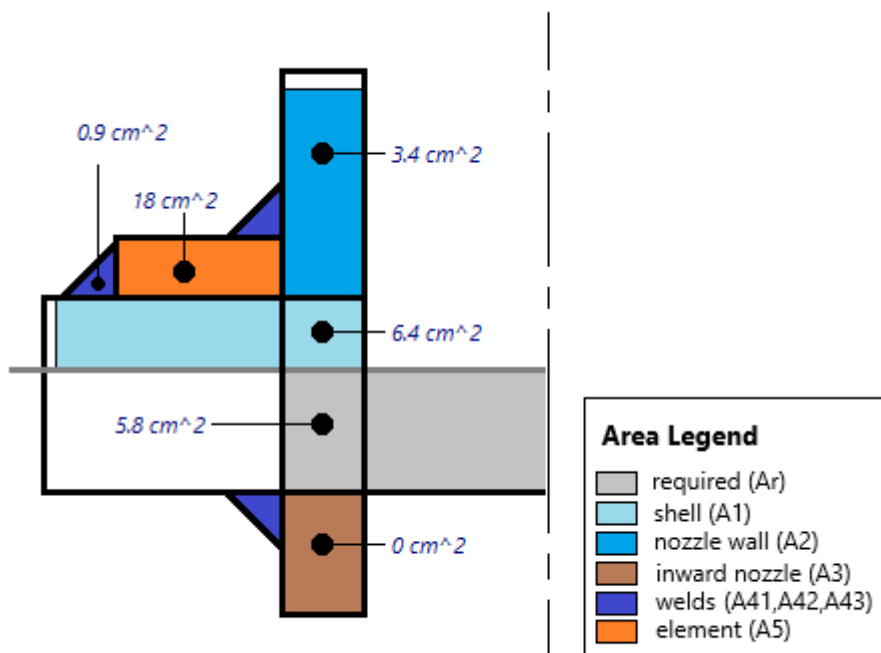


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	5.269	5.822	NA
Area in Shell	A1	12.595	6.397	NA
Area in Nozzle Wall	A2	3.496	3.408	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.855	0.855	NA
Area in Element	A5	18.027	18.027	NA
TOTAL AREA AVAILABLE	Atot	34.974	28.687	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1))$ per UG-37(d)
 $= 0.5(200 \cdot 5.75 \cdot 1 + 2 \cdot 9.7 \cdot 5.75 \cdot 1(1 - 0.86))$
 $= 5.822 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot tr - F \cdot tr) - 2 \cdot tn(E1 \cdot tr - F \cdot tr) \cdot (1 - fr1)$
 $= 200(1 \cdot 9 - 1 \cdot 5.75) - 2 \cdot 9.7$
 $(1 \cdot 9 - 1 \cdot 5.75) \cdot (1 - 0.86)$
 $= 6.397 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 22.5) \cdot (9.7 - 0.84) \cdot 0.86$
 $= 3.408 \text{ cm}^2$

Area Available in Welds [A41 + A42 + A43]:
 $= Wo^2 \cdot fr3 + (Wi\text{-}can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$
 $= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86 + 0^2 \cdot 1$

= 0.855 cm²

Area Available in Element [A5]:

= (min(Dp,DL)-(Nozzle OD))*(min(tp,Tlwp,te)) • fr4
 = (399 - 219) • 10 • 1
 = 18.027 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 3.8443 mm.
 Wall Thickness per UG16(b), tr16b = 4.5000 mm.
 Wall Thickness, shell/head, internal pressure trb1 = 5.6021 mm.
 Wall Thickness tb1 = max(trb1, tr16b) = 5.6021 mm.
 Wall Thickness, shell/head, external pressure trb2 = 3.4053 mm.
 Wall Thickness tb2 = max(trb2, tr16b) = 4.5000 mm.
 Wall Thickness per table UG-45 tb3 = 10.1600 mm.

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]
 = min[10.2, max(5.6, 4.5)]
 = 5.6021 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
 = max(3.84, 5.6)
 = 5.6021 mm.

Available Nozzle Neck Thickness = 0.875 * 12.700 = 11.113 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained :	30.4,	Allowable :	118.0 N./mm ²	Passed
Expansion :	0.0,	Allowable :	264.6 N./mm ²	Passed
Occasional :	3.3,	Allowable :	156.9 N./mm ²	Passed
Shear :	17.8,	Allowable :	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 11.1, tr = 0.61, c = 3 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.076, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A -5 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 10, c = 3 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.29, Temp. Reduction = 78 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 10, c = 3 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.29, Temp. Reduction = 78 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 10, c = 3 mm., E* = 1



Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.29$, Temp. Reduction = 78 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $tg = 11.1$, $tr = 0.61$, $c = 3$ mm., $E^* = 1$
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.076$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -26 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
Gov. MDMT of the Reinforcement Pad : -104 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $6.62/51.00 = 0.130$

Weld Size Calculations, Description: K3-8"

Intermediate Calc. for nozzle/shell Welds T_{min} 9.7000 mm.
Intermediate Calc. for pad/shell Welds T_{minPad} 9.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o$ mm.
Pad Weld	$4.5000 = 0.5 \cdot T_{minPad}$	$5.6560 = 0.7 \cdot W_p$ mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (5.82 - 6.4 + 2 \cdot 9.7 \cdot 0.86 \cdot (1 \cdot 9 - 5.75)) \cdot 138)$
 $= \max(0, -50.40) \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (Wi - Can / .707)^2 \cdot fr2) \cdot Sv$
 $= (3.41 + 18 + 0.86 - 0 \cdot 0.86) \cdot 138$
 $= 31365.64 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$
 $= (3.41 + 0 + 0.86 + (1.49)) \cdot 138$
 $= 8098.94 \text{ Kgf}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$
 $= (3.41 + 0 + 0.86 + 18 + (1.49)) \cdot 138$
 $= 33466.46 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 219 \cdot 10 \cdot 0.49 \cdot 118$
 $= 20289. \text{ Kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$



$$= (3.14/2.0) \cdot 400 \cdot 8 \cdot 0.49 \cdot 138$$

$$= 34658. \text{ Kg}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 105) \cdot (12.7 - 3) \cdot 0.7 \cdot 118$$

$$= 26870. \text{ Kg}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.14/2) \cdot 219 \cdot 10 \cdot 0.74 \cdot 138$$

$$= 35833. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 219 \cdot (8 - 0) \cdot 0.74 \cdot 138$$

$$= 28667. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (34658 + 26870) = 61528 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (20289 + 35833 + 28667 + 0) = 84789 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (34658 + 28667 + 0) = 63325 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 61527	Kgf, must exceed W = 0 Kg	or W1 = 31365 Kg
Path 2-2 = 84788	Kgf, must exceed W = 0 Kg	or W2 = 8098 Kg
Path 3-3 = 63325	Kgf, must exceed W = 0 Kg	or W3 = 33466 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6.4 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1.03 bars

The Drop for this Nozzle is : 11.2797 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 353.2797 mm.

This nozzle or its diameter limit intersects weld seam # [4] on this element. The distance between the nozzle and seam is, 71.13 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 199.67 mm.

Input Echo, WRC297 Item 1, Description: K3-8" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	1075.000	mm.
Shell Thickness	12.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	219.075	mm.

Nozzle Thickness		12.7000	mm.
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00	N./mm2
Nozzle Hot S.I. Allowable	SNmh	118.00	N./mm2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter of Reinforcing Pad	400.000	mm.
Thickness of Reinforcing Pad	10.0000	mm.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUStained loads.*

Design Internal Pressure	Dp	6.62	bars
Radial Load	P	1529.68	Kgf
Circumferential Shear	Vc	1019.78	Kgf
Longitudinal Shear	Vl	1019.78	Kgf
Circumferential Moment	Mc	570.86	Kg-m.
Longitudinal Moment	Ml	570.86	Kg-m.
Torsional Moment	Mt	815.51	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 219.075 + 2 \cdot 1.65 \cdot \sqrt{(550.0(12.0 - 3.0))}$
 $= 451.251 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1100.000	mm.
Nozzle Outside Diameter	(d) =	219.075	mm.
Vessel Thickness used	(T) =	19.000	mm.
Nozzle Thickness used	(t) =	9.700	mm.
T / t	=	1.959	
d / t	=	22.585	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.515$$

Nr/P	=	0.082
Mr/P	=	0.098
M0/P	=	0.056
N0/P	=	0.151
MrD/Mc	=	0.205
NrDL/Mc	=	0.131
M0d/Mc	=	0.138
N0DL/Mc	=	0.143
MrD/Ml	=	0.140
NrDL/Ml	=	0.090
M0D/Ml	=	0.088
N0DL/Ml	=	0.259

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	-6	-6	-6	-6
Outplane Bending (P)	-13	13	-13	13
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-18	-18	18	18



Outplane Bending (ML)	-37	37	37	-37
Normal Pressure Stress	18	19	18	19

Outplane Stress Summary	-57	46	54	7
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	-3	-3	-3	-3
Inplane Bending (P)	-24	24	-24	24
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-6	-6	6	6
Inplane Bending (ML)	-59	59	59	-59
Inplane Pressure Stress	9	9	9	9

Inplane Stress Summary	-84	83	47	-22
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	5	5	5	5

Shear Stress Summary	7	7	4	4
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	86	84	56	31
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Membrane (P)	-6	-6	-6	-6
Outplane Bending (P)	-13	13	-13	13
Outplane Membrane (Mc)	-10	-10	10	10
Outplane Bending (Mc)	-58	58	58	-58
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	9	9	9	9

Outplane Stress Summary	-79	65	57	-31
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	-3	-3	-3	-3
Inplane Bending (P)	-24	24	-24	24
Inplane Membrane (Mc)	-9	-9	9	9
Inplane Bending (Mc)	-86	86	86	-86
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	18	19	18	19

Inplane Stress Summary	-105	117	86	-37
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	5	5	5	5
Shear Stress Summary	4	4	7	7

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	106	118	88	42

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	-6	-6	-6	-6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-18	-18	18	18
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	7	8	7	8
Hoop Stress Summary	-16	-16	19	20

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	-2	-2	-2	-2
Axial Bending (P)	-54	54	-54	54
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-17	-17	17	17
Axial Bending (ML)	-155	155	155	-155
Axial Pressure Stress	3	3	3	3
Axial Stress Summary	-225	193	119	-81

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	10	10	10	10
Shear Stress Summary	13	13	7	7

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	226	211	120	103

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	-6	-6	-6	-6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-10	-10	10	10



Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	7	8	7	8

Hoop Stress Summary	-8	-7	11	12
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	-2	-2	-2	-2
Axial Bending (P)	-54	54	-54	54
Axial Membrane (Mc)	-17	-17	17	17
Axial Bending (Mc)	-226	226	226	-226
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3

Axial Stress Summary	-297	265	191	-153
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	10	10	10	10

Shear Stress Summary	7	7	13	13
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	297	273	192	167
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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 219.075 + 2 \cdot 1.65 \cdot \sqrt{545.0(12.0 - 3.0)}$$

$$= 450.193 \text{ mm.}$$

WRC 297 Curve Access Parameters:

$$\text{Vessel Mean Diameter (D)} = 1090.000 \text{ mm.}$$

$$\text{Nozzle Outside Diameter (d)} = 400.000 \text{ mm.}$$

$$\text{Vessel Thickness used (T)} = 9.000 \text{ mm.}$$

$$\text{Nozzle Thickness used (t)} = 100.162 \text{ mm.}$$

$$T / t = 0.090$$

$$d / t = 10.000$$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.039$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

$$N_r/P = 0.052$$

$$M_r/P = 0.083$$

$$M_0/P = 0.024$$

$$N_0/P = 0.033$$

$$M_rD/Mc = 0.252$$

$$N_rDL/Mc = 0.105$$

$$M_0d/Mc = 0.075$$

$$N_0DL/Mc = 0.058$$

$$M_rD/Ml = 0.060$$

$$N_rDL/Ml = 0.061$$

$$M_0D/Ml = 0.018$$



NØDL/M1 = 0.036

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses Normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	-6	-6	-6	-6
Outplane Bending (P)	-26	26	-26	26
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-6	-6	6	6
Outplane Bending (ML)	-18	18	18	-18
Normal Pressure Stress	39	40	39	40
Outplane Stress Summary	-17	72	32	47

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	-9	-9	-9	-9
Inplane Bending (P)	-92	92	-92	92
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-10	-10	10	10
Inplane Bending (ML)	-62	62	62	-62
Inplane Pressure Stress	19	19	19	19
Inplane Stress Summary	-155	154	-9	50

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	5	5	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	155	154	41	51

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	-6	-6	-6	-6
Outplane Bending (P)	-26	26	-26	26
Outplane Membrane (Mc)	-10	-10	10	10
Outplane Bending (Mc)	-78	78	78	-78
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	19	19	19	19
Outplane Stress Summary	-100	107	75	-28

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside



Inplane Membrane (P)	-9	-9	-9	-9
Inplane Bending (P)	-92	92	-92	92
Inplane Membrane (Mc)	-18	-18	18	18
Inplane Bending (Mc)	-261	261	261	-261
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	39	40	39	40

Inplane Stress Summary	-341	366	216	-120
------------------------	------	-----	-----	------

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		-1	-1	1	1
Torsional Shear (Mt)		3	3	3	3

Shear Stress Summary	1	1	5	5
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stress Intensities)		Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	341	366	217	120
------------------------	-----	-----	-----	-----

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	18	19	18	19	18	19	18	19
Circ. Pl (SUS)	-24	-24	12	12	-12	-12	6	6
Circ. Q (SUS)	-50	50	24	-24	-110	110	62	-62
Long. Pm (SUS)	9	9	9	9	9	9	9	9
Long. Pl (SUS)	-9	-9	3	3	-16	-16	4	4
Long. Q (SUS)	-83	83	35	-35	-71	71	45	-45
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	18.0	19.0	18.0	19.0	18.0	19.0	18.0	19.0
Pm+Pl (SUS)	6.3	5.4	30.1	31.1	13.2	14.1	24.1	25.1
Pm+Pl+Q (Total)	84.3	83.9	55.8	31.0	104.6	117.3	87.2	41.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	19.00	138.00	Passed
Pm+Pl (SUS)	31.05	207.00	Passed
Pm+Pl+Q (TOTAL)	117.30	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):**



Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	39	40	39	40	39	40	39	40
Circ. Pl (SUS)	-12	-12	0	0	-27	-27	9	9
Circ. Q (SUS)	-44	44	-8	8	-353	353	169	-169
Long. Pm (SUS)	19	19	19	19	19	19	19	19
Long. Pl (SUS)	-19	-19	1	1	-16	-16	4	4
Long. Q (SUS)	-154	154	-30	30	-104	104	52	-52
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	39.0	40.0	39.0	40.0	39.0	40.0	39.0	40.0
Pm+Pl (SUS)	27.1	28.1	39.1	40.0	12.1	13.1	48.0	49.0
Pm+Pl+Q (Total)	154.1	154.2	41.2	51.2	341.0	366.0	217.1	120.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	40.00	138.00	Passed
Pm+Pl (SUS)	49.04	207.00	Passed
Pm+Pl+Q (TOTAL)	366.02	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	7	8	7	8	7	8	7	8
Circ. Pl (SUS)	-24	-24	12	12	-16	-16	4	4
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. Pl (SUS)	-19	-19	15	15	-19	-19	15	15
Long. Q (SUS)	-209	209	101	-101	-280	280	172	-172
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	10	10	10	10	10	10	10	10
Pm (SUS)	7.0	8.0	7.0	8.0	7.0	8.0	7.0	8.0
Pm+Pl (SUS)	18.6	18.0	20.6	21.2	16.5	16.5	18.5	18.6
Pm+Pl+Q (Total)	225.7	210.4	119.6	104.2	296.2	272.5	190.8	167.7

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	8.00	118.00	Passed
Pm+Pl (SUS)	21.24	177.00	Passed
Pm+Pl+Q (TOTAL)	296.22	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

PV Elite 28 U1 Licensee: SL User

FileName : HP Steam Blowdown Drum (186-D)_Rev0- 7%

Nozzle Calcs.: K3-8"

Page 139 of 163

Noz1: 27 2:19pm Sep 16,2026



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Input, Nozzle Desc: N2-8"

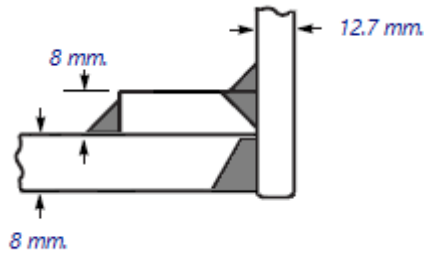
From: 40

Pressure for Reinforcement Calculations	P	6.624	bars
Temperature for Internal Pressure	Temp	200	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	1075.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	8.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm2
Allowable Stress At Ambient	Sna	118.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		8.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		XS	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	330.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	8.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	350.0000	mm.
Thickness of Pad	te	8.0000	mm.
Weld leg size between Pad and Shell	Wp	6.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	8.0000	mm.
Reinforcing Pad Width		65.4625	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Overall MAWP.

**Reinforcement CALCULATION, Description: N2-8"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	8.625 in.
Actual Thickness Used in Calculation	0.500 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 6.62 \cdot 1081 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 6.62)$
 $= 2.3240$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 6.62 \cdot 110 / (118 \cdot 1 + 0.4 \cdot 6.62)$
 $= 0.6136$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.8443 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	399.3500 mm.
Parallel to Vessel Wall, opening length	d	199.6750 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		12.5000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118/138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118/138)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, Sp/Sv)$
 $= \min(1, 138/138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$

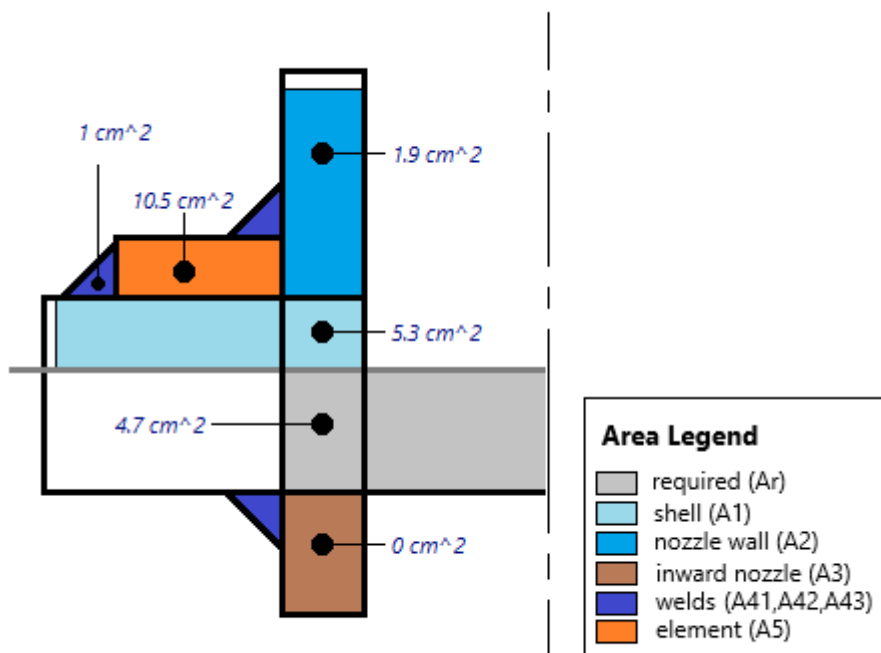


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	4.706	2.906	NA
Area in Shell	A1	5.268	4.192	NA
Area in Nozzle Wall	A2	1.942	1.893	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.956	0.956	NA
Area in Element	A5	10.474	10.474	NA
TOTAL AREA AVAILABLE	Atot	18.641	17.515	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (200 \cdot 2.32 \cdot 1 + 2 \cdot 9.7 \cdot 2.32 \cdot 1 \cdot (1 - 0.86)) \\
 &= 4.706 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 200 (1 \cdot 5 - 1 \cdot 2.32) - 2 \cdot 9.7 \\
 &\quad (1 \cdot 5 - 1 \cdot 2.32) \cdot (1 - 0.86) \\
 &= 5.268 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\
 &= (2 \cdot 12.5) \cdot (9.7 - 0.61) \cdot 0.86 \\
 &= 1.942 \text{ cm}^2
 \end{aligned}$$

Area Available in Welds [A41 + A42 + A43]:

$$\begin{aligned}
 &= (Wo^2 - Ar \text{ Lost}) \cdot Fr3 + ((Wi - can / 0.707)^2 - Ar \text{ Lost}) \cdot fr2 + Wp^2 \cdot fr4 \\
 &= (0.7) \cdot 0.86 + (0) \cdot 0.86 + 6^2 \cdot 1
 \end{aligned}$$

= 0.956 cm²

Area Available in Element [A5]:

= (min(Dp,DL)-(Nozzle OD))*(min(tp,Tlwp,te)) • fr4
 = (350 - 219) • 8 • 1
 = 10.474 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 3.8443 mm.
 Wall Thickness per UG16(b), tr16b = 4.5000 mm.
 Wall Thickness, shell/head, internal pressure trb1 = 5.5768 mm.
 Wall Thickness tb1 = max(trb1, tr16b) = 5.5768 mm.
 Wall Thickness, shell/head, external pressure trb2 = 3.4021 mm.
 Wall Thickness tb2 = max(trb2, tr16b) = 4.5000 mm.
 Wall Thickness per table UG-45 tb3 = 10.1600 mm.

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]
 = min[10.2, max(5.58, 4.5)]
 = 5.5768 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
 = max(3.84, 5.58)
 = 5.5768 mm.

Available Nozzle Neck Thickness = 0.875 * 12.700 = 11.113 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained :	30.4,	Allowable :	118.0 N./mm ²	Passed
Expansion :	0.0,	Allowable :	264.6 N./mm ²	Passed
Occasional :	3.3,	Allowable :	156.9 N./mm ²	Passed
Shear :	17.8,	Allowable :	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 11.1, tr = 0.61, c = 3 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.076, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A -5 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 8, c = 3 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.46, Temp. Reduction = 37 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 8, c = 3 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.46, Temp. Reduction = 37 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 8, tr = 2.32, c = 3 mm., E* = 1



Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.46$, Temp. Reduction = 37 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $tg = 8$, $tr = 2.32$, $c = 3$ mm., $E^* = 1$

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.46$, Temp. Reduction = 37 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Gov. MDMT of the Nozzle : -48 °C
Gov. MDMT of the Reinforcement Pad : -48 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $6.62/51.00 = 0.130$

Weld Size Calculations, Description: N2-8"

Intermediate Calc. for nozzle/shell Welds T_{min} 8.0000 mm.
Intermediate Calc. for pad/shell Welds T_{minPad} 7.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.6000 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm.
Pad Weld	$3.5000 = 0.5 \cdot T_{minPad}$	$4.2420 = 0.7 \cdot W_p$ mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (4.71 - 5.27 + 2 \cdot 9.7 \cdot 0.86 \cdot (1 \cdot 5 - 2.32)) \cdot 138)$
 $= \max(0, -166.80) \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$= (A2 + A5 + A4 - (W_i - Can / .707)^2 \cdot f_{r2}) \cdot S_v$
 $= (1.94 + 10.5 + 0.96 - 0 \cdot 0.86) \cdot 138$
 $= 18817.63 \text{ Kgf}$

Weld Load [W2]:

$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (1.94 + 0 + 0.86 + (0.83)) \cdot 138$
 $= 5103.59 \text{ Kgf}$

Weld Load [W3]:

$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (1.94 + 0 + 0.96 + 10.5 + (0.83)) \cdot 138$
 $= 19984.76 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$= (\pi/2) \cdot D_o \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 219 \cdot 10 \cdot 0.49 \cdot 118$
 $= 20289. \text{ Kgf}$

Shear, Pad Element Weld [Spew]:

$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.14/2.0) \cdot 350 \cdot 6 \cdot 0.49 \cdot 138$



= 22745. Kgf

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 105) \cdot (12.7 - 3) \cdot 0.7 \cdot 118$$

$$= 26870. Kgf$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.14/2) \cdot 219 \cdot 8 \cdot 0.74 \cdot 138$$

$$= 28667. Kgf$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 219 \cdot (8 - 3) \cdot 0.74 \cdot 138$$

$$= 17917. Kgf$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (22745 + 26870) = 49614 Kgf$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (20289 + 28667 + 17917 + 0) = 66872 Kgf$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (22745 + 17917 + 0) = 40661 Kgf$$

Summary of Failure Path Calculations:

Path 1-1 = 49614 Kgf, must exceed W = 0 Kgf	or W1 = 18817 Kgf
Path 2-2 = 66872 Kgf, must exceed W = 0 Kgf	or W2 = 5103 Kgf
Path 3-3 = 40661 Kgf, must exceed W = 0 Kgf	or W3 = 19984 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6.4 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1.03 bars

The Drop for this Nozzle is : 6.1796 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 344.1796 mm.

Input Echo, WRC107/537 Item 1, Description: N2-8" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm.
Vessel Diameter	Dv	1935.000	mm.
Vessel Thickness	Tv	8.000	mm.
Design Temperature	T1	200.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm2
Vessel Hot S.I. Allowable	Smh	138.00	N./mm2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round
WRC107 Attachment Classification	Holsol	Hollow

Diameter Basis for Nozzle	Nbasis	OD
Corrosion Allowance for Nozzle	Can	3.0000 mm.
Nozzle Diameter	Dn	219.075 mm.



Nozzle Thickness	Tn	12.700	mm.
Nozzle Material		SA-106 B	
Nozzle UNS Number		K03006	
Nozzle Cold S.I. Allowable	SNmc	118.00	N./mm2
Nozzle Hot S.I. Allowable	SNmh	118.00	N./mm2
Thickness of Reinforcing Pad	Tpad	8.000	mm.
Diameter of Reinforcing Pad	Dpad	350.000	mm.
Design Internal Pressure	Dp	6.624	bars
Include Pressure Thrust		No	

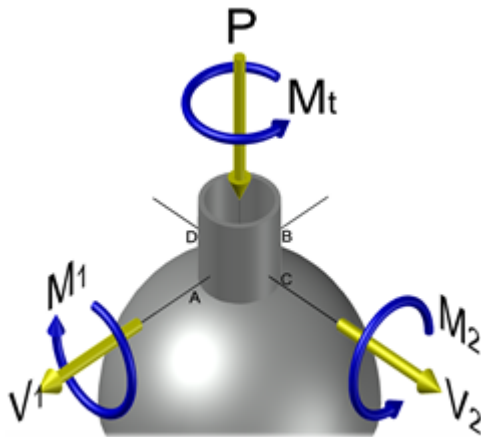
External Forces and Moments in WRC 107/537 Convention:

Radial Load	(SUS)	P	1529.7	Kgf
Longitudinal Shear	(SUS)	(V1) V1	1019.8	Kgf
Circumferential Shear	(SUS)	(Vc) V2	1019.8	Kgf
Circumferential Moment	(SUS)	(Mc) M1	570.9	Kg-m.
Longitudinal Moment	(SUS)	(Ml) M2	570.9	Kg-m.
Torsional Moment	(SUS)	Mt	815.5	Kg-m.

Use Interactive Control No
WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No
Compute Pressure Stress per WRC-368 No
Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))} \\
 &= 219.075 + 2 \cdot 1.65 \cdot \sqrt{(973.0 (8.0 - 3.0))} \\
 &= 449.249 \text{ mm.}
 \end{aligned}$$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load		P	1529.7	Kgf
Circumferential Shear	(VC)	V2	1019.8	Kgf
Longitudinal Shear	(VL)	V1	1019.8	Kgf
Circumferential Moment	(MC)	M1	570.9	Kg-m.
Longitudinal Moment	(ML)	M2	570.9	Kg-m.
Torsional Moment		MT	815.5	Kg-m.

Dimensionless Parameters: U = 0.97 TAU = 10.79 RHO = 1.34

Dimensionless Loads for Spherical Shells at Attachment Junction:



Curves read for 1979	Figure	Value	Location
N(x) * T / P	SP 2	0.03432	(A,B,C,D)
M(x) / P	SP 2	0.03837	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SM 2	0.05047	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SM 2	0.06586	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SM 2	0.05047	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SM 2	0.06586	(A,B,C,D)
N(y) * T / P	SP 2	0.10543	(A,B,C,D)
M(y) / P	SP 2	0.01663	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SM 2	0.10934	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SM 2	0.03459	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SM 2	0.10934	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SM 2	0.03459	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		-3.0	-3.0	-3.0	-3.0	-3.0	-3.0	-3.0	-3.0
Rad. Bend. P		-20.4	20.4	-20.4	20.4	-20.4	20.4	-20.4	20.4
Rad. Memb. MC		0.0	0.0	0.0	0.0	-14.8	-14.8	14.8	14.8
Rad. Bend. MC		0.0	0.0	0.0	0.0	-116.1	116.1	116.1	-116.1
Rad. Memb. ML		-14.8	-14.8	14.8	14.8	0.0	0.0	0.0	0.0
Rad. Bend. ML		-116.1	116.1	116.1	-116.1	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-154.5	118.7	107.5	-83.9	-154.5	118.7	107.5	-83.9
Tang. Memb. P		-9.4	-9.4	-9.4	-9.4	-9.4	-9.4	-9.4	-9.4
Tang. Bend. P		-8.9	8.9	-8.9	8.9	-8.9	8.9	-8.9	8.9
Tang. Memb. MC		0.0	0.0	0.0	0.0	-32.1	-32.1	32.1	32.1
Tang. Bend. MC		0.0	0.0	0.0	0.0	-61.0	61.0	61.0	-61.0
Tang. Memb. ML		-32.1	-32.1	32.1	32.1	0.0	0.0	0.0	0.0
Tang. Bend. ML		-61.0	61.0	61.0	-61.0	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-111.4	28.4	74.9	-29.4	-111.4	28.4	74.9	-29.4
Shear VC		2.2	2.2	-2.2	-2.2	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-2.2	-2.2	2.2	2.2
Shear MT		8.2	8.2	8.2	8.2	8.2	8.2	8.2	8.2
Tot. Shear		10.4	10.4	5.9	5.9	5.9	5.9	10.4	10.4
Str. Int.		156.8	119.9	108.5	84.6	155.3	119.1	110.5	85.8

Unitless Prm: U = 2.51 TAU = 0.00 (17.54) RHO = 0.00 (0.52)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SR 2	0.02175	(A,B,C,D)
M(x) / P	SR 2	0.00990	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SR 3	0.01691	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SR 3	0.01053	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SR 3	0.01691	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SR 3	0.01053	(A,B,C,D)
N(y) * T / P	SR 2	0.00650	(A,B,C,D)
M(y) / P	SR 2	0.00300	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SR 3	0.00507	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SR 3	0.00317	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SR 3	0.00507	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SR 3	0.00317	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		-13.1	-13.1	-13.1	-13.1	-13.1	-13.1	-13.1	-13.1
Rad. Bend. P		-35.6	35.6	-35.6	35.6	-35.6	35.6	-35.6	35.6
Rad. Memb. MC		0.0	0.0	0.0	0.0	-54.3	-54.3	54.3	54.3
Rad. Bend. MC		0.0	0.0	0.0	0.0	-202.8	202.8	202.8	-202.8
Rad. Memb. ML		-54.3	-54.3	54.3	54.3	0.0	0.0	0.0	0.0
Rad. Bend. ML		-202.8	202.8	202.8	-202.8	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-305.8	171.1	208.4	-125.9	-305.8	171.1	208.4	-125.9
Tang. Memb. P		-3.9	-3.9	-3.9	-3.9	-3.9	-3.9	-3.9	-3.9
Tang. Bend. P		-10.8	10.8	-10.8	10.8	-10.8	10.8	-10.8	10.8
Tang. Memb. MC		0.0	0.0	0.0	0.0	-16.3	-16.3	16.3	16.3
Tang. Bend. MC		0.0	0.0	0.0	0.0	-61.2	61.2	61.2	-61.2
Tang. Memb. ML		-16.3	-16.3	16.3	16.3	0.0	0.0	0.0	0.0
Tang. Bend. ML		-61.2	61.2	61.2	-61.2	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-92.1	51.8	62.7	-38.0	-92.1	51.8	62.7	-38.0
Shear VC		3.6	3.6	-3.6	-3.6	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-3.6	-3.6	3.6	3.6
Shear MT		8.3	8.3	8.3	8.3	8.3	8.3	8.3	8.3
Tot. Shear		12.0	12.0	4.7	4.7	4.7	4.7	12.0	12.0
Str. Int.		306.4	172.3	208.5	126.1	305.9	171.3	209.4	127.5

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		24.7	24.7	24.7	24.7	24.7	24.7	24.7	24.7
Rad. Pl (SUS)		-17.9	-17.9	11.8	11.8	-17.9	-17.9	11.8	11.8
Rad. Q (SUS)		-136.6	136.6	95.7	-95.7	-136.6	136.6	95.7	-95.7
Long. Pm (SUS)		24.7	24.7	24.7	24.7	24.7	24.7	24.7	24.7
Long. Pl (SUS)		-41.5	-41.5	22.8	22.8	-41.5	-41.5	22.8	22.8
Long. Q (SUS)		-69.9	69.9	52.1	-52.1	-69.9	69.9	52.1	-52.1
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		2.2	2.2	-2.2	-2.2	-2.2	-2.2	2.2	2.2
Shear Q (SUS)		8.2	8.2	8.2	8.2	8.2	8.2	8.2	8.2
Pm (SUS)		24.7	24.7	24.7	24.7	24.7	24.7	24.7	24.7
Pm+Pl (SUS)		24.0	24.0	47.9	47.9	24.0	24.0	47.9	47.9
Pm+Pl+Q (Total)		132.1	144.6	133.3	59.8	130.5	143.8	135.3	61.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	24.73	138.00	Passed
Pm+Pl (SUS)	47.95	207.00	Passed
Pm+Pl+Q (TOTAL)	144.61	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	64.3	64.3	64.3	64.3	64.3	64.3	64.3	64.3
Rad.	Pl (SUS)	-67.4	-67.4	41.3	41.3	-67.4	-67.4	41.3	41.3
Rad.	Q (SUS)	-238.4	238.4	167.1	-167.1	-238.4	238.4	167.1	-167.1
Long.	Pm (SUS)	64.3	64.3	64.3	64.3	64.3	64.3	64.3	64.3
Long.	Pl (SUS)	-20.2	-20.2	12.4	12.4	-20.2	-20.2	12.4	12.4
Long.	Q (SUS)	-72.0	72.0	50.4	-50.4	-72.0	72.0	50.4	-50.4
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	3.6	3.6	-3.6	-3.6	-3.6	-3.6	3.6	3.6
Shear	Q (SUS)	8.3	8.3	8.3	8.3	8.3	8.3	8.3	8.3
Pm (SUS)		64.3	64.3	64.3	64.3	64.3	64.3	64.3	64.3
Pm+Pl (SUS)		47.7	47.7	106.0	106.0	47.7	47.7	106.0	106.0
Pm+Pl+Q (Total)		242.2	236.5	272.8	88.4	241.6	235.5	273.7	91.1

Vessel Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	64.29	138.00	Passed
Pm+Pl (SUS)	106.00	207.00	Passed
Pm+Pl+Q (TOTAL)	273.66	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: K6-2"

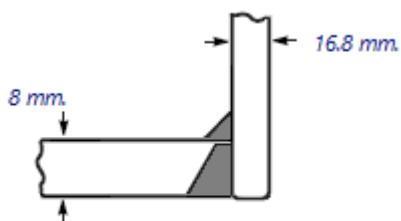
From: 40

Pressure for Reinforcement Calculations	P	6.624	bars
Temperature for Internal Pressure	Temp	200	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	1075.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	8.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	350.0000	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	480.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		8.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Overall MAWP.



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: K6-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 6.62 \cdot 1081 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 6.62)$
 $= 2.3240$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.62 \cdot 28.4 / (138 \cdot 1 - 0.6 \cdot 6.62)$
 $= 0.1367$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.5729 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	123.5572	mm.
Parallel to Vessel Wall, opening length	d	61.7786	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	12.5000	mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138/138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138/138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

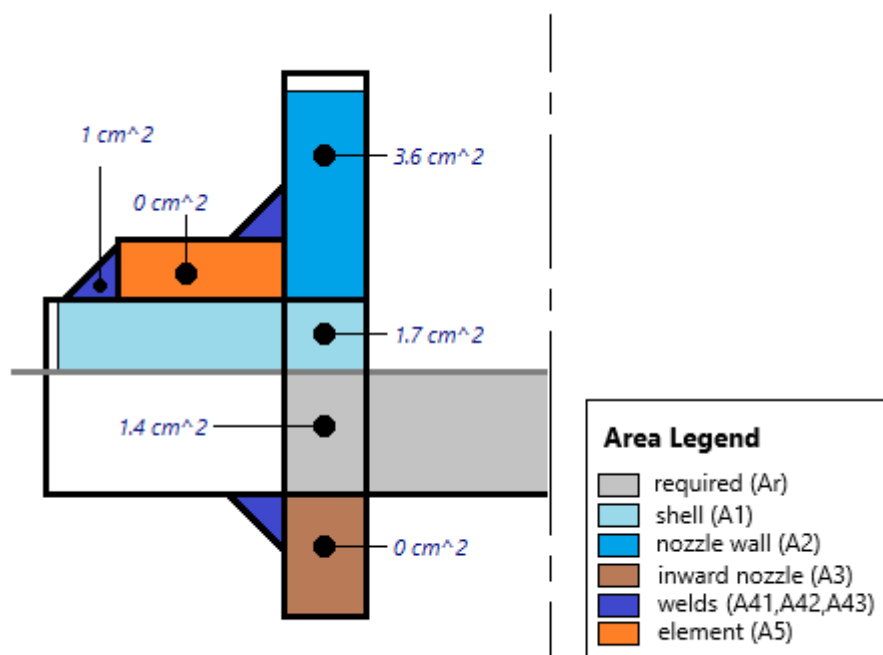


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)



AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	1.436	0.887	NA
Area in Shell	A1	1.653	1.315	NA
Area in Nozzle Wall	A2	3.627	3.511	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	6.280	5.826	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 66.84 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (61.8 \cdot 2.32 \cdot 1 + 2 \cdot 13.8 \cdot 2.32 \cdot 1 \cdot (1 - 1))$$

$$= 1.436 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 61.8 (1 \cdot 5 - 1 \cdot 2.32) - 2 \cdot 13.8$$

$$(1 \cdot 5 - 1 \cdot 2.32) \cdot (1 - 1)$$

$$= 1.653 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp) (tn - trn) fr2 / \sin(\alpha3)$$

$$= (2 \cdot 12.5) (13.8 - 0.14) / \sin(70)$$

$$= 3.627 \text{ cm}^2$$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 8, tr = 2.32, c = 3 mm., E* = 1

Thickness Ratio = tr * E* / (tg - c) = 0.46, Temp. Reduction = 37 °C

Min Metal Temp. w/o impact per UCS-66, Curve A -8 °C

Min Metal Temp. at Required thickness (UCS 66.1) -45 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -45 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

$$\text{Design Pressure/Ambient Rating} = 6.62/51.00 = 0.130$$

Weld Size Calculations, Description: K6-2"

Intermediate Calc. for nozzle/shell Welds Tmin 7.0000 mm.

Results Per UW-16.1:

Required Thickness Actual Thickness
Nozzle Weld 4.9000 = 0.7 * tmin. 7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (1.44 - 1.65 + 2 \cdot 13.8 \cdot 1 \cdot (1 \cdot 5 - 2.32)) \cdot 138)$$

$$= 730.52 \text{ Kg}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (3.63 + 0 + 1 - 0 \cdot 1) \cdot 138$$

$$= 6510.34 \text{ Kg}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (3.63 + 0 + 1 + (1.38)) \cdot 138$$

$$= 8447.16 \text{ Kg}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (3.63 + 0 + 1 + 0 + (1.38)) \cdot 138$$

$$= 8447.16 \text{ Kg}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 91.7 \cdot 10 \cdot 0.49 \cdot 138$$

$$= 9934. \text{ Kg}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 38.4) \cdot (16.8 - 3) \cdot 0.7 \cdot 138$$

$$= 16345. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 91.7 \cdot (8 - 3) \cdot 0.74 \cdot 138$$

$$= 7501. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (9934 + 16345) = 26279 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (9934 + 0 + 7501 + 0) = 17435 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (9934 + 7501 + 0) = 17435 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 26278	Kgf, must exceed W = 730 Kg	or W1 = 6510 Kg
Path 2-2 = 17435	Kgf, must exceed W = 730 Kg	or W2 = 8447 Kg
Path 3-3 = 17435	Kgf, must exceed W = 730 Kg	or W3 = 8447 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6.4 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1.03 bars

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K6-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 6.62 \cdot 1081 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 6.62)$
 $= 2.3240$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.62 \cdot 28.4 / (138 \cdot 1 - 0.6 \cdot 6.62)$
 $= 0.1367$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.5729 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 113.6000 mm.
 Parallel to Vessel Wall, opening length d 56.8000 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 12.5000 mm.

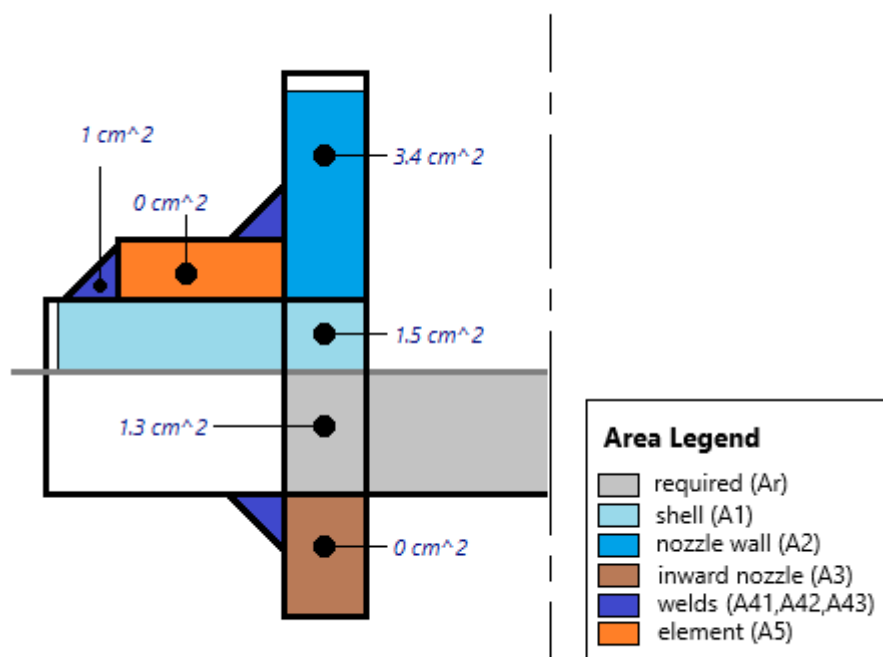


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	1.320	0.815	NA
Area in Shell	A1	1.520	1.209	NA
Area in Nozzle Wall	A2	3.407	3.298	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	5.927	5.507	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (56.8 \cdot 2.32 \cdot 1 + 2 \cdot 13.8 \cdot 2.32 \cdot 1 \cdot (1 - 1))$
 $= 1.320 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 56.8(1 \cdot 5 - 1 \cdot 2.32) - 2 \cdot 13.8$
 $(1 \cdot 5 - 1 \cdot 2.32) \cdot (1 - 1)$
 $= 1.520 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tnp)(tn - trn)fr2$
 $= (2 \cdot 12.5)(13.8 - 0.14)1$
 $= 3.407 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 1 + (0)^2 \cdot 1$
 $= 1.000 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.5729 mm.
Wall Thickness per UG16(b),	tr16b = 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 5.5768 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 5.5768 mm.
Wall Thickness, shell/head, external pressure	trb2 = 3.4021 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm.
Wall Thickness per table UG-45	tb3 = 7.8000 mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(5.58, 4.5)]$
 $= 5.5768 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.57, 5.58)$
 $= 5.5768 \text{ mm.}$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Weld Size Calculations, Description: K6-2"

Intermediate Calc. for nozzle/shell welds Tmin 7.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.9000 = 0.7 \cdot tmin.$	$7.0700 = 0.7 \cdot Wo \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr))Sv)$
 $= \max(0, (1.32 - 1.52 + 2 \cdot 13.8 \cdot 1 \cdot$
 $(1 \cdot 5 - 2.32))138)$
 $= 755.19 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (Wi - Can/.707)^2 \cdot fr2) \cdot Sv$



$$= (3.41 + 0 + 1 - 0 \cdot 1) \cdot 138$$

$$= 6201.09 \text{ Kg}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$

$$= (3.41 + 0 + 1 + (1.38)) \cdot 138$$

$$= 8137.90 \text{ Kg}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (3.41 + 0 + 1 + 0 + (1.38)) \cdot 138$$

$$= 8137.90 \text{ Kg}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.14/2.0) \cdot 84.3 \cdot 10 \cdot 0.49 \cdot 138$$

$$= 9133. \text{ Kg}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 35.3) \cdot (16.8 - 3) \cdot 0.7 \cdot 138$$

$$= 15028. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 84.3 \cdot (8 - 3) \cdot 0.74 \cdot 138$$

$$= 6897. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (9133 + 15028) = 24161 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (9133 + 0 + 6897 + 0) = 16030 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (9133 + 6897 + 0) = 16030 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 24160	Kgf, must exceed W = 755 Kg	or W1 = 6201 Kg
Path 2-2 = 16029	Kgf, must exceed W = 755 Kg	or W2 = 8137 Kg
Path 3-3 = 16029	Kgf, must exceed W = 755 Kg	or W3 = 8137 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6.4 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1.03 bars

The Drop for this Nozzle is : 17.3820 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 505.9496 mm.

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Input, Nozzle Desc: K7-2"

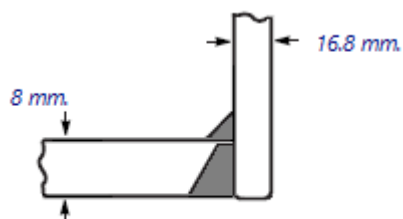
From: 40

Pressure for Reinforcement Calculations	P	6.624	bars
Temperature for Internal Pressure	Temp	200	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	1075.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	8.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	350.0000	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	480.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		8.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Overall MAWP.



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: K7-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 6.62 \cdot 1081 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 6.62)$
 $= 2.3240$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.62 \cdot 28.4 / (138 \cdot 1 - 0.6 \cdot 6.62)$
 $= 0.1367$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.5729 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	123.5572	mm.
Parallel to Vessel Wall, opening length	d	61.7786	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	12.5000	mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138/138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138/138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

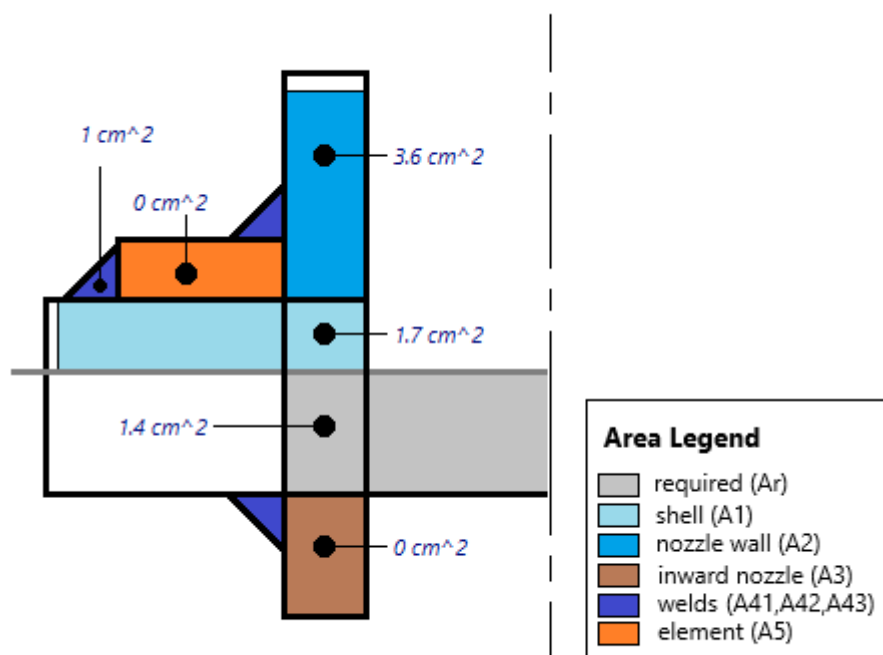


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)



AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	1.436	0.887	NA
Area in Shell	A1	1.653	1.315	NA
Area in Nozzle Wall	A2	3.627	3.511	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	6.280	5.826	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 66.84 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (61.8 \cdot 2.32 \cdot 1 + 2 \cdot 13.8 \cdot 2.32 \cdot 1 \cdot (1 - 1))$$

$$= 1.436 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 61.8 (1 \cdot 5 - 1 \cdot 2.32) - 2 \cdot 13.8$$

$$(1 \cdot 5 - 1 \cdot 2.32) \cdot (1 - 1)$$

$$= 1.653 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp) (tn - trn) fr2 / \sin(\alpha3)$$

$$= (2 \cdot 12.5) (13.8 - 0.14) / \sin(70)$$

$$= 3.627 \text{ cm}^2$$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 8, tr = 2.32, c = 3 mm., E* = 1

Thickness Ratio = tr * E* / (tg - c) = 0.46, Temp. Reduction = 37 °C

Min Metal Temp. w/o impact per UCS-66, Curve A -8 °C

Min Metal Temp. at Required thickness (UCS 66.1) -45 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -45 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

$$\text{Design Pressure/Ambient Rating} = 6.62/51.00 = 0.130$$

Weld Size Calculations, Description: K7-2"

Intermediate Calc. for nozzle/shell Welds Tmin 7.0000 mm.

Results Per UW-16.1:

Required Thickness Actual Thickness
Nozzle Weld 4.9000 = 0.7 * tmin. 7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (1.44 - 1.65 + 2 \cdot 13.8 \cdot 1 \cdot (1 \cdot 5 - 2.32)) \cdot 138)$$

$$= 730.52 \text{ Kg}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (3.63 + 0 + 1 - 0 \cdot 1) \cdot 138$$

$$= 6510.34 \text{ Kg}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (3.63 + 0 + 1 + (1.38)) \cdot 138$$

$$= 8447.16 \text{ Kg}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (3.63 + 0 + 1 + 0 + (1.38)) \cdot 138$$

$$= 8447.16 \text{ Kg}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 91.7 \cdot 10 \cdot 0.49 \cdot 138$$

$$= 9934. \text{ Kg}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 38.4) \cdot (16.8 - 3) \cdot 0.7 \cdot 138$$

$$= 16345. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 91.7 \cdot (8 - 3) \cdot 0.74 \cdot 138$$

$$= 7501. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (9934 + 16345) = 26279 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (9934 + 0 + 7501 + 0) = 17435 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (9934 + 7501 + 0) = 17435 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 26278	Kgf, must exceed W = 730 Kg	or W1 = 6510 Kg
Path 2-2 = 17435	Kgf, must exceed W = 730 Kg	or W2 = 8447 Kg
Path 3-3 = 17435	Kgf, must exceed W = 730 Kg	or W3 = 8447 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6.4 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1.03 bars

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K7-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 6.62 \cdot 1081 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 6.62)$
 $= 2.3240$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.62 \cdot 28.4 / (138 \cdot 1 - 0.6 \cdot 6.62)$
 $= 0.1367$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.5729 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 113.6000 mm.
 Parallel to Vessel Wall, opening length d 56.8000 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 12.5000 mm.

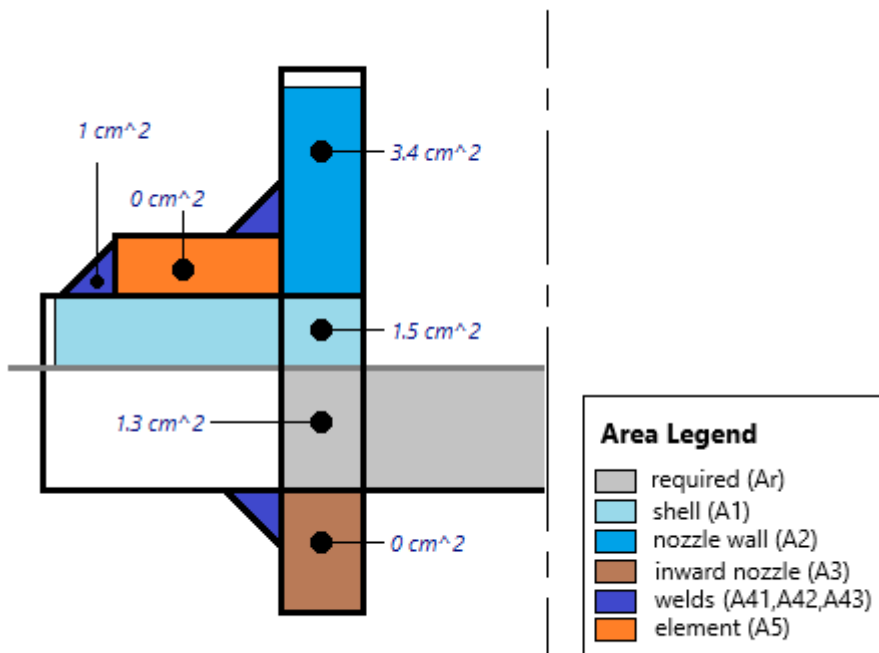


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	1.320	0.815	NA
Area in Shell	A1	1.520	1.209	NA
Area in Nozzle Wall	A2	3.407	3.298	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	5.927	5.507	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (56.8 \cdot 2.32 \cdot 1 + 2 \cdot 13.8 \cdot 2.32 \cdot 1 \cdot (1 - 1))$
 $= 1.320 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 56.8(1 \cdot 5 - 1 \cdot 2.32) - 2 \cdot 13.8$
 $(1 \cdot 5 - 1 \cdot 2.32) \cdot (1 - 1)$
 $= 1.520 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tnp)(tn - trn)fr2$
 $= (2 \cdot 12.5)(13.8 - 0.14)1$
 $= 3.407 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 1 + (0)^2 \cdot 1$
 $= 1.000 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.5729 mm.
Wall Thickness per UG16(b),	tr16b = 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 5.5768 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 5.5768 mm.
Wall Thickness, shell/head, external pressure	trb2 = 3.4021 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm.
Wall Thickness per table UG-45	tb3 = 7.8000 mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(5.58, 4.5)]$
 $= 5.5768 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.57, 5.58)$
 $= 5.5768 \text{ mm.}$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Weld Size Calculations, Description: K7-2"

Intermediate Calc. for nozzle/shell welds Tmin 7.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.9000 = 0.7 \cdot tmin.$	$7.0700 = 0.7 \cdot Wo \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr))Sv)$
 $= \max(0, (1.32 - 1.52 + 2 \cdot 13.8 \cdot 1 \cdot$
 $(1 \cdot 5 - 2.32))138)$
 $= 755.19 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (Wi - Can/0.707)^2 \cdot fr2) \cdot Sv$



$$= (3.41 + 0 + 1 - 0 \cdot 1) \cdot 138$$

$$= 6201.09 \text{ Kg}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$

$$= (3.41 + 0 + 1 + (1.38)) \cdot 138$$

$$= 8137.90 \text{ Kg}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (3.41 + 0 + 1 + 0 + (1.38)) \cdot 138$$

$$= 8137.90 \text{ Kg}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.14/2.0) \cdot 84.3 \cdot 10 \cdot 0.49 \cdot 138$$

$$= 9133. \text{ Kg}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 35.3) \cdot (16.8 - 3) \cdot 0.7 \cdot 138$$

$$= 15028. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 84.3 \cdot (8 - 3) \cdot 0.74 \cdot 138$$

$$= 6897. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (9133 + 15028) = 24161 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (9133 + 0 + 6897 + 0) = 16030 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (9133 + 6897 + 0) = 16030 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 24160	Kgf, must exceed W = 755 Kg	or W1 = 6201 Kg
Path 2-2 = 16029	Kgf, must exceed W = 755 Kg	or W2 = 8137 Kg
Path 3-3 = 16029	Kgf, must exceed W = 755 Kg	or W3 = 8137 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6.4 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1.03 bars

The Drop for this Nozzle is : 17.3820 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 505.9496 mm.

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PV Elite® Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: